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VOLCOMM: VOLUNTEER AND COMMUNITY SERVICE HUB FOR NON-GOVERNMENTAL ORGANIZATIONS (NGOS)

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SUPERVISOR APPROVAL

VOLCOMM: VOLUNTEER ANDCOMMUNITY SERVICE HUB FOR NON-GOVERNMENTAL ORGANIZATIONS (NGOS)

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This proposal was prepared under the supervision of the project supervisor, Dr. Nur Suhailayani Binti Suhaimi. It was submitted to the College of Computing, Informatics and Mathematics and was accepted in partial fulfilment of the requirements for the degree of Bachelor of Information Systems (Hons.) Information System Engineering.

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STUDENT DECLARATION

I certify that this proposal and the project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practices of the discipline.

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ABSTRACT

VolComm is a platform that helps NGOs and volunteers manage activities more easily and effectively. It allows volunteers to register, create profiles, and find events that match their skills and interests. NGOs can organize and promote events, manage volunteers, and track participation. The system includes a smart matching feature that connects volunteers with suitable activities and an automated digital certificate system to recognize their contributions. It also has a communication tool for instant updates and a feedback system to improve event quality. Based on conducted survey, 87% of users found VolComm helpful in organizing events, while 92% said it made finding volunteer opportunities easier. To improve the platform, more volunteer categories will be added, making it easier for people with different skills to participate. A geolocation feature will help volunteers find nearby events, making it more convenient to get involved. Additionally, a donation and crowdfunding system will allow NGOs to raise funds directly through the platform to support their activities. These improvements aim to make VolComm a more user-friendly and effective tool for both volunteers and NGOs.

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CHAPTER 1

INTRODUCTION

This system is entitled the VolComm: Volunteer and Community Service Hub For Non-Governmental Organizations (NGO). The detailed information and motivation for the VolComm are explained in this chapter. The project background and problem statement are presented at the beginning of this chapter, followed by the aim and objectives. This chapter also includes VolComm's significance, scopes, thesis outline, and project summary.

1.1 Background of Study

Group Volunteer Malaysia (GVM), led by En Nor Izzuddin Shah, is a Facebook group created to make volunteer recruitment in Malaysia easier. The group connects people who want to offer their time and skills with various Non-Governmental Organizations (NGOs) in the country. NGOs use this platform to share their needs for volunteers, from short-term to long-term projects. Other than that, people interested in volunteering can easily find opportunities that match their skills, interests, and availability. The GVM group plays an important role in bringing together volunteers and NGOs, helping to create a strong sense of community. Volunteers can participate in activities like sharing food with the homeless or helping communities affected by disasters. This collaboration helps raise awareness about important social issues and encourages more people to get involved. Group Volunteer Malaysia supports these connections, making it easier for both volunteers and NGOs to work together for positive change. The platform also helps create a space for sharing experiences, learning new skills, and building meaningful relationships. By fostering such a network, GVM strengthens the volunteer culture in Malaysia and promotes the values of empathy, cooperation, and service to others. Figure 1.1 shows some of the organizations in Group Volunteer Malaysia and the services they promoted.

Table 1.1 List of Organizations and Their Volunteer Services in Group Volunteer Malaysia

Organization	Services Promoted
Kembara Kitchen	Distributing food to the homeless, disaster relief, and community service efforts
Bekind	Distributing food to the homeless, disaster relief, and community service efforts
Bevital	Distributing food to the homeless, disaster relief, and community service efforts
Hopes Malaysia	Distributing food to the homeless, disaster relief, and community service efforts
Teach For Malaysia	Education programs for underprivileged students, community service efforts
The Lost Food Project	Distributing food to the homeless, disaster relief, and community service efforts

Figure 1.1 illustrates how organizations and volunteers can link up via Facebook. Organizations post volunteer opportunities on the group, and volunteers search through these listings to identify suitable matches. Once the volunteer found an opportunity that catches their attention, they fill the application form that are stated on the post. The organization evaluates the application and determines if they will approve or reject it. If the volunteer is approved, the organization will get in touch to move forward. If the volunteer is not accepted, the process will stop for them. Volunteers who are accepted may also be contacted by the organizer.

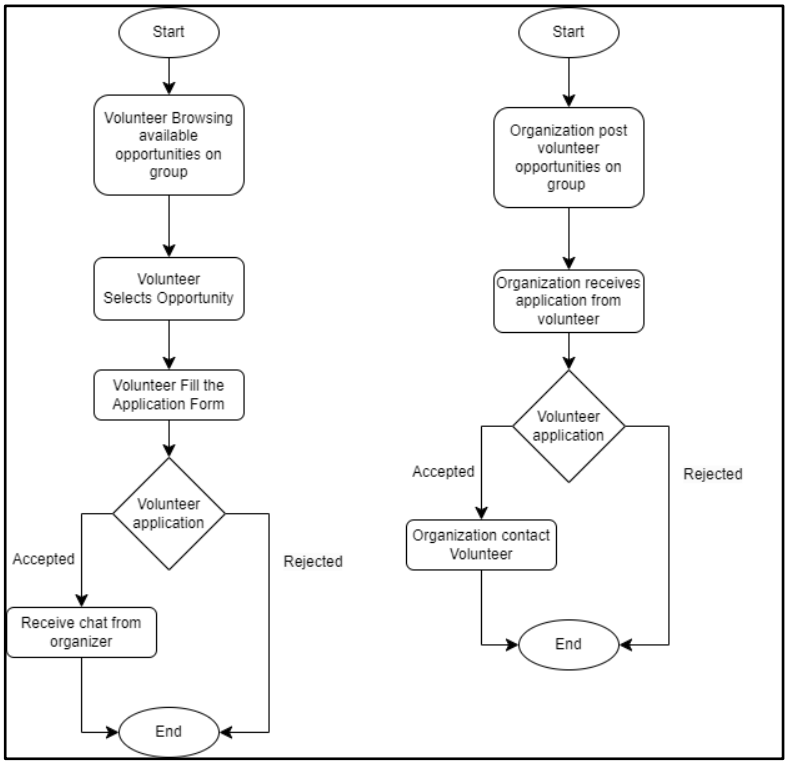


Figure 1.1 Flowchart of current process

1.2 Problem Statement

Generally, the GVM is managed mainly by doing things manually. This leads to three main problems which is volunteers find it difficult to search for suitable activities, there is a lack of a centralized platform to manage volunteers and events, and lack of feature to generate and manage volunteers certificate records. Further elaboration on each of these problems are as follows:

a) Volunteer difficult to search suitable activities based on preferences

According to En Nor Izzuddin Shah, volunteers face difficulties in finding suitable group activities because there are no filtering features available. In current platform, volunteers often spend a lot of time and effort searching for activities that match their skills and interests since they must go through many individual postings manually. This cumbersome process can lead to frustration and reduces the chances of volunteers finding activities they genuinely enjoy. Additionally, the lack of filtering options makes it hard to identify events based on specific criteria. Information overload and the absence of detailed information about events can also demotivate volunteers. Furthermore, there is a lack of matching features to help align volunteer preferences with available opportunities.

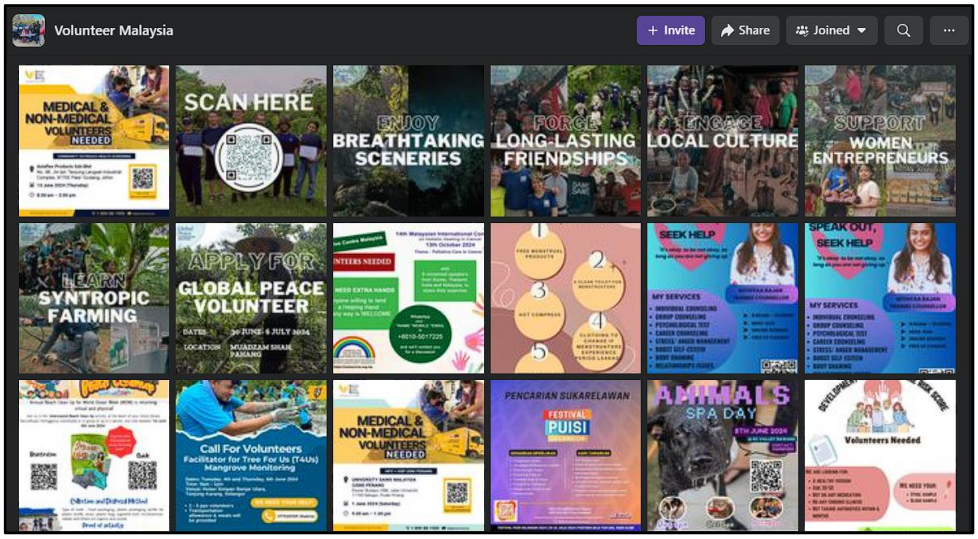


Figure 1.2 Volunteer opportunities available on the group

Figure 1.2 shows the posts available in the group about the volunteer opportunities, but the posts are not organized properly.

b) Lack of centralized platform to manage volunteers

Without a central platform, volunteers have to go to different places like Google Forms or WhatsApp to sign up or contact organizations. This creates confusion and inconvenience as volunteers frequently switch between platforms. It also leads to redundancy because volunteers often have to fill in the same information on multiple Google Forms. Additionally, this situation reduces the efficiency and effectiveness of volunteer recruitment for both individuals and organizations, as there is no single place to manage everything. Redundancy can result in missed deadlines and events being poorly managed, especially when volunteers face last-minute cancellations.

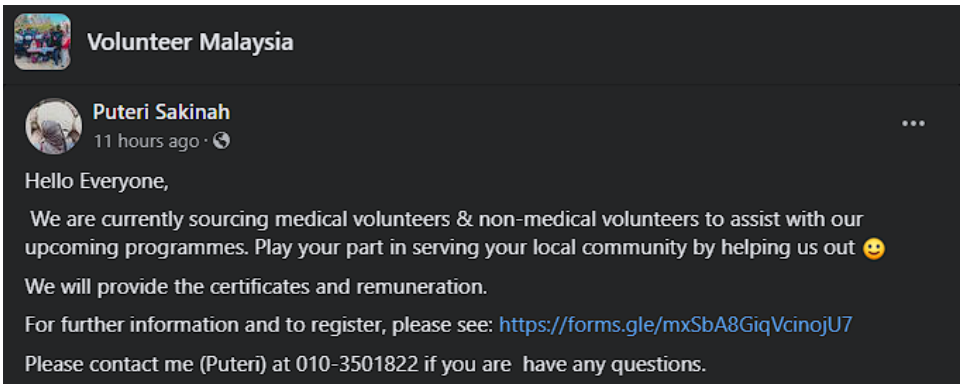


Figure 1.3 Volunteer opportunities post on group

Figure 1.3 shows a volunteer opportunity post in the group. Each post includes a link that directs users to a Google Form to fill out their details and register.

c) Lack of feature to generate and keep volunteers' certificate record

According to En Nor Izzudin Shah, the admin of Group Volunteer Malaysia, volunteers often find it hard to keep track of their certificates because there is no centralized system. Based on his own experience, this

makes it difficult for volunteers to show their skills and achievements, and for organizations to recognize and reward their efforts. Additionally, some organizers do not provide certificates because creating them can be troublesome.

1.3 Project Aim

VolComm aims to enhance the volunteer experience by streamlining the process of connecting volunteers with organizations, offering a user-friendly interface for browsing and signing up for volunteer opportunities, and implementing a certificate management feature to enable volunteers to easily access and showcase their achievements. The platform also provides a centralized space for volunteers to find opportunities that match their skills and interests, encouraging greater participation. Additionally, VolComm helps build a sense of community among volunteers, fostering collaboration and creating a network of individuals dedicated to positive social impact.

1.4 Project Objectives

The three objectives of this project are:

a) To gather data and analyze requirements of Volunteer and Community Service Hub for Non-Governmental Organizations (NGOs).
b) To design the system of Volunteer and Community Service Hub for Non-Governmental Organizations (NGOs).
c) To develop a system of Volunteer and Community Service Hub for Non-Governmental Organizations (NGOs) with Email Notification feature.

1.5 Project Scopes

This system would include volunteer opportunities throughout Peninsular Malaysia, making it accessible to everyone. Key features will include email notifications to inform volunteers when their applications are accepted and to

remind them when events are approaching. This helps ensure that volunteers stay updated and can prepare for their commitments. Additionally, the system will suggest events based on volunteers' preferences, making it easier for them to find activities that match their interests and skills.

It is important to note that the system would not provide any monetary benefits, including crowdfunding options, as its main focus is to connect volunteers and organizations in a straightforward way.

1.6 Project Significances

a) Significance to volunteers:

   i) Centralized Access: The VolComm system offers a centralized platform for volunteers to access various volunteering opportunities across Peninsular Malaysia.
   ii) Efficient Registration: Volunteers can quickly search for and register for activities that match their skills and interests, saving time and effort.
   iii) Certificate Management: The system includes a certificate management feature, allowing volunteers to easily track and display their accomplishments.

b) Significance to event or charity organiser:

   i) Simplified Recruitment: The VolComm system simplifies volunteer recruitment for event and charity organizers by providing a centralized platform across Peninsular Malaysia.
   ii) Effective Acknowledgment: The certificate management feature enables organizers to acknowledge and reward volunteers efficiently.

iii) Increased Volunteer Engagement: By offering an easy-to-use platform for finding and connecting with volunteers, organizers can attract more participants and increase engagement.

1.7 Outline of the thesis

This thesis is divided into five (5) main chapters. Below is an outline of each chapter:

a) The introduction, as previously outlined, has been covered in the Chapter 1
b) Summary of related study on the Volunteer and Community Service Hub for Non-Governmental Organizations (NGOs) are covered in Chapter 2.
c) Research and interview questions, as well as the methodology phases and approach will be used in this project, are covered in Chapter 3. The methodology study designs that have been used to learn more about this project are included in this chapter.
d) Chapter 4 presents the results found during the project based on the methods previously selected. This section highlights key discoveries and their implications.
e) Chapter 5 concludes the thesis by discussing challenges we faced and suggesting ways to improve the project in the future. This chapter emphasizes lessons learned and how they can inform future work.

1.8 Summary

The context of the study and the problem statement of the Volunteer and Community Service Hub for Non-Governmental Organizations (NGOs) were highlighted in this chapter. Also, the project aim and the objective have also been explained. Finally, the project scope is discussed, along with its significances.

CHAPTER 2

LITERATURE REVIEW

This chapter discusses the literature review of VolComm: Volunteer and Community Service Hub for Non-Governmental Organizations (NGOs). The information gathered in this chapter is based on various sources such as articles, books, journals, and other reliable materials related to this project. These sources provide a comprehensive foundation and understanding for the development and implementation of VolComm, ensuring that all aspects are thoroughly researched and supported by credible evidence.

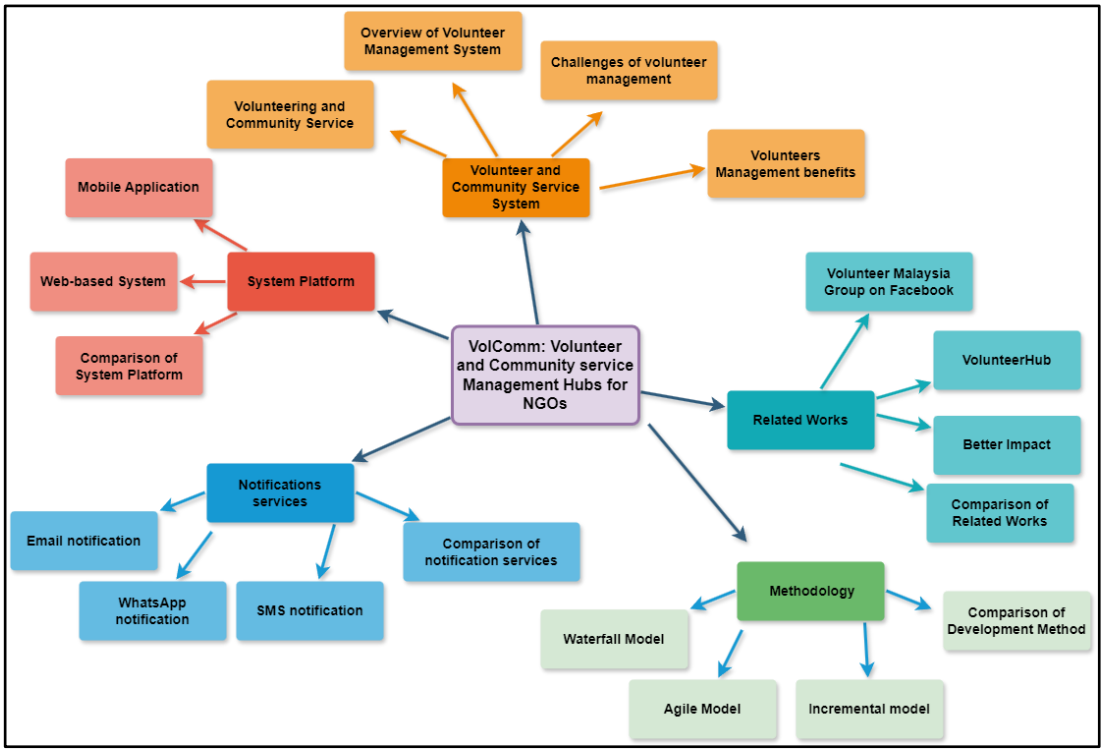


Figure 2.1 Overview of Literature Review for VolComm

Figure 2.1 gives an overview of the topics and subtopics explored in this chapter. The research topics are directly relevant to the project and include five distinct subtopics which are Volunteer and Community Service systems, system platforms, notification services, methodologies, and related works. Additionally, this chapter includes a summary to provide a concise review of the covered content

2.1 Volunteering and Community Services

According to Eyler and Giles (1999), community service happens when people help others or their community without getting paid. Different types of services, like charity care, health promotion, research and education, and service accessibility, are offered by these organizations (Ginn, Shen, & Moseley, 2009). Singh, Young, Loomer, and Madison (2018) stated that Nonprofit hospitals have a notable impact on communities by providing benefits like charity care, and performing community health needs assessments.

Volunteering refers to various activities undertaken by individuals during their leisure time to support others, whether through informal assistance or formal organizational involvement (Tucker & Gearhart, 2022). According to Lum and Lightfoot (2005), volunteering has been associated with positive effects on physical and mental health, particularly among older individuals. Mustafa, Radzi, and Hamzah (2020) outlined four primary motivations behind individuals' engagement in volunteering, which are the desire to contribute to the community, the opportunity to acquire new skills, heightened awareness of societal issues, and the expansion of social networks.

Community service programs have been discovered to be advantageous for not just the beneficiaries but also the volunteers and the organizations participating. Volunteers frequently receive both internal and external rewards from their involvement, which enhances their personal development and overall wellness (Haski-Leventhal, Hustinx, & Handy, 2011).

2.1.1 Volunteer Management System (VMS)

Volunteer management has evolved to focus on providing tailored practices that suit each organization's unique context, much like how human resource management operates (Piatak & Carman, 2023). Volunteer Management Systems (VMS) play a crucial role in this evolution by simplifying the process

of connecting volunteers with opportunities, organizing tasks, facilitating communication, and implementing evaluation and motivation strategies (Schonbock, Raab, Altmann, Kapsammer, Kusel, Proll, Retschitzegger, & Schwinger, 2016). According to Chui and Chan (2019), VMS also help organizations expand and diversify their volunteer pool by enabling efficient recruitment, screening, and onboarding of volunteers. VMS are essential across various sectors, including emergency and disaster management, global health training, and nonprofit organizations. Informal volunteers are a valuable resource during emergencies and disasters, underscoring the importance of understanding their roles and challenges (Whittaker, McLennan, & Handmer, 2015). According to MacNairn (2019), in global health programs VMS can foster mutually beneficial partnerships and optimize benefits while minimizing costs or harms. Organizational factors significantly impact volunteer coordination, necessitating a comprehensive approach to enhance volunteer engagement and effectiveness (Studer & Schnurbein, 2012).

2.1.2 Challenges of Volunteer Management

Jacob (2016) highlighted that some volunteers may prioritize their personal satisfaction above organizational objectives. Consequently, they may engage in disruptive behavior, such as breaking rules, particularly when rules or their enforcement impede their enjoyment or personal aspirations. According to Bilan (2019), volunteers frequently encounter challenges stemming from ambiguity surrounding their roles and the corresponding rules, which can result in inconsistent behavior and difficulties in meeting organizational expectations. Moreover, the uncertainty surrounding roles can contribute to high turnover rates, as volunteers disengage due to unclear expectations or dissatisfaction with their volunteer experience. This turnover not only disrupts organizational continuity but also poses challenges in maintaining a committed volunteer community. According to Cho, Wong, & Chiu (2020), these issues of role ambiguity and turnover continue to be significant challenges in volunteer management.

2.1.3 Volunteer Management Benefits

According to Cho, Wong, and Chiu (2020), effective volunteer management not only boosts volunteers' commitment but also bolsters their intention to remain engaged. This reduces the necessity for frequent recruitment efforts, consequently fostering organizational stability and continuity. A Volunteer Management System (VMS) serves as a crucial tool in this regard, offering enhanced coordination and communication between volunteers and staff members (Cnaan & Cascio, 1999). Such systems facilitate smoother collaboration and teamwork, thereby enhancing overall organizational effectiveness. Piatak and Carman (2023) also mentioned that good volunteer management practices, such as providing training, fostering an inclusive organizational climate, and offering organizational support, significantly enhance volunteers' intentions to continue volunteering for the organization.

2.2 System Platform

A System Platform refers to a comprehensive software framework that provides essential tools, services, and infrastructure for developing and managing complex systems efficiently (Smith & Johnson, 2019). It offers standardized components, such as operating systems, databases, and middleware, to support various applications and streamline development processes.

2.2.1 Mobile Application

Mobile applications, often referred to as apps, are software programs designed specifically for smartphones, tablets, and similar mobile devices to efficiently address user needs (Sørensen, De Reuver, & Basole, 2015). The development of mobile applications has seen significant growth due to the increasing popularity of smartphones among users (Hsiao, 2017). The main mobile platforms, iOS by Apple and Android by Google, have transformed customer expectations for mobile devices, fostering a diverse ecosystem of app

developers, from individuals to large companies (Iskren & Ivailo, 2022). Apps have become essential tools for various tasks, offering convenience and accessibility across different areas, from social networking to productivity and entertainment. According to Dobrica and Pietraru (2017), mobile apps offered advantages such as portability, accessibility, and enhanced value and quality when integrated with a website.

2.2.2 Web-based System

A web-based system is a software application or platform that can be accessed via a web browser on the internet, enabling users to reach it from any location globally (Murugeswari, Patan, Ravella, & Sanikireddy, 2022). Users do not need to install software or applications on their computers because everything is stored and managed on the server. Lo, Chan, and Yeh (2012) stated that delivering information to users is more efficient and effective with web-based systems.

One benefit of web-based systems is that they store all data in one place and allow access from any computer connected to the internet, as multiple users share a server that can be accessed simultaneously from anywhere (Jin, 2016). It is also advantageous in terms of compatibility, as it can operate on various operating systems and devices with minimal difficulty, ultimately reducing development costs (Rashid, Lim, Eng, Ping, Zainol, & Majid, 2016).

2.2.3 Comparison of System Platform

Mobile applications and web-based platforms are both utilized in software development, each offering unique strengths and weaknesses. Mobile applications provide accessibility and portability, while web-based platforms offer versatility and ease of maintenance. The choice between the two depends on the specific requirements and objectives of the project.

Table 2.1 Comparison between mobile application and web-based platforms

Aspect	Mobile Application	Web-based Platform
Accessibility	Accessible via mobile devices	Accessible via web browsers
Portability	Can be used offline	Requires internet connection
Development Approach	Native or hybrid development	Web development technologies

Table 2.1 (Cont.)

Installation	Requires download and installation	Accessible via URL
User Experience	Tailored for mobile devices	Consistent across devices
Performance	Generally faster and smoother	May vary based on browser and connection
Offline Functionality	Can function offline with limited features	Typically requires internet connection
Updates	Updates through app stores	Instant updates for all users
Maintenance	Requires updates for each platform	Centralized updates for all users

Table 2.1 compares mobile applications and web-based platforms. Mobile apps are designed for mobile devices, offering faster performance and offline functionality, but require installation from app stores and manual updates. They are optimized for specific devices, ensuring a tailored user experience. In contrast, web-based platforms can be accessed via any device with a web browser, require an internet connection, and offer a consistent experience across devices. They do not require installation and automatically apply updates, but their performance can varied based on internet connection and browser.

2.3 Notifications services

A notification service refers to a system that electronically sends messages to users' devices to convey information securely, aiming to avoid issues like excessive mail, privacy exposure, and delivery errors (Kim, 2021). Notifications, in the form of visual pop-ups or sounds, have been widely recognized as a useful method for assisting users in staying informed (Iqbal & Horvitz, 2010). On the other hand, Pielot, Dingler, Pedro, and Oliver (2015) defined notifications as cues generated by applications to relay information to users outside their current focus of attention, encompassing visual, auditory, or haptic alerts.

2.3.1 Email notification

Email notifications have emerged as a prevalent means of communication, allowing for the exchange of information across various contexts (Halder, Maheshwari, & Suresh, 2021). It is also a common and straightforward method of communication (Liu, Guo, Chou, & Wong, 2003). They are widely used for various purposes due to their ease of use and accessibility. Research has delved into the impact of notifications on user behavior, indicating that notifications from communication apps like email are generally considered important, while those from other types of apps may be ignored or even perceived as bothersome (Pielot et al., 2015). Moreover, the impact of email notifications on work habits has been investigated, revealing that individuals with higher levels of trait self-control are better equipped to resist or ignore new email notifications, enabling them to achieve task goals amidst work-email demands (Russell, Daniels, Jackson, & Fullman, 2021).

2.3.2 WhatsApp notification

WhatsApp, a mobile messaging app, facilitates message exchange without the need for SMS fees (Christiansen & Piekarz, 2019). Users can send messages manually by hitting the send button or automatically through a computerized system called a WhatsApp gateway. These gateways, such as notification systems, can transmit alerts and updates directly to recipients (Yulianto, Setia, & Atmaja, 2021). These notifications enable users to receive timely information without actively initiating the message sending process. For instance, in the context of disaster management, online social media networks like WhatsApp can provide traceable information and notifications to disaster victims (Çelebi, Tuna, & Çalişkan, 2024). Moreover, during the COVID-19 pandemic, notifications were manually shared through WhatsApp until the integration of COVID-19 into electronic surveillance systems (Almayahi et al., 2023). WhatsApp notifications offer real-time interactions and quick responses, making them suitable for immediate communication needs (Demirel, Öğüt, & Eşme, 2022).

2.3.3 SMS notification

SMS, or Short Message Service, is a popular mobile communication technology that lets users send and receive short text messages between mobile devices (Delia, Liyanage, Chen, Gyorodi, & He, 2018). Users can easily send short text messages between mobile phones, making it a convenient and effective mode of communication for different reasons.

Pandithurai, Suganya, Naresh, Deva, and Suresh (2013) claimed that SMS has the advantage of being compatible with all GSM mobile devices, and many mobile service providers have cost-effective SMS options available. According to Zefferer, Tauber, and Zwattendorfer (2013), SMS technology is user-friendly, making it accessible to those with limited technical skills who can send and receive messages without any setup or configuration.

2.3.4 Comparison of notification services

Email notification, WhatsApp notification, and SMS notification services are all utilized for communication purposes, each presenting its own advantages and drawbacks. Email notifications offer versatility and can convey detailed information, while WhatsApp notifications prioritize real-time interaction and multimedia sharing. On the other hand, SMS notifications provide simplicity and reliability, often reaching users even in areas with limited internet connectivity. The most suitable notification method depends on requirements and objectives of the project. Table 2.2 presents a comparison of email, WhatsApp, and SMS notification services.

Table 2.2 Comparison between Email, WhatsApp, and SMS notification services

Aspect	Email Notification	WhatsApp Notification	SMS Notification
Delivery Speed	Varies, can be delayed	Real-time	Nearly instant
Accessibility	Requires internet connection	Requires internet connection	Works with or without internet
Message Length	Typically longer	Limited characters	Limited characters
Multimedia Support	Yes, supports attachments	Yes, supports multimedia files	No

Table 2.2 (Cont.)

User Interaction	Limited interactivity	High interactivity	Limited interactivity
Cost	Often free or low-cost	Free	Often free or low-cost
Reliability	Subject to spam filters	Dependent on internet connection	Generally reliable

Table 2.2 outlined the differences between email, WhatsApp, and SMS notifications. Email notifications can take longer to reach you and might be delayed, while WhatsApp notifications come in real-time, meaning they arrive instantly. SMS notifications also come almost instantly and work even without internet. Emails allow longer messages and support attachments, but WhatsApp and SMS have limited character space. WhatsApp offers high interactivity with features like messaging and calling, while emails and SMS have limited interaction. WhatsApp and SMS notifications are usually free, but email services might have costs or limits. However, emails might face spam filters, while WhatsApp and SMS rely on internet connection for reliability, with SMS being generally dependable.

2.4 Methodology for VolComm Development

There are several methodologies were considered to be used for VolComm development. Methodology in software development refers to the set of principles, practices, and processes that guide the development of software systems (Tegegne, Seppänen, & Ahmad, 2019). Software development methodologies are essential for guiding developers on best practices and values throughout the development process to ensure project success (Winter, 2014). Gbaranwi, Precious, Ojekudo, and Akpofure (2021) stated that the choice of software development methodology depends on various factors, such as the project's complexity, the team's experience, the organization's culture, and the customer's needs. Essentially, they serve as roadmaps that outline the steps to be taken throughout the software development lifecycle, facilitating coordination, organization, and ultimately, the delivery of high-quality software products.

2.4.1 Waterfall Model

The traditional waterfall methodology in software development follows a sequential approach, involving distinct phases of analysis, design, implementation, testing, and maintenance (Cheatham & Crenshaw, 1991). This method is commonly employed in software development companies, although it presents challenges such as difficulty in accommodating changes and detecting defects late in the process (Petersen, Wohlin, & Baca, 2009). While Nugraha, Usman, and Tiawan (2021) agreed that the waterfall approach is most suitable for developing generic systems or software, as it entails identifying all requirements upfront through a general specification.

The advantage of the waterfall model is that it gives a clear plan for managing a software project (Kannan & Jhajharia, 2014). It helps spot design issues and mistakes early on, saving time during development. Ben-Zahia and Jaluta (2014) mentioned that the waterfall model is well-suited for software development projects that have well-defined requirements and are less user-focused, such as compilers and operating systems. The waterfall model's sequential nature makes it easier to document and manage the software development process, as each phase is completed before moving to the next (Ruparelia, 2010).

According to Pargaonkar (2023), the Waterfall method has some disadvantage which are the strict step-by-step process can make it hard to make changes once a phase is done, disrupting the whole project and also the customers usually only get involved early on, so there is not much feedback later which can makes it tough to adjust to what the customer wants as things change. The disadvantage is supported by Thummadi, Shiv, Berente, and Lyytinen (2011) who highlight that in the waterfall model, the final product is not validated until the end of the development process. This means that any issues or problems are not identified until late in the development cycle, which can lead to significant rework and increased costs.

2.4.2 Agile Model

Agile model are development techniques that prioritize human-centered approaches to software production, with the goal of delivering high-quality products quickly to meet customer needs (Ceschi, Sillitti, Succi, & De Panfilis, 2005). Examples of agile methods include Extreme Programming (XP) and Scrum, both of which tackle the challenges of an unpredictable environment by emphasizing the importance of skilled individuals and their relationships in the software development process (Dyba & Dingsoyr, 2009). Hobbs and Petit (2017) mentioned that these methods are commonly adopted in large projects within large organizations, with key features such as the utilization of Scrum methodology and the presence of agile coaches to facilitate the process.

According to Butt, Piñeres-Espitia, Ariza-Colpas, and Tariq (2022), Agile model promotes faster development cycles by breaking projects into smaller iterative cycles, enabling teams to deliver working software quickly and receive frequent customer feedback. The agile approach also allows for changes in requirements and priorities at any stage of the development process, enabling teams to respond quickly to evolving customer needs (Jogu & Reddy, 2013).

One key disadvantage is the lack of extensive documentation required in agile, as it can create challenges for training, support, and knowledge transfer (Binder, Aillaud, & Schilli, 2014). The uncertainty and ambiguity inherent in iterative approaches can also be problematic, especially for organizations accustomed to more predictable, plan-driven development. Another disadvantage is the difficulty in applying agile in distributed software development projects (Layman, Williams, Damian, & Bures, 2006). This disadvantage is also highlighted by Paasivaara, Durasiewicz, and Lassenius (2008) that stated since Agile methods rely heavily on face-to-face communication, which is difficult to achieve in globally distributed teams.

2.4.3 Incremental Model

The incremental model methodology, as described by Larman and Basili (2003), involves breaking down the project into small, manageable increments or iterations, each of which delivers a part of the overall functionality. Each part, or increment, adds new features while keeping the ones already done, as explained by Pressman and Maxim (2014). Moreover, it helps mitigate the risks associated with large-scale software projects by breaking them down into smaller, more manageable pieces (Braude, 2000).

The incremental model has several advantages that make it a popular choice for software development. Firstly, it is allows for the delivery of working software in smaller, more manageable increments rather than waiting for the entire system to be completed (Ibrahim, 2020). This model also can be more easily scaled and maintained over time, as the software is developed in smaller, modular components (Mahmood, Strüber, Berger, Lämmel, & Mukelabai, 2021).

The incremental model, while offering advantages, also comes with its own set of drawbacks. According to Singh et al., (2017), there is one limitation which is each phase of an iteration is rigid and does not overlap with the others. This means that once a phase is completed, it must be finished before moving on to the next, potentially causing delays if there are any issues. Additionally, problems can arise related to system architecture because not all requirements are gathered upfront for the entire software life cycle. This lack of comprehensive planning upfront may lead to inconsistencies or inefficiencies in the system's design, making it harder to ensure that all parts work together smoothly.

2.4.4 Comparison of Development Methodology

Agile, waterfall, and incremental models are all used in software development, but each has its strengths and weaknesses. Agile is flexible and focuses on

customer collaboration, while waterfall follows a linear approach with distinct phases. Incremental combines elements of both, breaking the project into smaller parts. Choosing the best method depends on the project's needs and goals. Table 2.3 shows the comparison of method which are agile, waterfall, and incremental.

Table 2.3 Comparison between Agile, Waterfall, and Incremental

Aspect	Agile	Waterfall	Incremental
Approach	Flexible and iterative	Sequential and linear	Combines elements of Agile and Waterfall
Flexibility	High	Low	Moderate
Customer Involvement	High, frequent collaboration	Limited involvement after initial requirements	Moderate involvement throughout the development process
Phases	Multiple short cycles	Distinct phases (Requirements, Design, Implementation, Testing, Deployment)	Phased approach with increments
Iterative	Yes	No	Yes

The choice between methodologies depends on the requirement for VolComm. For example, Agile is best suited for most projects where requirements may change frequently, as it allows for flexibility and iterative development. This methodology works well for projects with high levels of customer involvement and where collaboration is essential for success. On the other hand, Waterfall is more suitable for projects with well-defined requirements and where a sequential approach is preferred. It's ideal for projects where changes are unlikely to occur once the requirements are finalized, as it follows a structured and linear process. Incremental methodology is a middle ground, combining elements of both Agile and Waterfall. It works well for projects that can be broken down into smaller parts or increments, allowing for flexibility while still following a phased approach. This methodology is suitable for projects with moderate levels of uncertainty and where some level of customer involvement is desired throughout the development process.

2.5 Related Works

There is a strong focus on theoretical studies of current volunteering platform and system in use. It is hoped that new ideas on volunteerism and contribution to the community will help enhance this system that currently exists. For instance, Facebook groups of volunteering such as Volunteer Malaysia, VolunteerHub, and BetterImpact have been explored concerning their significance and nexus in volunteering. In this case, studying these platforms enables one to improve existing ways of organizing volunteering in a bid to achieve a higher level of effectiveness.

2.5.1 Volunteer Malaysia Group on Facebook

The 'Volunteer Malaysia' Facebook page shares information about volunteering opportunities in Malaysia and allows people to post about events. It is a platform where individuals can learn about events or projects and find organizations seeking volunteers. Users can browse available volunteer posts, express interest, and ask questions about volunteering.

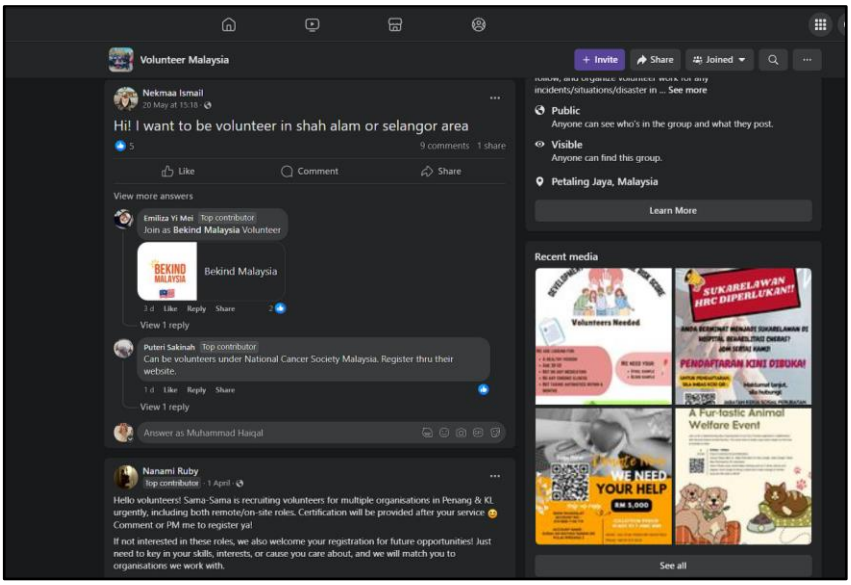


Figure 2.2 Volunteer Malaysia group on Facebook

2.5.2 VolunteerHub

VolunteerHub is a utility to work with volunteers if they are needed in the functioning of some organization. It allows them to publish volunteering opportunities on the Internet. Possible it can be added that volunteers have option to register and to indicate when they are willing to help. It also enables the organizers to know when to assign volunteers and schedule them for various tasks, and even send them some notifications. It is also useful when marking how much the volunteers contribute and all the reports. It also interacts with other software apart from databases and has integrations with them. It can be made to suit the organization's practice and palliative care can easily be integrated into it. Additionally, there is an application that can be installed both on phones that is helpful for volunteers and organizers.

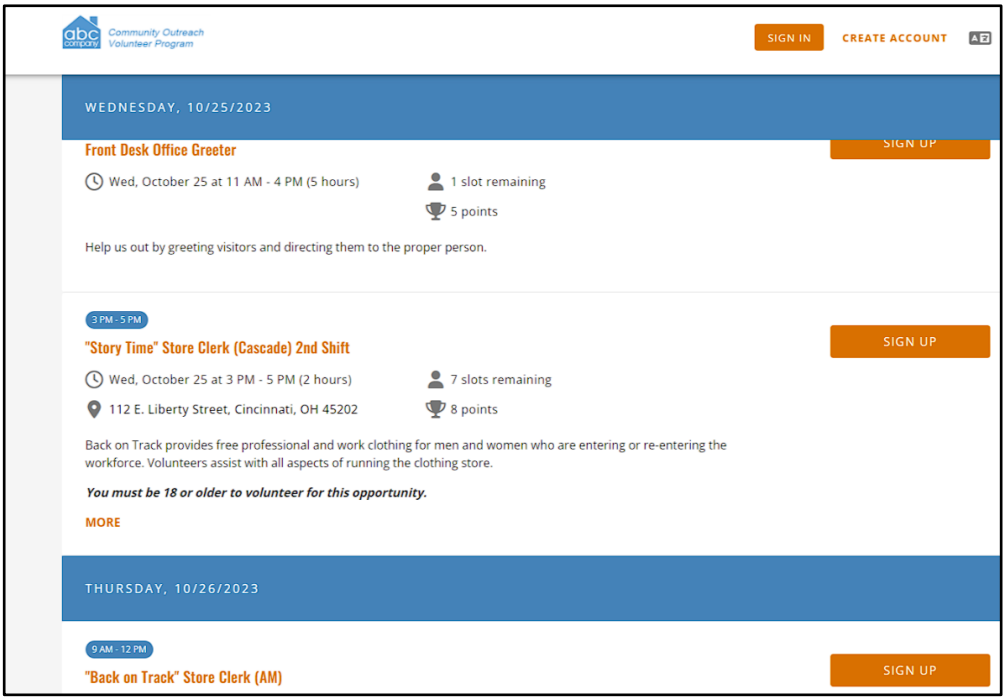


Figure 2.3 VolunteerHub Volunteer Opportunities Page

Figure 2.3 also demonstrates the VolunteerHub's interface which is quite simple and easy to navigate, and it is not overwhelmed with excessive icons or buttons. The design is simple and good for that users so that they find it easy to navigate on the design. Several areas of the main menu have descriptive

titles, and all the keys are organized and easily accessible. The appearance of this software improves upon VolunteerHub's aesthetic and usability while prioritizing efficiency and ease of use by removing unnecessary buttons and toolboxes that would overwhelm less tech-savvy users.

2.5.4 BetterImpact

The software is known as BetterImpact and the idea behind it is to assist organizations in managing their volunteers. It can assist them in recruiting people for various positions it also allows them to track volunteer hours. This program also permits organizations to relay messages to volunteers and also document those who helped in what. BetterImpact, for instance, helps organizations to manage volunteer schedules and all other relateds items more efficiently as most of the data is captured in one place.



Figure 2.4 BetterImpact Volunteer Management Page

Figure 2.4 displays the BetterImpact layout, which is not very friendly to the user though easy to comprehend. One has to deal with unnaturally large amounts of information on the screen, which hinders convenient navigation. Organization of sections within the layout is a major problem here – it is not unified or clear. It becomes exhausting and immensely frustrating to wander around and explore an unfamiliar place. Interfactional subcategories of

importance are hidden under numerous layers of pull-down menus, while the whole experience is disorganized and unwieldy.

2.5.5 Comparison of Related Works

Volunteer Malaysia Group on Facebook, VolunteerHub, and BetterImpact are all platforms and websites established with the purpose of increasing volunteerism, but the way they work and the capabilities are different. Volunteer Malaysia Group on Facebook, Join this FB group is to help connect all the volunteers and organisations, providing a wider coverage and rather a more casual communication. On the other hand, BetterImpact provides an elaborate volunteer management software with application such as; tracking volunteers, analytics, and communication among others. In each of the platforms described above there are pros and cons and therefore deciding which one to use depends on the objectives and approximately on the nature of the activity of an organization or a project. Table 2.4 describes these platforms and their attributes wherein users can compare and contrast to make their decisions.

Table 2.4 Comparison between Volunteer Malaysia Group on Facebook, VolunteerHub, and BetterImpact

Feature	**Volunteer Malaysia Group on Facebook**	**VolunteerHub**	**BetterImpact**
Volunteer Opportunity Listings	Yes	Yes	No
Online Volunteer Registration	No	Yes	Yes
Volunteer Scheduling	No	Yes	Yes
Communication Tools	Limited (Comments, Messenger)	Yes (Emails, Notifications)	Yes (Emails, Messages)
Reporting and Analytics	No	Yes	Yes (Advanced Reporting)
Integration Capabilities	No	Yes (CRM, Email Marketing, etc)	Yes (CRM, Integration)
Customization Options	No	Yes	Yes (Extensive Customization)
Mobile Accessibility	Yes (Facebook App)	Yes (Mobile App)	Yes (Mobile App)

The Table 2.4 shows comparison of volunteer management platforms: Volunteer Malaysia Group on Facebook, Volunteer Hub and BetterImpact, respectively. Volunteer Malaysia Group on Facebook enables the user to post volunteer opportunities however, it does not support online volunteer registration and scheduling. VolunteerHub and BetterImpact are more complex solutions for managing volunteers through the web registration and using scheduling tools. Different platforms have different modes of communication: in Volunteer Malaysia Group, they can only communicate through the post comments in the Messenger while VolunteerHub and BetterImpact offer a wider array of communication methods such as emails and messages to mention but a few. To be more specific, BetterImpact is stronger in the aspect of reporting, while VolunteerHub possesses better compatibility with other systems such as CRM and Email marketing. There are some aspects that may be customized in both VolunteerHub and BetterImpact, nonetheless, BetterImpact has the most extensive customization features. Of course, mobility is present and characteristic in all three, and VolunteerHub, and BetterImpact even have their own applications that you can use on your smartphone.

2.6 Discussion

This chapter discusses is explores the best methodology, notification method, and system platform for VolComm by considering stakeholder preferences and requirements. The waterfall methodology is chosen because stakeholder prefer a clear detailed plan. This method works well when all project details are known from the beginning and remain stable, helping to keep the project organized and on schedule, despite the difficulty in making changes later. Due to the limitation of development timeframe, agile methodology is less suitable to be implemented. For notification, email is chosen based on stakeholder requirements. It is a reliable way to send updates directly to volunteers' inboxes because it is a common tool and volunteers likely to check it regularly. A web-based system is chosen due to stakeholders preference due to scalability. Operating through a web-based system, this platform allows user to access

information and updates from anywhere with an internet connection. This convenience method make sure that volunteers and organizers can easily access and stay informed.

2.7 Summary

This chapter covers the essential reviews and details of the project. It elaborate the literature related to developing the Volunteer Management System (VMS) and describes similar projects, the system platform, notification methods, and the SDLC used in this project. This information will help the developer tackle any issues that come up during system development.

CHAPTER 3

METHODOLOGY

This chapter explains the VolComm: Volunteer and Community Service Hub for Non-Governmental Organizations (NGOs) methodology. It will discuss the system development requirements, the necessary procedures, and each step of the project's development. The Waterfall methodology will be used to achieve the project's goals. Additionally, this chapter will give an overview of each stage and task based on the chosen system development life cycle (SDLC) model.

3.1 Overview of SDLC Methodology

Methodology can be defined as a set of approaches that one uses in order to conduct research, in the case of a dissertation, or to put into practice plans and solving problems in an area of study or a particular profession. This method provides a systematic way of planning the most effective manner and pattern of accomplishing the tasks, decision making and realizing set goals. In a given context, methodology offers a prescription on which actions should be taken, which other action to consider employing in the performance of the intended study or project, and how to evaluate the consequences of the study or project.

The SDLC refers to a well-articulated pattern that is employed in software engineering for designing as well as implementing software systems. This structure is composed of various phases where specific activities occur and deliver certain outcomes to assure the overall success of a software product. The widely utilized phases of SDLC include the planning phase, analyzing phase, designing phase, implementation phase, testing phase, and the maintenance phase.

During the planning stage of a project, objectives are set, requirements, and constraints are identified, the feasibility of the project is also assessed

determines the chances of success of the project. The phase of analysis includes the gathering and assessment of information about users' needs in the course of definition of functionality and scope of the systems.

This stage involves the planning of the optimum system structure in detail and more specific specifications for the required software pieces are made. This involves developing the software code the construction of the software in relation to the design specifications.

3.2 Waterfall Model for VolComm

Waterfall Model is an ideal linear model having one phase after another fully completed before proceeding with the next. It is chosen because of its simplicity in architecture and clear incremental changes; thus, a suitable choice for projects that allow little to no alteration to the General requirements set.

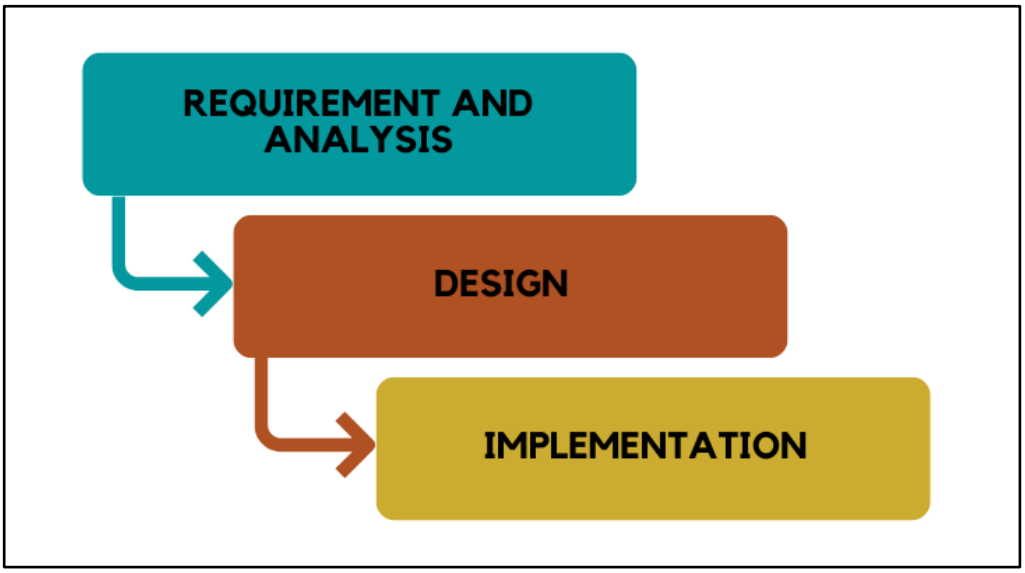


Figure 3.1 Waterfall Model

Figure 3.1 explains a waterfall model followed while developing the VolComm. The specification requirement and the analysis phases are incorporated in one in the model to integrate it. This approach was adopted to try and improve efficiency and also to ensure that it is understood what needs to be achieved before going to the design and implementation stage.

3.3 Phases of VolComm Development

For VolComm development, it consists of three main stages which are Requirement Gathering and Analysis, Design, and Implementation. Every stage of a project comes with its own set of tasks that need to be successfully finished in order to reach the goals prior to advancing to the following stage. The tasks for each phase are carefully stated in Table 3.1, offering a detailed reference for stakeholders to monitor progress and keep a structured approach throughout the software development life cycle.

Table 3.1 The project framework using SDLC approach

Phase	Activities	Deliverables	Objective
Requirement Gathering and Analysis	a)Write interview questions for stakeholders and students.	Requirement gathered	To gather and understand the needs for the VolComm: Volunteer and Community Service Hub For Non-Governmental Organizations (NGO)
	b)Interview stakeholders via WhatsApp and direct message.		
	c)Analyze the information from the interviews.		
	d)Create diagrams: i)Use case diagram ii) Domain class diagram iii) Activity diagram iv) Sequence diagram	d) Requirement diagrams: i) Use case diagram ii) Domain class diagram iii) Activity diagram iv) Sequence diagram	
	e) Compile all the information into a document.	e) Software Requirement Specification (SRS).	
Design	a) Create the user interface.	a) Storyboard showing the flow of the system.	To design the web application system for the VolComm: Volunteer and Community Service Hub For NGO

Table 3.1 (Cont.)

	b) Make design diagrams.	b) Design diagrams: i) Design class diagram ii) Domain class diagram (DCD) iii) Entity relationship diagram (ERD)	
	c) Write down all the design details.	c) Software Design Documentation (SDD).	
Implementation	a) Create the database tables.	a) Database created using MySQL.	To develop the VolComm: Volunteer and Community Service Hub For Non-Governmental Organizations (NGO)
	b) Develop the system.	b) VolComm: Volunteer and Community Service Hub For Non-Governmental Organizations (NGO)	
	c) Add email notification feature to the system.	c) Email notification feature added to the evaluation system.	

Phase 1 : Requirement Gathering and Analysis for Volcomm

The first phase in the methodology is requirement gathering and analysis. This phase starts by creating detailed interview questions to gather important information from Encik Nor Izzudin Shah, the Admin of Volunteer Malaysia group on Facebook, who acts as the stakeholder. These questions aim to understand their specific needs, expectations, and current challenges. Interviews are conducted through WhatsApp for flexible and easy participation. Table 3.2 displays some of these questions and the corresponding answers received during the interview process.

Table 3.2 Question and answer from the interview session

Interview Question	Stakeholder Answer
How do you manage your volunteer group, and what specific tasks do you undertake within the group?	We use the group as our platform for connecting volunteers with opportunities. Members in that group can post about the volunteer. My task is just to handle the group only.

Table 3.2 (Cont.)

Could you tell me through the process of how organizations share volunteer opportunities and how volunteers can get involved?	Members can post the volunteer opportunities in that group, and people who are interested in the activity can reach out to the one who posted it or click any given link.
On a scale of 1 to 10, how challenging is it for you to ensure smooth operations within the group?	3/10. It is not that challenging since there is not much job for me to do. I just need to remove any posts that are not related to volunteering, so I need to check one by one.
Which tasks do you find particularly challenging?	Checking posts one by one to remove any posts that are not related to volunteering.
What suggestions do you have for improving or adding new features on the future system to streamline the process for the organization and volunteers?	I think you can make volunteers also sign up through the system, so there is no need to fill out another form.

After completing the interviews, the collected data is carefully analyzed to identify key requirements and objectives. This analysis helps understand the current process, pinpoint problems, and determine necessary features for the new system.

To visualize and communicate these requirements, several diagrams are created. Use case diagrams (UCD) display how users and the system interact, emphasizing required functions. Domain class diagrams (DCD) provide a structured perspective of the system's data and the connections among various data entities. Activity diagrams outline the flow of work and procedures inside the system, whereas sequence diagrams show the connections between various system components as time progresses.

All the gathered and analyzed information is then compiled into a comprehensive Software Requirement Specification (SRS) document. The SRS acts as a comprehensive guide for the development team, specifying all functional and non-functional requirements. It guarantees that all parties involved have a clear comprehension of the system's functions and implementation, establishing a strong base for the upcoming project stages.

Phase 2 : Design of Volcomm

In the methodology's second phase, the focus is on designing the blueprint for the web application system of VolComm, an NGO hub. We start by designing the user interface, using Canva to create a storyboard. The utilization of a storyboard is crucial in the system production process, serving as a visual guide. It designs a user interface according to stakeholder requirements, ensuring user satisfaction. This storyboard illustrates how users will navigate the system, providing detailed screen layouts and interactive elements. Following interface design, a variety of diagrams are created to organize the system structure. The design class diagram (DCD) defines classes and their relationships, while the domain class diagram (DCD) gives an overview of the data entities and their connections. The entity relationship diagram (ERD) outlines how data entities relate in the database. Finally, all design details are documented in the Software Design Documentation (SDD), providing a guide for the development team to build the system according to the design.

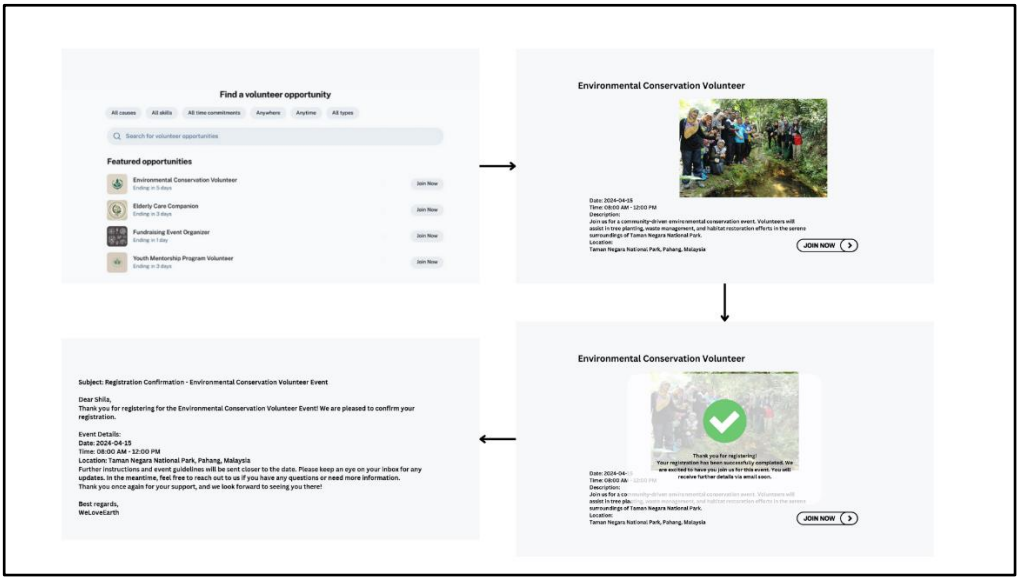


Figure 3.2 Storyboard for registration volunteer opportunity

Figure 3.2 shows a storyboard that illustrates the steps a user takes to register for a volunteer opportunity and receive confirmation. In the first section, the user is presented with a volunteer opportunity search page that lists featured opportunities. Each opportunity includes a "Join Now" button, allowing users

to easily register for their selected event. The user clicks on the "Join Now" button for the "Environmental Conservation Volunteer" event. Afterward, the user is redirected to the event details page, which includes information such as the date, time, and location of the volunteer opportunity. The description outlines the activities involved in the event. The user is then given the option to confirm their registration by clicking the "Join Now" button again. Finally, the user receives an email confirmation for their registration. The email provides the event details, and informs them that further instructions and guidelines will be sent closer to the event date.

Phase 3 : Implementation of Volcomm

Finally, the implementation phase involves executing the VolComm: Volunteer and Community Service Hub for Non-Governmental Organizations (NGOs), following the collected requirements and the developed design. The first step in system development is creating simple programs. After that, the code is implemented using the PHP programming language, including an email notification system. To make this system work, several essential hardware and software components are needed, as listed in Table 3.3 and Table 3.4.

Table 3.3 Hardware specifications

Hardware	Specification
Laptop	Brand: Acer Aspire3 15 Processor: AMD Ryzen 5 7520U with Radeon Graphics 2.80 GHz Installed RAM: 8.00 GB System Type: 64-bit operating system, x64-based processor OS Name: Windows 11 Home Single Language
Smartphone	Brand: Apple Model: iPhone 13

The most important hardware component is the laptop, used for writing the report and building the system. It has a good CPU and enough RAM, making it suitable for developing and running the expert system. The smartphone, a crucial piece of modern technology, is used mainly for communication. It

serves as the primary device for accessing and reviewing all communications from the supervisor, including receiving phone calls.

Table 3.4 Software specifications

Software	Description
Microsoft® Word for Microsoft 365 MSO (Version 2404 Build 16.0.17531.20152) 64-bit	Report and documentation
Google Chrome	Internet Browser
Draw.io and Canva	Create Diagram
PhpStorm	Integrated Development Environment (IDE)
MySQL	Database Management System (DBMS)
GitHub	Software Development Platform

The software requirements outline the necessary components for developing the expert system. Microsoft Word is utilized for creating reports and documentation. Google Chrome serves as the internet browser for research purposes, offering access to a wealth of relevant information. Draw.io and Canva are used for creating diagrams. PhpStorm functions as the Integrated Development Environment (IDE) for programming and coding the system. MySQL is a Database Management System (DBMS) for storing user data. GitHub is being use as the software development platform for deploying the system to the cloud.

3.5 Summary

This chapter focuses on the development of the VolComm system and the methodology involved in its development which is intended to be a platform for volunteering and various charitable works among NGOs, featuring email notifications. It explains what is done in each phase and the outcomes of actions done of the Software Development Life Cycle (SDLC) model and its flexibility as it can be adapted to include features of other models.

CHAPTER 4

RESULTS AND FINDINGS

The purpose of this chapter is to outline the VolComm system. This chapter covers each of the three stages. Project development is the general term used to describe the requirements collection and analysis, system design, and project implementation process. It will include a descriptive analysis of every project stage, focusing function to the actions done to achieve the project objectives. By giving a brief overview of the project's evolution, this chapter provides information on how the project has progressed and its capacity to meet stakeholder expectations.

4.1 Requirements Gathering Results

In the first stage of the waterfall design process, requirement collection and analysis, stakeholders' input is gathered and examined to determine the system's requirements. Determining the project's scope is dependent on the data acquired, with the goal of achieving common understanding and agreement among all stakeholders. By describing the system's operations and relationships, the requirement diagram acts as an accurate blueprint that development teams can follow to. System Sequence Diagram, Business Activity Diagram, Use Case Diagram (UCD), and Software Requirement Specification (SRS) are all included in this diagram. The System Sequence Diagram shows communications flows for use cases, whereas the Use Case Diagram (UCD) illustrates user-system interactions. Together with the Business Activity Diagram, these diagrams help the Software Requirement Specification (SRS) make sure that development is in line with the requirements and needs.

4.1.1 Findings from Interview

The main goal of the interview was to get all the information that was needed from the participants. In order to gather the information and requirements, the necessity was brought up during the stakeholder interview. Table 4.1 lists the different types of stakeholder-related queries, their answers, and the findings of each inquiry's study.

Table 4.1 Requirements Gathering Results

Question	Answer	Findings
Sometimes organizers forget to let volunteers know if their application for an opportunity has been accepted or not. Would it help if the system sent notifications automatically when there's an update?	Sometimes organizers forget to update volunteers about their application status. Automatic notifications would be very helpful to keep them informed through channels like email or Telegram.	Use Case: Notify Volunteer
Currently on Facebook, anyone can post opportunities. Do you think there should be any control over this?	Yes, I think it would be better if the system allowed only verified organizations to post. Admins could approve organizations first to ensure credibility and maintain the quality of opportunities posted.	Use Case: Approve Organization Manage Opportunity Posting
As a volunteer, what challenge do you face when choosing an opportunity?	There are so many different opportunities, which can be overwhelming. It would be helpful to sort them by interest.	Use Case: Sort Opportunities Based on Interest
What role do you see administrators playing in managing the platform?	Admins should be able to oversee everything, like managing accounts for both organizations and volunteers. They should also approve organizations to make sure only verified organizations can post opportunities.	Use Case: Manage Accounts & Approve Organization
How important are certificates for volunteers?	Certificates are beneficial, especially since some organizations don’t have time to prepare them. The platform could generate them automatically after events.	Use Case: Generate Certificate

Table 4.1 describes the identification of the interview questions conducted to the stakeholders. The Use Case was developed following the identification of the stakeholder problems.

4.2 Findings of Requirement Analysis

The waterfall process's analysis phase examines the requirements gathered via in-depth conversations and stakeholder questioning. Maintaining a tight relationship with stakeholders can be important during this phase. Make sure the data makes logic and supports the project's goals by thoroughly analyzing it. The findings of the analysis phase are crucial for guiding the subsequent stages of the waterfall model and for providing guidance for the development of complex project plans and design specifications. The analysis process involves an in-depth review of the needs obtained by speaking with stakeholders. It's important to stay in touch with stakeholders throughout this time. Make sure everything makes sense and fits with the goals of our project by carefully reviewing it all. The information gained from analysis affects further actions, such as creating thorough designs and design blueprints.

4.2.1 Use Case Diagram

The use case diagram represents three key actors in the system which are Organization, Volunteer, and Admin, each with distinct roles and access to various system functions.

Organization actor is responsible for managing volunteer events and volunteer participation. They have the ability to create an organization account, and organize volunteer events. Organizations can view and manage these events, accept volunteer applications, and track participation history. Additionally, they have the authority to generate certificates for volunteers who complete events. Organizations can also view the list of volunteers who have registered for their events and manage their accounts and event details.

Next, volunteer actor represents individuals who participate in events organized by NGOs. Volunteers can create their own accounts, view and

apply for volunteer events, and track their participation history. They can also view and request certificates once they have completed an event. Volunteers are notified regarding their applications.

Finally, the Admin actor holds higher-level authority within the system. Admins can approve organizations, add new administrators, and manage both their own and others' administrative accounts. They have the ability to oversee the overall system, including verifying organizations and ensuring smooth operation, but do not directly manage events or volunteer participation.

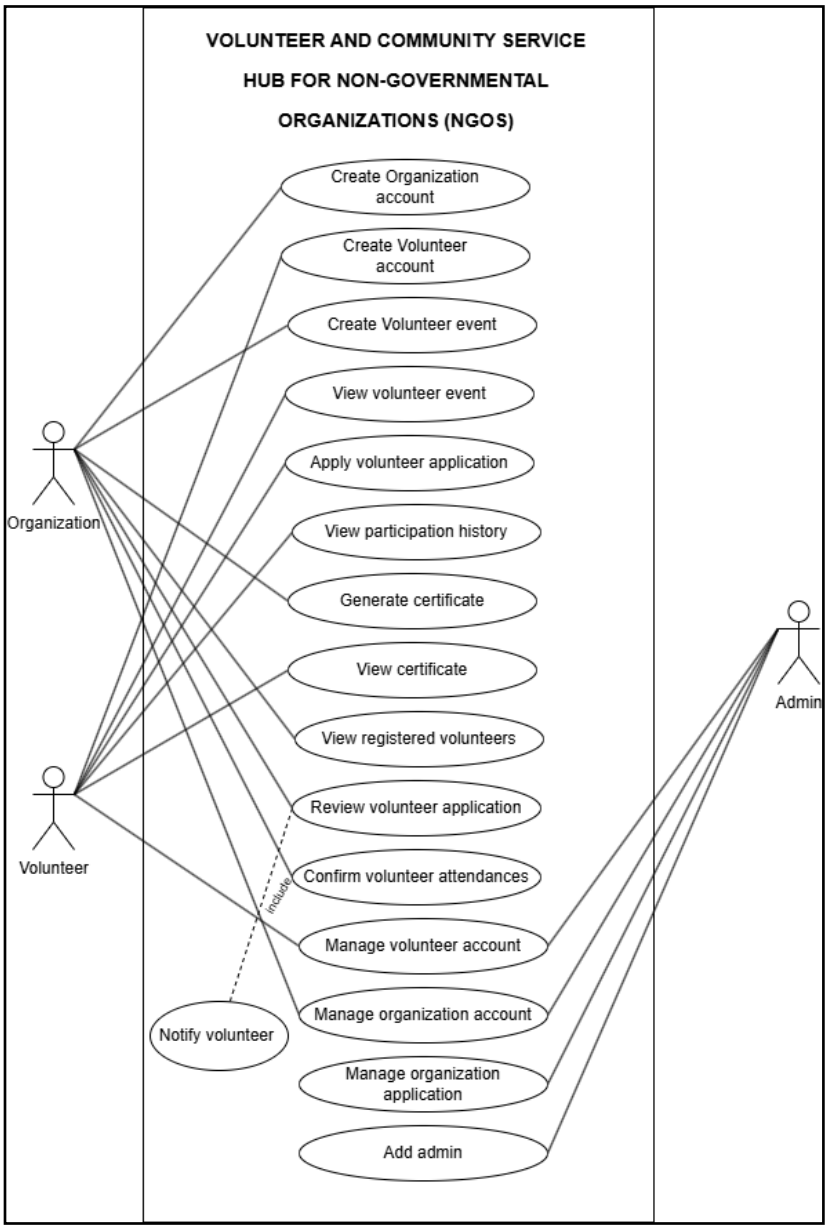


Figure 4.1 Use Case diagram VolComm

The use case diagram in Figure 4.1 shows a total of 15 use cases, with each actor having respective use case functions. Table 4.2 explains the function of each use case.

Table 4.2 Use Case Functionality

Actor	Use Case No.	Use Case	Description
Organization	UCD001	Create Organization Account	Allows an organization to register and create an account on the platform.
Organization	UCD002	Create Volunteer Event	Enables an organization to create a new volunteer event or opportunity.
Organization	UCD003	View Volunteer Event	Allows an volunteer to view details of opportunity posted on the platform.
Organization	UCD004	View Registered Volunteers	Allows an organization to view the list of volunteers who have registered for their events.
Organization	UCD005	Accept Volunteer Application	Enables an organization to review and accept volunteer applications for their events.
Organization	UCD006	Confirm Volunteer Attendance	Allows an organization to confirm attendance of volunteers for completed events.
Organization	UCD007	Manage Organization Account	Enables an organization to edit or update their account details on the platform.
Volunteer	UCD008	Create Volunteer Account	Allows a volunteer to register and create an account on the platform.
Volunteer	UCD009	Apply for Volunteer Application	Enables a volunteer to apply for volunteer opportunities or events.
Volunteer	UCD010	View Participation History	Allows a volunteer to view their history of completed volunteer events.
Volunteer	UCD011	Generate Certificate	Allows a volunteer to generate a certificate for completed volunteer events.
Volunteer	UCD012	View Certificate	Enables a volunteer to view and download their certificates.
Volunteer	UCD013	Manage Volunteer Account	Allows a volunteer to edit or update their account details on the platform.
Admin	UCD014	Approve Organization	Enables an admin to review and approve organization accounts before they can post opportunities.
Admin	UCD015	Add Admin	Allows an admin to add new admins to the platform.

The use cases describe how these actors work with the system to accomplish different goals. Each use case is thoroughly described in the table, which also shows the relationships and procedures that are involved.

4.2.2 Activity Diagram

In the analysis phase, an activity diagram is utilized to visually represent the sequence of actions, tasks, and procedures within a system or a specific educational process. Figure 4.2 presents the sequential steps that a student needs to follow successfully logging into the system to complete the registration process.

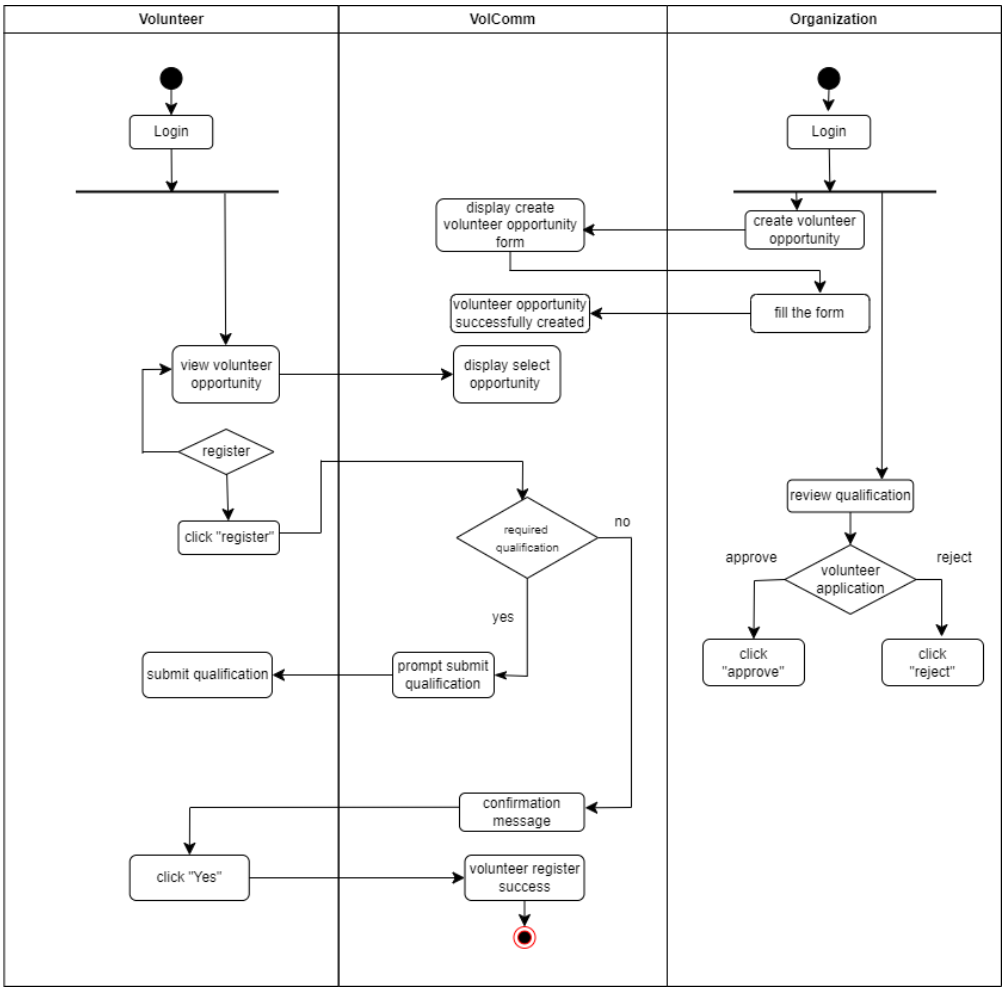


Figure 4.2 Activity diagram for volunteer registering an opportunity

Figure 4.2 illustrates the activity diagram for a volunteer registering for an opportunity. The process begins with the volunteer logging into the system. After logging in, the volunteer views the available opportunities. Upon selecting an opportunity, the volunteer clicks on the "register" option to apply. If the opportunity has required qualifications, the system prompts the volunteer to submit their qualifications. Once the volunteer submits the

required qualifications, the organization reviews them. If the organization approves the qualifications, the volunteer is successfully registered, and a confirmation message is displayed. If the qualifications are rejected, the volunteer will receive an email notification, and the process ends.

4.2.3 Domain Class Diagram

The structure of this system is detailed in the Domain Class Diagram (DCD) for VOLCOMM, which shows all relevant classes and their properties. It includes all essential components to understand how the system functions and interacts within its environment.

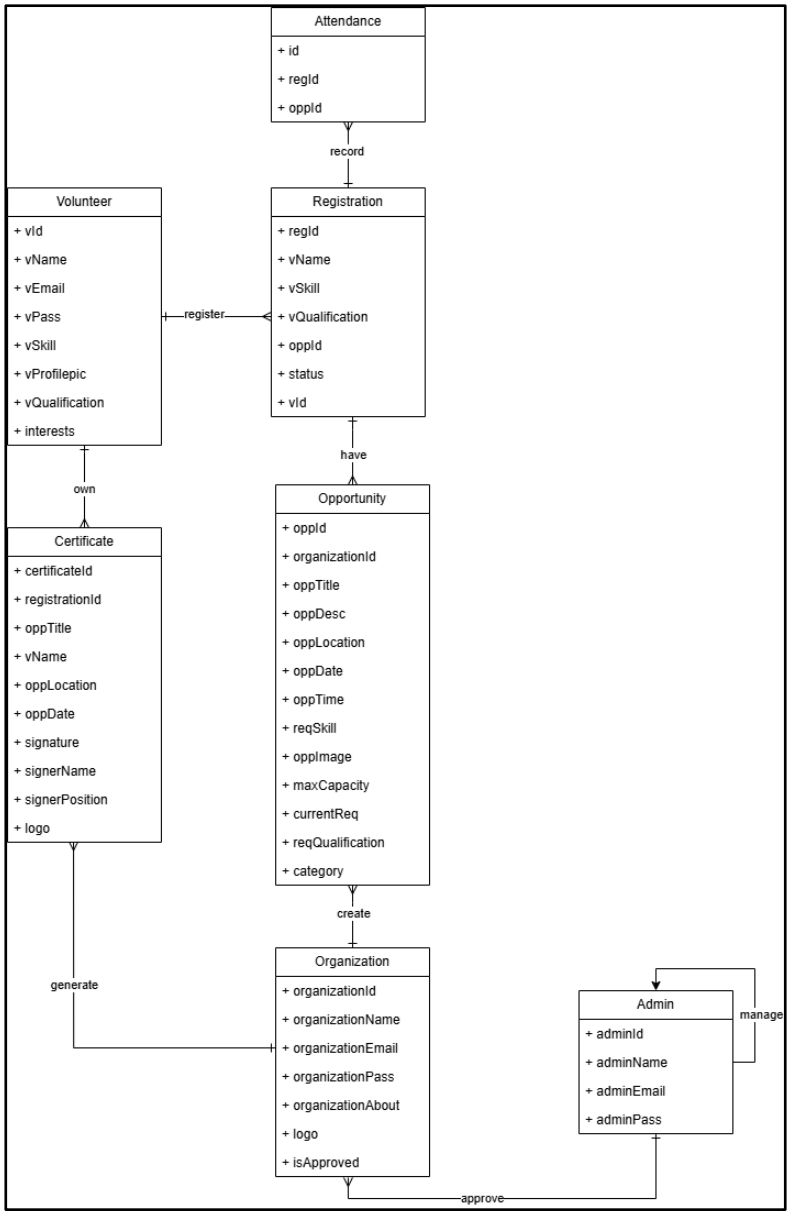


Figure 4.3 Domain Class Diagram for VOLCOMM.

Figure 4.3 illustrates the relationships between different entities in the volunteer registration system. The Volunteer entity registers for an Opportunity through the Registration entity, which stores the volunteer's skills and qualifications. The Registration entity is linked to the Opportunity entity, showing which event the volunteer has signed up for. A Volunteer can also own a Certificate for completing an opportunity, which is associated with their registration. Each Opportunity is created by an Organization, and the Admin is responsible for approving the organizations. The Admin also manages the system to ensure it runs smoothly. Additionally, the Attendance entity records which volunteers attended the opportunities, linking it to the Registration and Opportunity entities.

4.2.4 System Sequence Diagram

A sequence diagram is like a story map showing how different things work together in order. Consider it a detailed manual that outlines the tasks and their respective deadlines. It's similar to creating a line to indicate how events occur in a particular order. Since the main focus is on displaying events in the correct order, it's sometimes referred to as event scenarios or event diagrams.

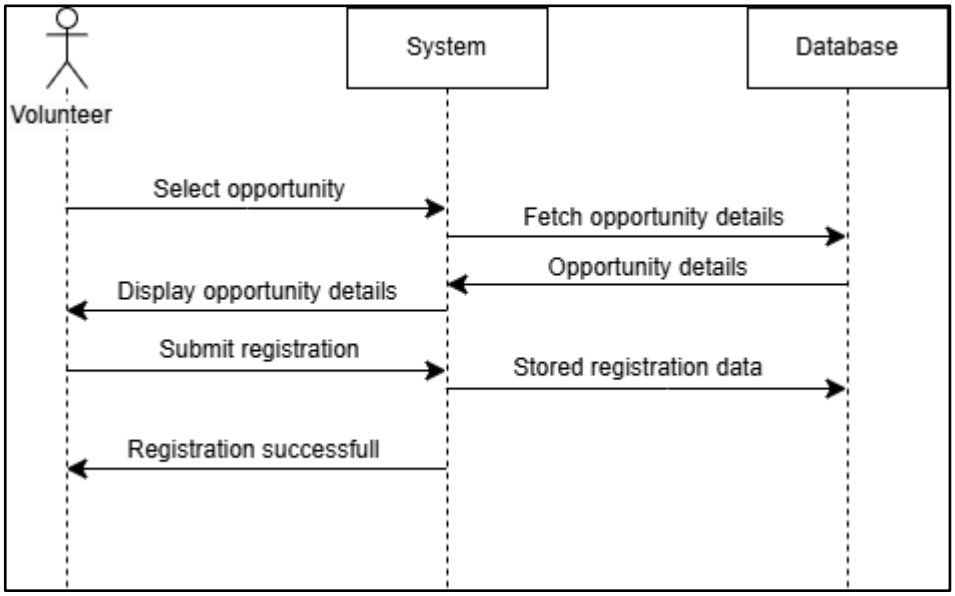


Figure 4.4 System sequence diagram for register opportunity

Figure 4.4 illustrates the system sequence diagram for a volunteer registering for an opportunity. The process begins when the Volunteer selects an opportunity they wish to apply for. The system then fetches the details of that opportunity from the Database. Once the opportunity details are displayed to the volunteer, they can proceed by submitting their registration. After the volunteer submits their registration, the system stores the registration data in the database. Finally, the system confirms the successful registration and notifies the volunteer of the successful registration process. This sequence highlights the flow of actions between the volunteer, system, and database during the registration process.

4.3 Findings of System Design

Following the collection and analysis of requirements, comes the design phase. This phase consists of three main tasks: creating the system's initial storyboard, creating the system's diagrams, and combining all of the system's designs into a single document known as the Software Design Documentation. The system's diagrams comprise a storyboard, package diagram, entity relationship diagram, and multilayer sequence diagram.

4.3.1 Multilayer Sequence Diagram

The study, design, and stakeholder communication of complex systems with several interacting components benefit greatly from the use of multilayer sequence diagrams. It accomplishes this by assisting in the multi-layered visualization of the data or control flow. Furthermore, these kinds of diagrams facilitate the process of converting design concepts into functional components and confirming their interrelationships during the testing and implementation stages.

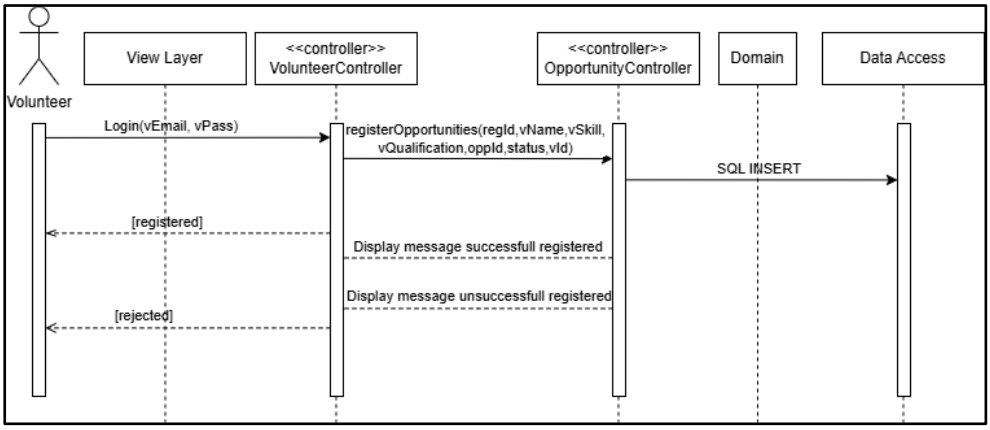


Figure 4.5 Multilayer sequence diagram for volunteer to register opportunity.

Figure 4.5 illustrates the process of a volunteer registering for an opportunity in a multilayer system. The volunteer first logs in through the View Layer by entering their email and password. The VolunteerController processes this login and sends the registration data to the OpportunityController. The OpportunityController then sends the registration details, such as the volunteer's name, skills, qualifications, and opportunity ID, to the Domain Layer. From there, the data moves to the Data Access Layer, where an SQL INSERT query is used to store the volunteer's registration information in the database. Once the registration is processed, the system sends a success or failure message back to the View Layer, notifying the volunteer of the result. This diagram shows how data flows through the system, from the volunteer's login to the registration process and its storage in the database.

4.3.2 Entity Relationship Diagram

An entity-relationship diagram (ERD) is a graphic depiction of a data model that shows how entities in a system relate to one another. It is frequently used in database design to show the relationships and interactions between various elements. Three main stages are usually involved in creating an ERD: conceptual modeling, which identifies entities and their relationships; logical modeling, which defines characteristics and constraints; and physical modeling, which implements the design into a database schema. Figure 4.6 below shows the ERD diagram for VolComm.

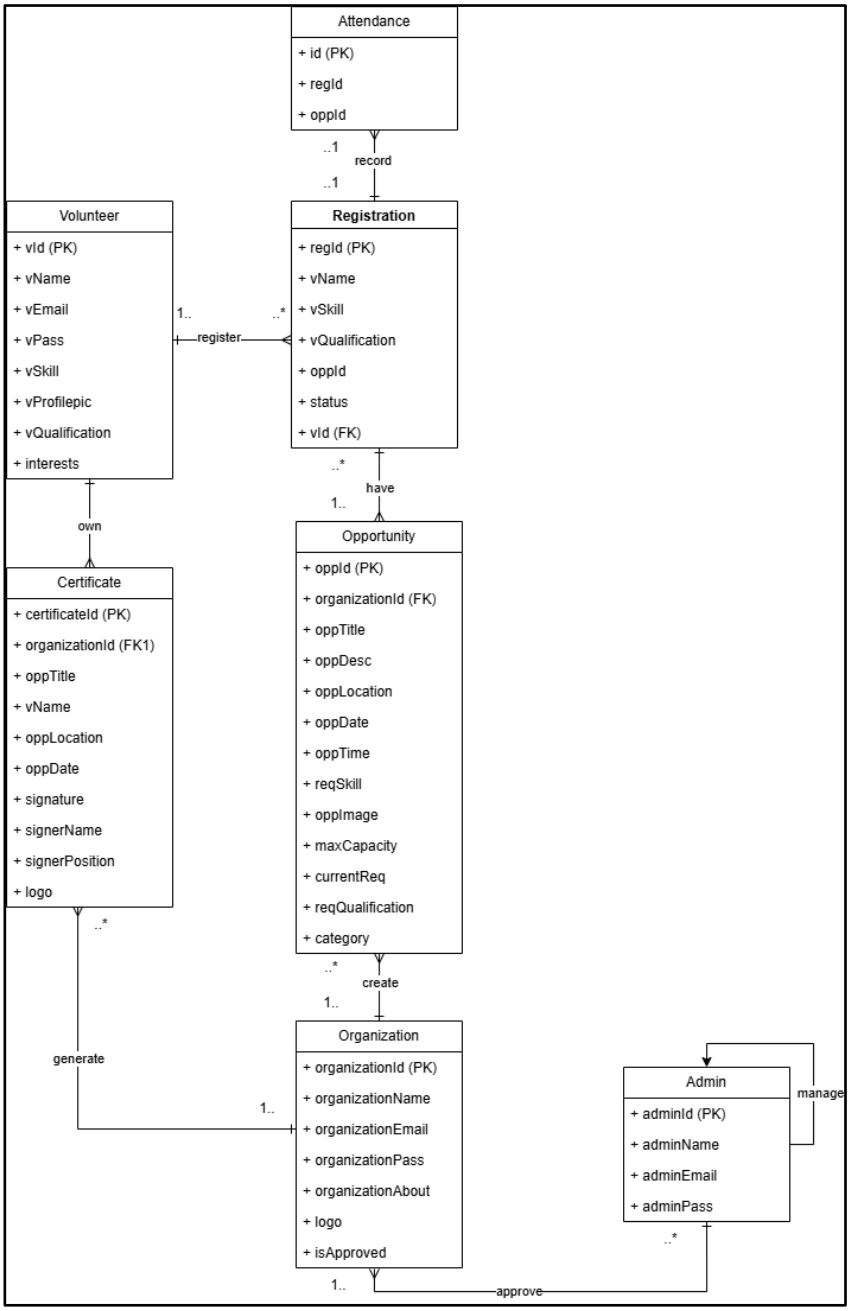


Figure 4.6 ERD of VolComm

Figure 4.6 shows the Entity-Relationship Diagram (ERD) of the VOLCOMM system. It consists of several entities, including Volunteer, Registration, Opportunity, Organization, Admin, Certificate, and Attendance. Volunteers can register for multiple opportunities, and each registration is linked to one specific opportunity (one-to-many, Volunteer to Registration). Each opportunity is created by one organization, and an organization can offer many opportunities (one-to-many, Organization to Opportunity). Volunteers can attend multiple opportunities, and each

attendance record is associated with one volunteer and one opportunity (many-to-many, Volunteer to Attendance to Opportunity). Once volunteers complete an opportunity, they may receive a certificate issued by the organization (one-to-many, Organization to Certificate). Admins manage and approve organizations, overseeing their participation in the system (one-to-many, Admin to Organization).

4.3.3 Package Diagram

In UML (Unified Modeling Language), a package diagram provides a high-level visual representation of the system's structure. It groups related elements, such as classes, interfaces, or use cases, into a package to form a unified unit. The VolComm package diagram is shown in Figure 4.7 below, with an arrow indicating the dependency relationship between each package.

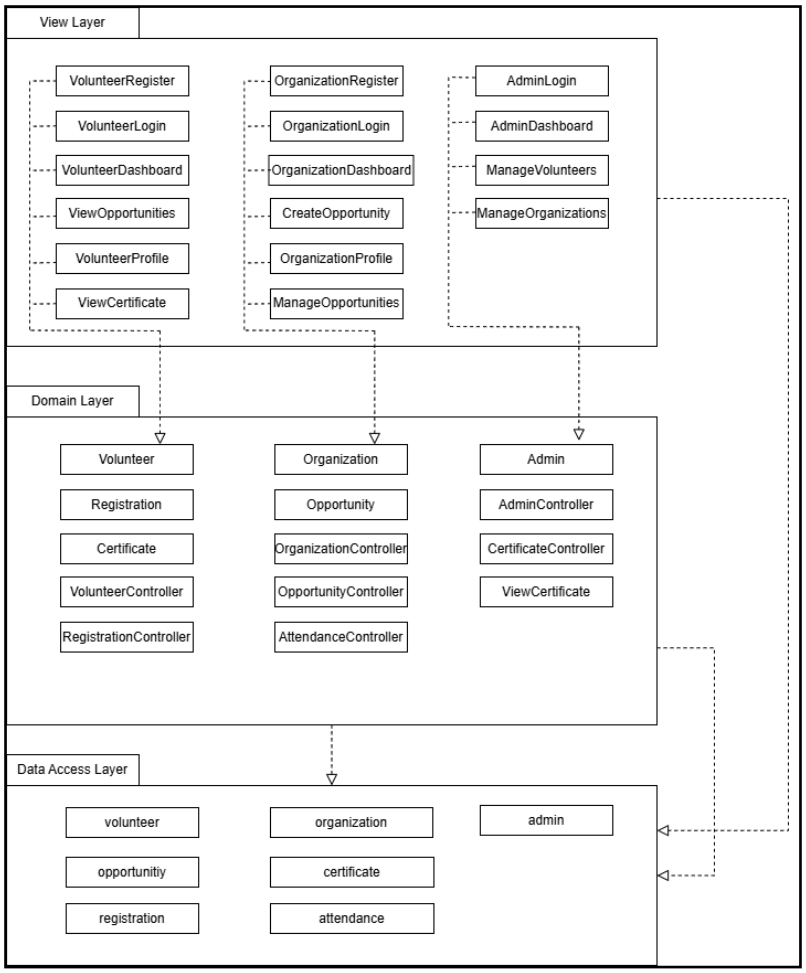


Figure 4.7 Package diagram for VOLCOMM

Figure 4.7 shows how the system is organized into different layers and packages. The View Layer handles user interactions, with packages for volunteers and organizations to manage their profiles, register, log in, view opportunities, and handle certificates. The Domain Layer processes the core data, with entities like Volunteer, Organization, Registration, Certificate, and Opportunity, each linked to controllers that manage the data flow. For example, the VolunteerController and OpportunityController handle actions related to volunteers and opportunities. The AdminController oversees admin tasks, while the ViewCertificate controller manages certificate displays. The Data Access Layer stores the actual data, with packages for volunteers, organizations, admins, opportunities, certificates, and registrations.

4.4 Result of Implementation Phase

The last phase of the Waterfall Methodology is the Implementation phase. In this stage, the system is built and improved over time. Various features and parts of the system are developed as it gradually takes shape and gets better. This phase includes writing the actual code and creating a database to store important information.

4.4.1 User Interface of the System

In this first phase, the system's user interfaces are developed based on the use case diagram for VOLCOMM, designed to match its specific functions. The user interface is important because it acts as a bridge between the user and the system, enabling easy communication and understanding. It is the main way users interact with the system and access its features. The design of the interface follows the storyboard created earlier, keeping the visual and functional elements clear and consistent. Additionally, the design includes user feedback and applies usability principles to improve the overall experience.

The interfaces that will be shown next are listed below:

a) Landing Page

The landing page serves as the entry point of the system. It provides an overview of the platform's purpose and guides users to various sections, such as volunteer registration, opportunities, and login options.



Figure 4.8 Landing page

b) Volunteer Login

The volunteer login page as shown in Figure 4.9 allows registered users to access their accounts. Volunteers can enter their credentials to view available opportunities, manage their profile, and track their history.

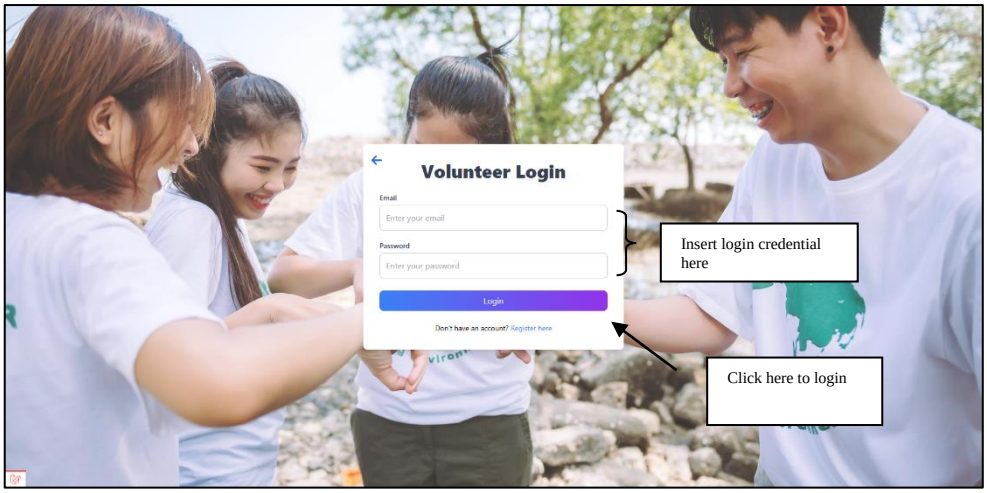


Figure 4.9 Volunteer login page

c) Create Volunteer Account

Figure 4.10 illustrates a registration page designed for new volunteers to sign up on the platform. Users are required to enter their personal details, including their name, email, and password, to complete the registration process.

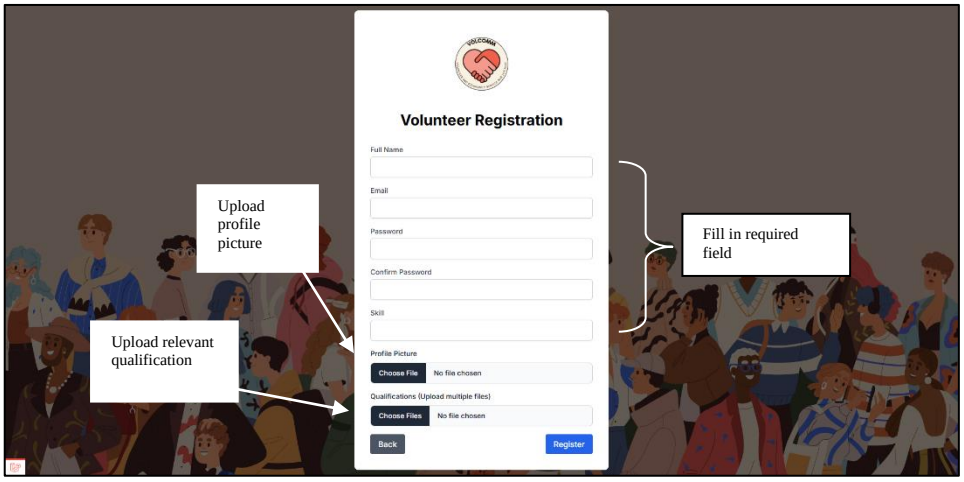


Figure 4.10 Volunteer create account page

d) Select Interest

Figure 4.11 shows the "Select Interest" page, where volunteers can choose areas of interest that align with their skills and passions. This feature helps personalize their volunteer experience by matching them with suitable opportunities.

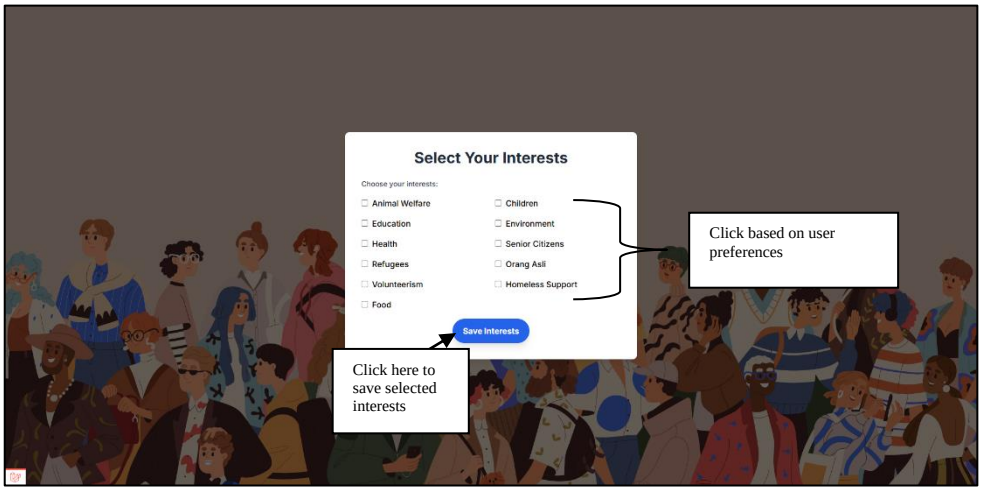


Figure 4.11 Select interest page

e) View opportunity page

Figure 4.12 shows the opportunity page displays available volunteer opportunities. Volunteers can browse through various opportunities, view details such as location, date, and interest, and apply for positions that match their interests.

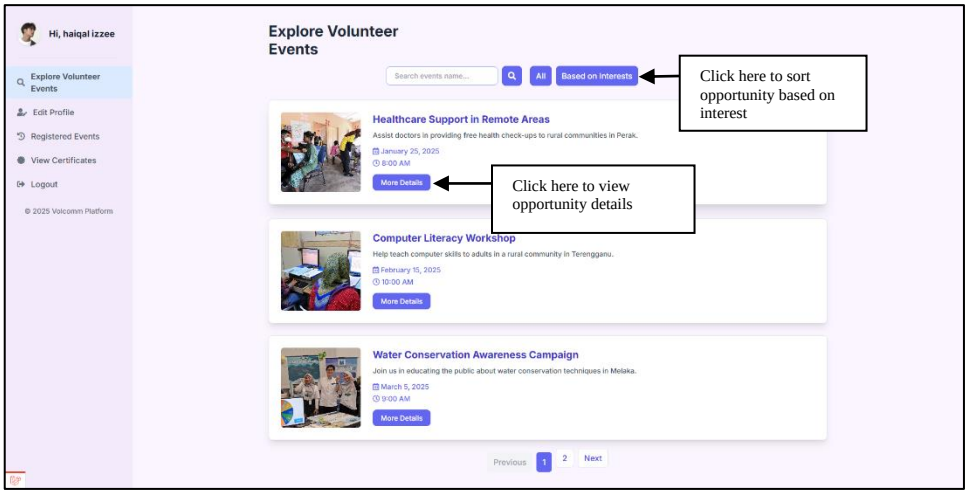


Figure 4.12 View opportunity page

f) Filtered by interest (Environment)

Figure 4.13 opportunities are filtered based on a specific area of interest (environment). Volunteers can see only those roles that align with their selected interest, making it easier to find relevant opportunities.

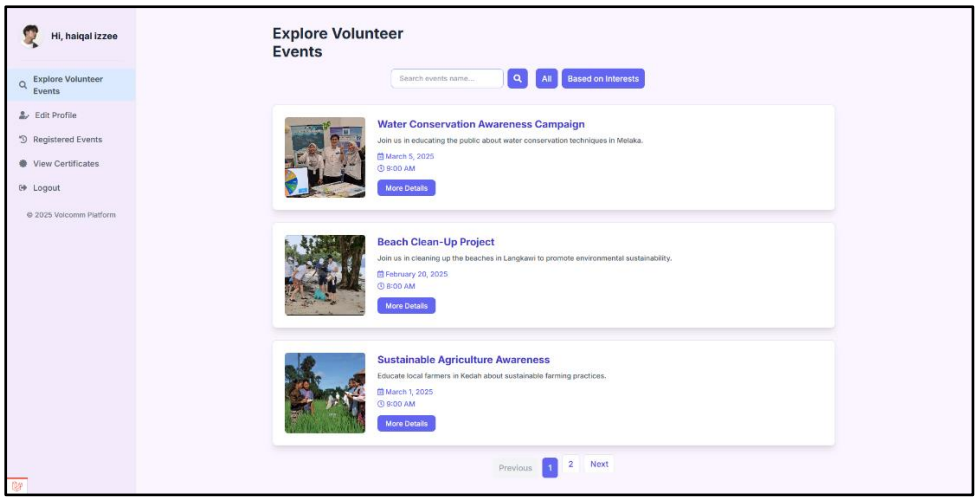


Figure 4.13 Filtered by interest (Environment)

g) Filtered by interest (Health and education)

Figure 4.14 shows the available opportunities by filtering them based on two interests (Health and education). It allows volunteers to find roles that match both of their areas of interest.

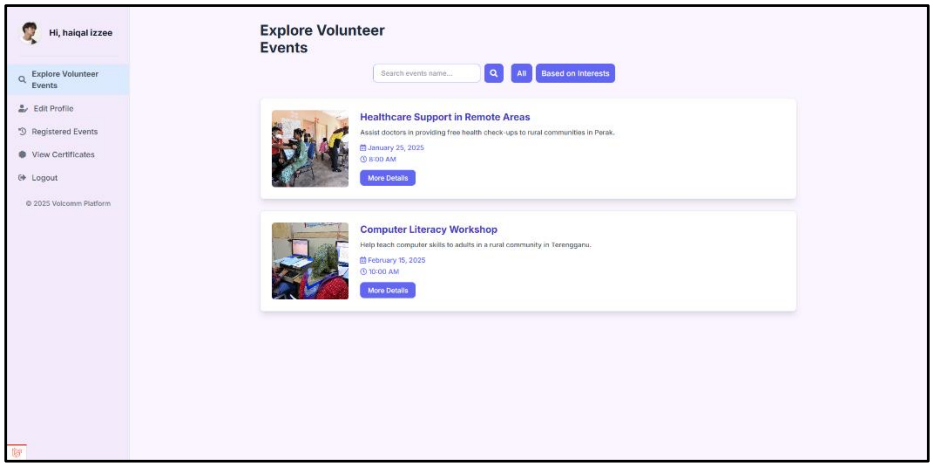


Figure 4.14 Filtered by interest (Health and education)

h) Filtered by interest but did not match

Figure 4.15 shows the volunteer opportunities that match the selected interest but do not meet all the interests.



Figure 4.15 Filtered by interest but did not match

i) Display selected opportunity

Figure 4.16 shows the selected opportunity details after the user clicks the "More Details" button. It displays information such as the current number of registered volunteers, required skills, and other relevant details.

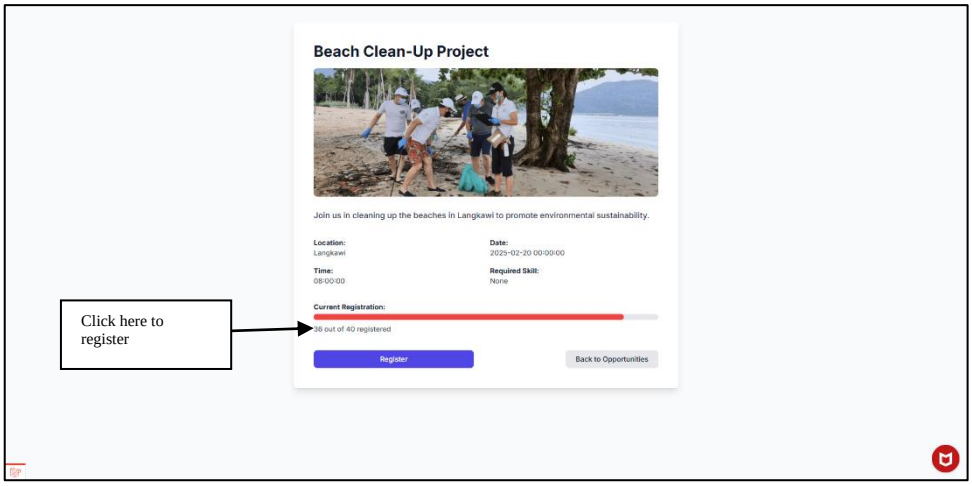


Figure 4.16 Display selected opportunity page

j) Edit Profile

Figure 4.17 shows the "Edit Profile" page, where users can update their name, password, qualifications, and interests using a toggle to switch sections.

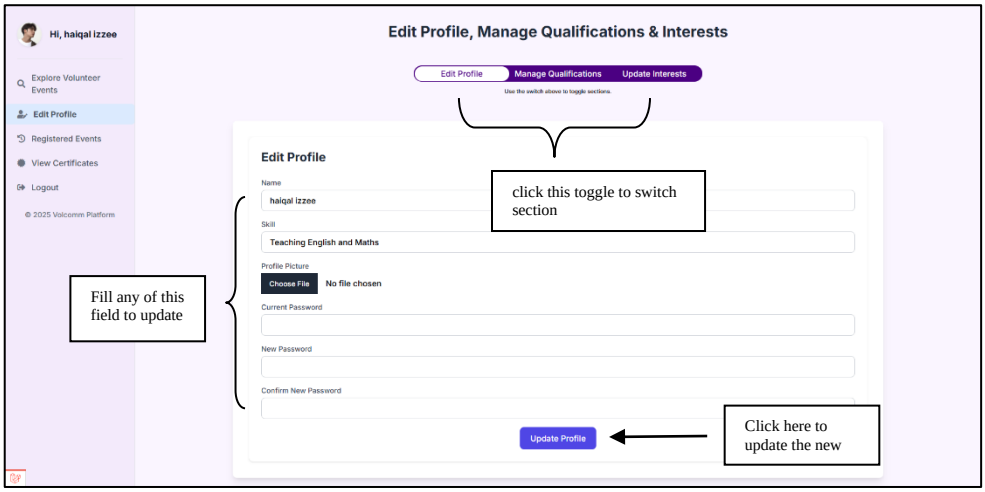


Figure 4.17 Edit profile page

k) Manage Qualification

Figure 4.18 shows page where user can manage their qualifications on this page. They can add, update, or delete qualifications that are important for applying to specific volunteer roles.

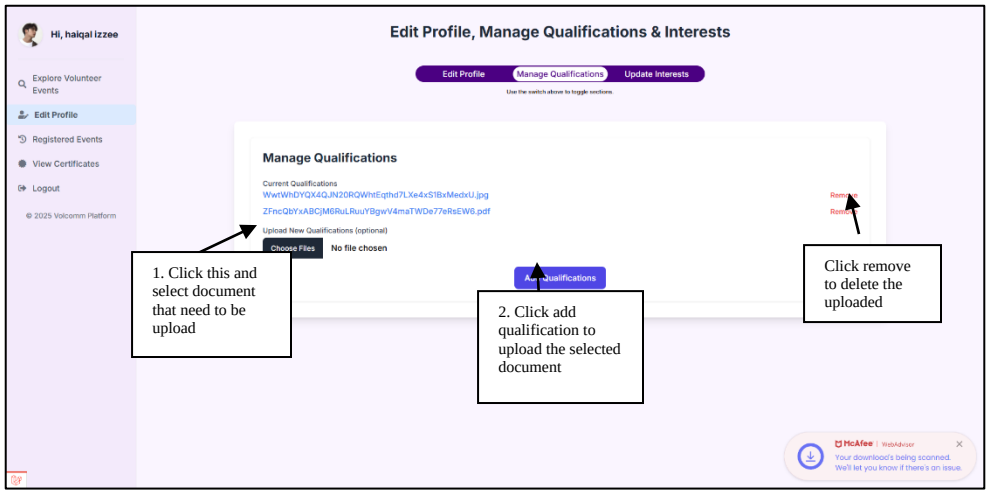


Figure 4.18 Manage qualification page

l) Update Interest

Figure 4.19 shows page that allow volunteers to update or change their areas of interest, ensuring they are matched with the most relevant opportunities

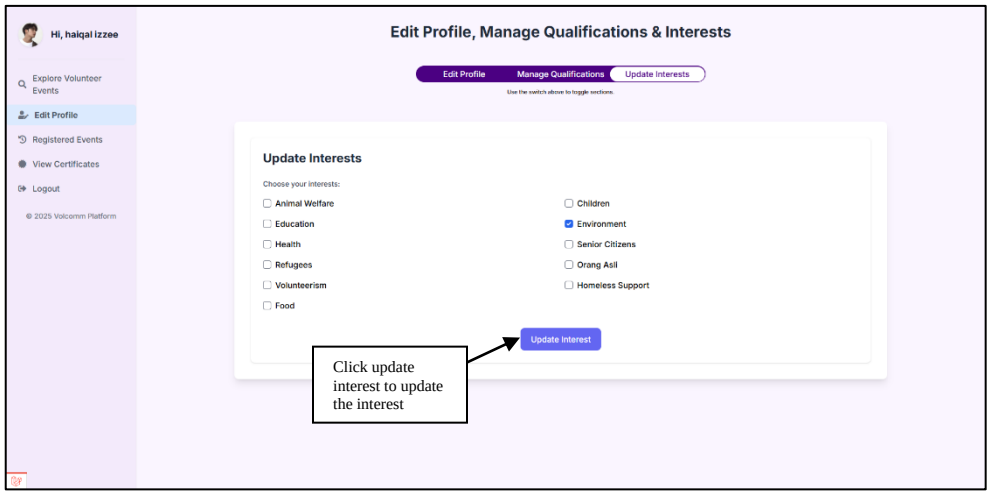


Figure 4.19 Update interest page

m) View participation history

Figure 4.20 shows "View History" page that displays a log of the volunteer's past activities. This provides volunteers with a record of their involvement.

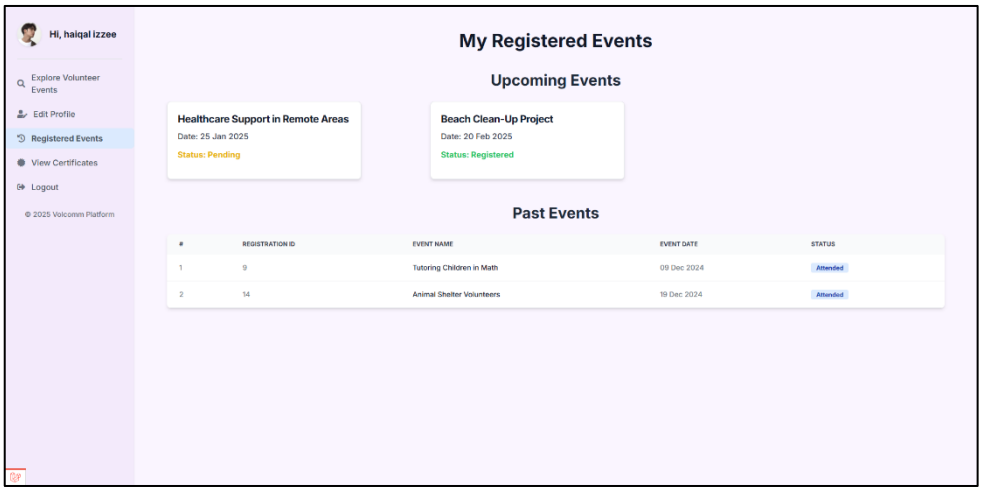


Figure 4.20 View participation history page

n) View Certificate

Figure 4.21 shows the "View Certificate" page, displaying certificates earned by volunteers after completing their service. They can click "View Certificate" to access it.

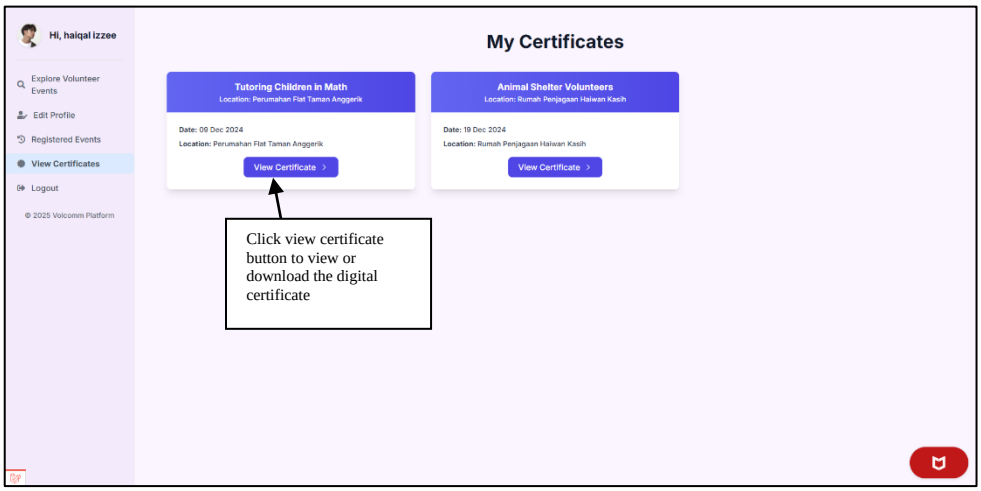


Figure 4.21 View certificate page

o) Display certificate

Figure 4.22 shows the page displayed after the volunteer clicks the "View Certificate" button. It presents the certificate with details such as the opportunity name and completion date. Users can also download it.



Figure 4.22 Display certificate page

p) Organization register page

Figure 4.23 shows organization register page where new organizations can create an account on the platform.

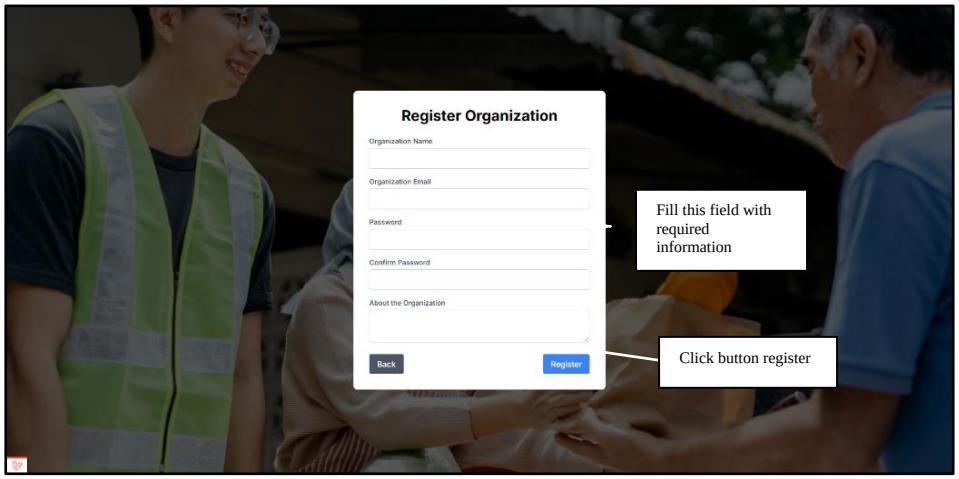


Figure 4.23 Organization register page

q) Organization login page

Figure 4.24 shows Organizations login page where they can log into their accounts through this page. Using their credentials, they gain access to their dashboard and can manage volunteer opportunities, registrations, and more.

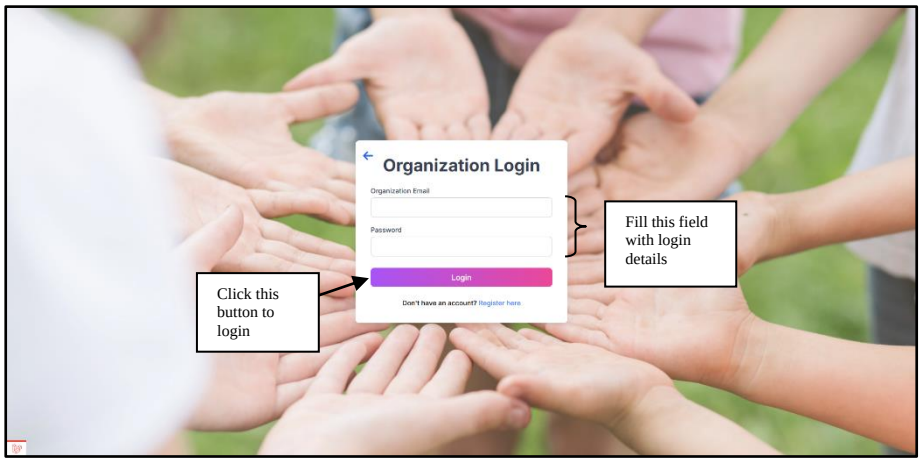


Figure 4.24 Organization login page.

r) Organization dashboard page

Figure 4.28 shows the dashboard page for organization, it serves as the main control center for organizations. It provides an overview of their volunteer activities, including upcoming events, volunteer registrations, and opportunities.

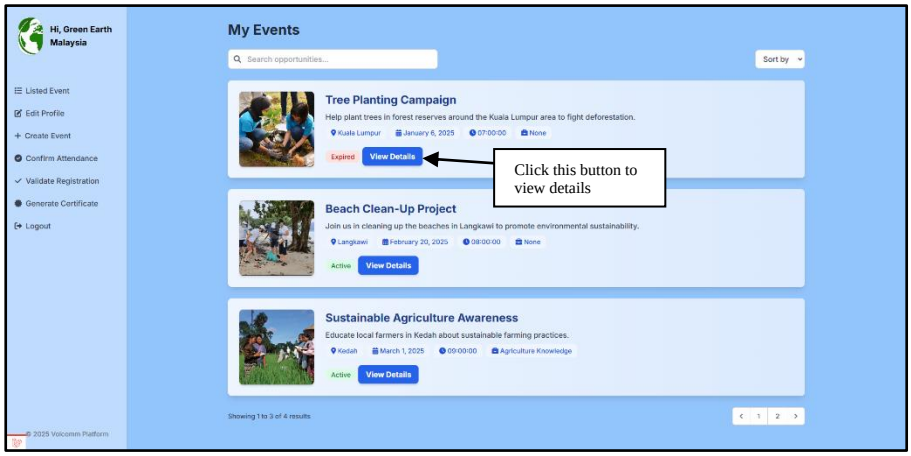


Figure 4.25 Organization dashboard page

s) Edit Profile (Organization)

Figure 4.26 shows the edit profile page for organization. They can update their profile details, including contact information, mission statement, and any other relevant data.



Figure 4.26 Edit Profile (Organization)

t) Create volunteer opportunity

Figure 4.27 shows page that allows organizations to create and post volunteer opportunities. They can provide details such as role descriptions, required skills, and the number of volunteers needed.

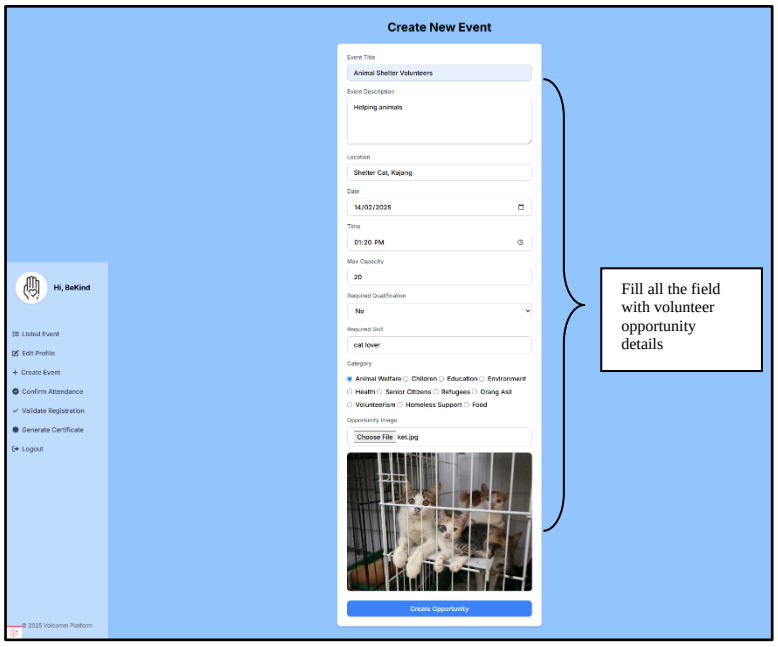


Figure 4.27 Create volunteer opportunity

u) Confirm attendance

Figure 4.28 shows the process of confirming volunteer attendance. Organization can only confirm their attendance on the volunteering day.

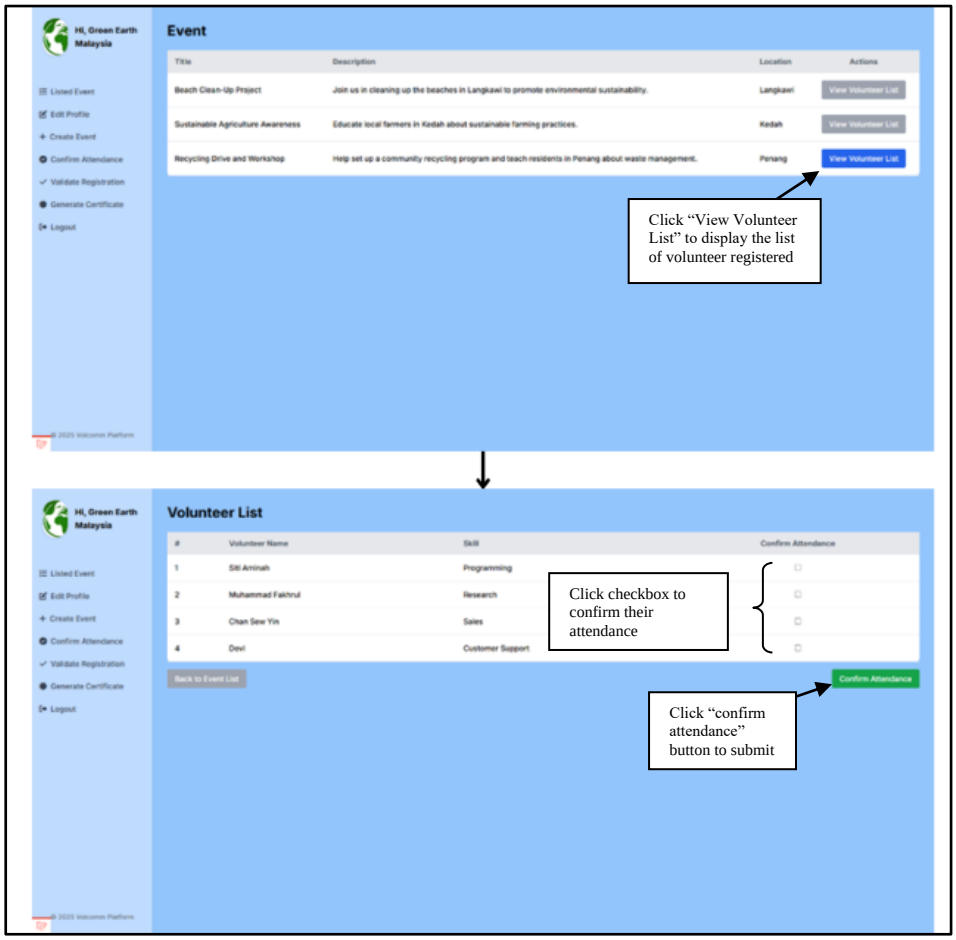


Figure 4.28 Storyboard for Confirm attendance

v) Review volunteer registration

Figure 4.29 shows the page used by organizations to validate volunteer registrations for specific opportunities. Before approving or rejecting, organizations can click the qualification file to view the submitted qualifications. Volunteers can then be approved or rejected based on the organization's requirements.



Figure 4.29 Review volunteer registration

w) Generate certificate

Figure 4.30 shows the storyboard on how organization generate certificates for volunteers who have completed an opportunity. First, select the completed opportunity, click "Select All" to tick all volunteers, upload the logo and signature, and enter the signer's name and position.

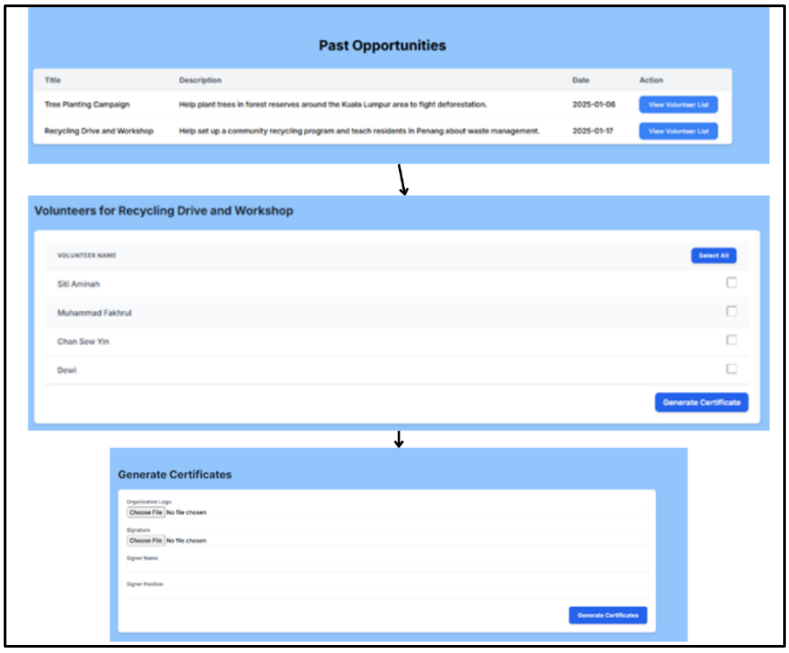


Figure 4.30 Generate certificate storyboard

x) Admin login page

Figure 4.35 shows admin login page that allows system administrators to access the platform.

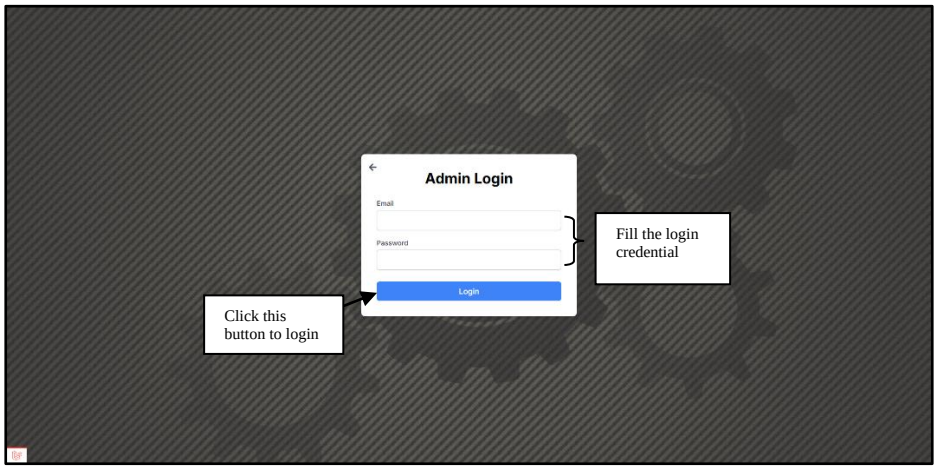


Figure 4.31 Admin login page

y) Admin dashboard

Figure 4.32 shows the admin dashboard, which serves as the main interface for system administrators. The sidebar includes functions for managing volunteers and organizations.

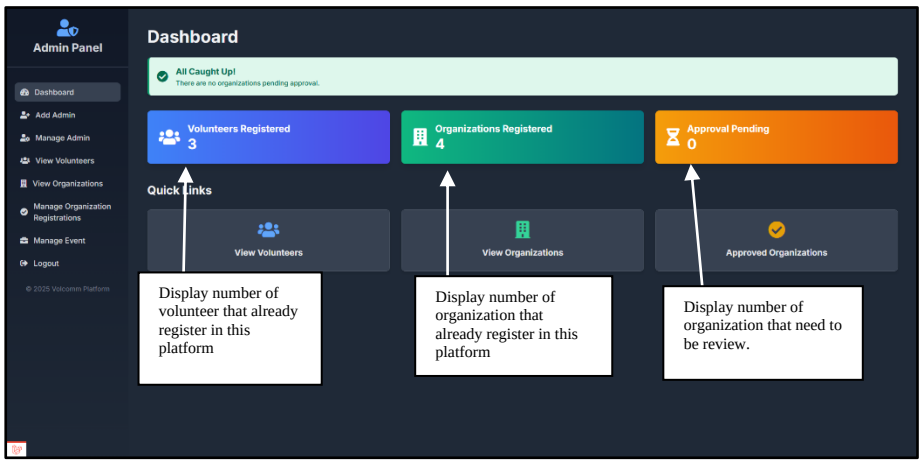


Figure 4.32 Admin dashboard

z) Add admin

Figure 4.33 shows page that allows an existing admin to add new administrators to the system.

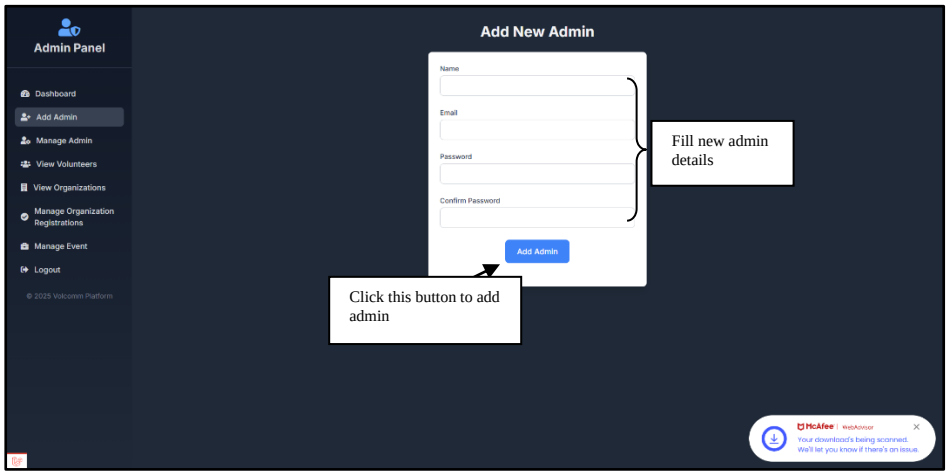


Figure 4.33 Add admin

aa) Manage admin

Figure 4.34 displays a page that allows administrators to edit or delete the profiles of other administrators. However, not all administrators have the authority to access this page.

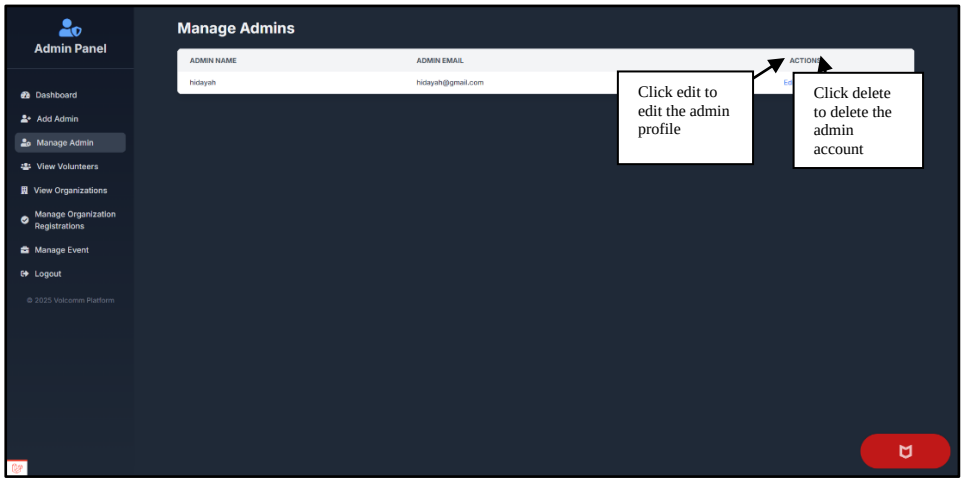


Figure 4.34 Manage admin

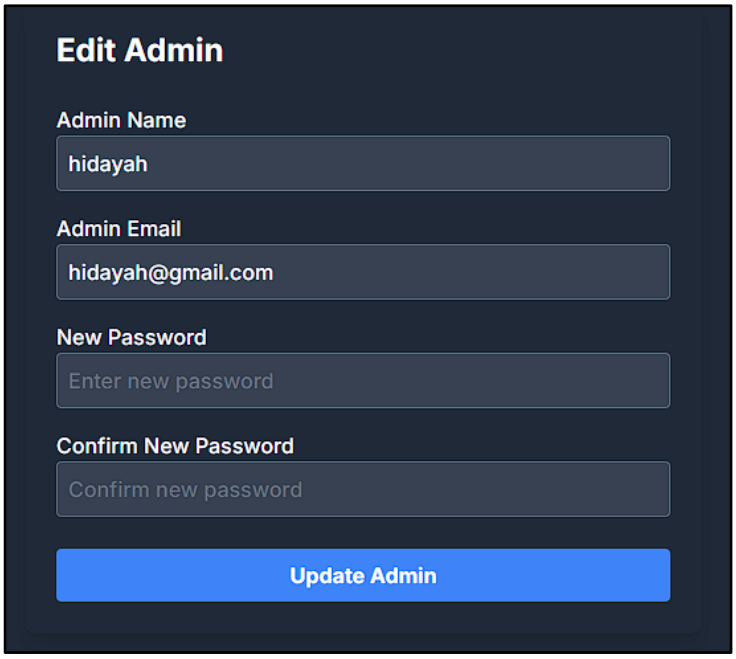


Figure 4.35 Edit Admin

Figure 4.35 shows page allows admin that have authority to edit another admin credential such as password and email.

bb) Manage volunteer

Figure 4.36 displays a page that allows administrators to manage volunteer accounts, including the option to delete an account. Figure 4.37 shows the profile page that appears when the user clicks the "View" button.

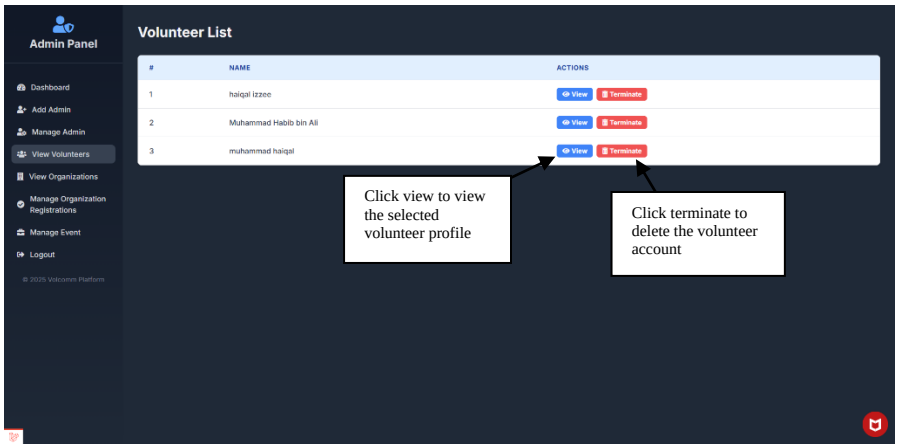


Figure 4.36 Manage volunteer

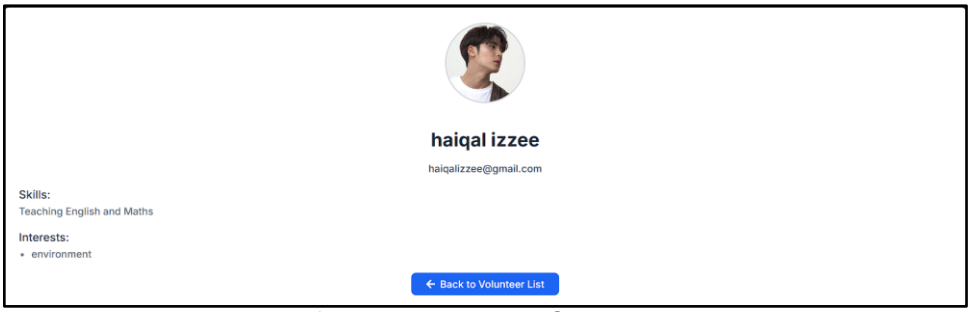


Figure 4.37 Manage volunteer

cc) Manage organization

Figure 4.38 displays a page where admin can manage organization profile. It allows them to remove profiles. Figure 4.39 shows the details of a selected organization when the user clicks the "View" button.

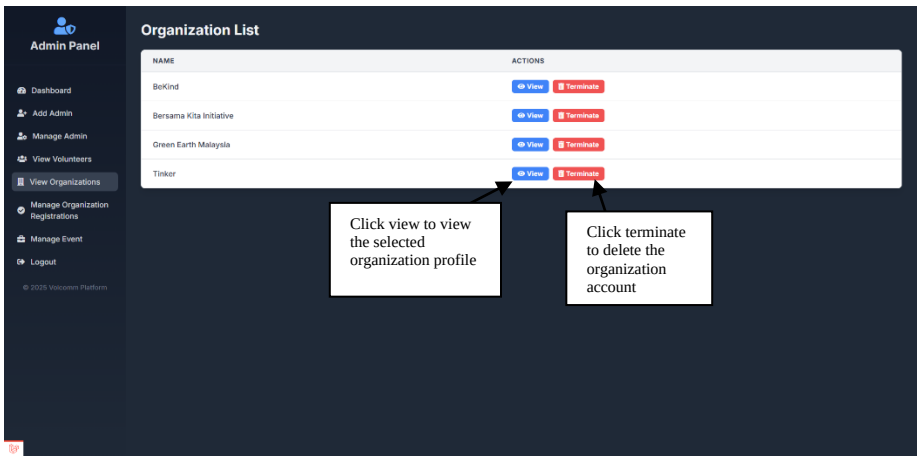


Figure 4.38 Manage organization

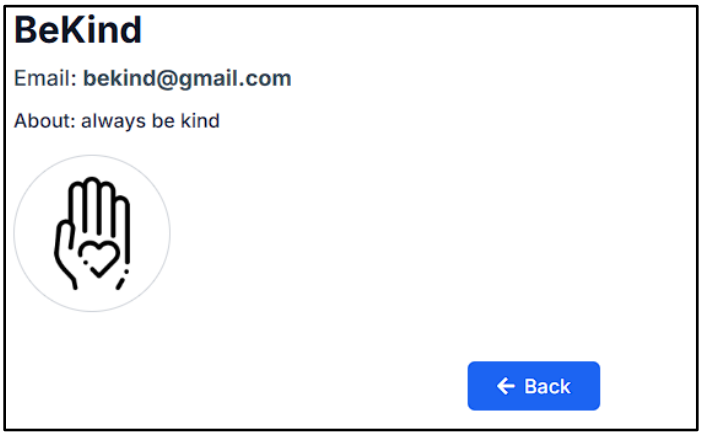


Figure 4.39 View organization page

dd) Manage organization registration

Figure 4.40 displays a page where administrators can approve or reject registration requests submitted by new organizations. They can make a decision to accept or reject the request using the available buttons.

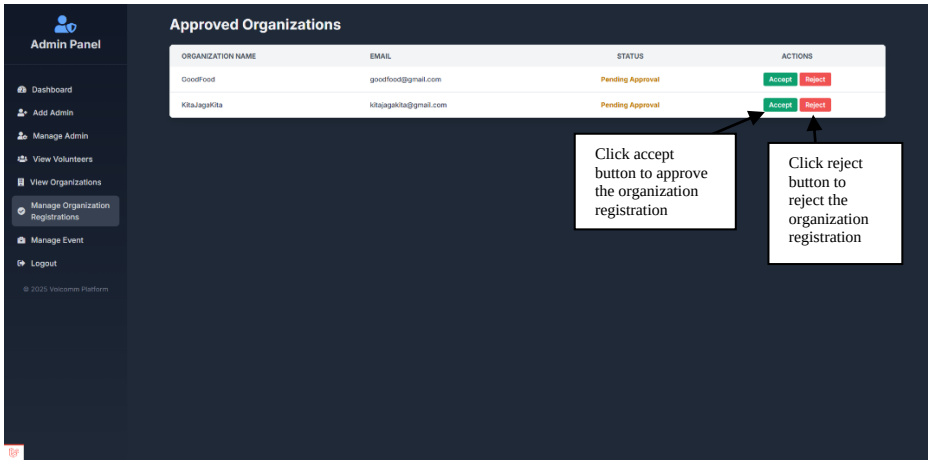


Figure 4.40 Manage organization registration

ee) Manage volunteer opportunity

Figure 4.41 shows a page where admins manage volunteer opportunities. They can review opportunities by clicking "View" and delete those that do not meet platform standards as shown on Figure 4.42 where it displays the selected opportunity details.

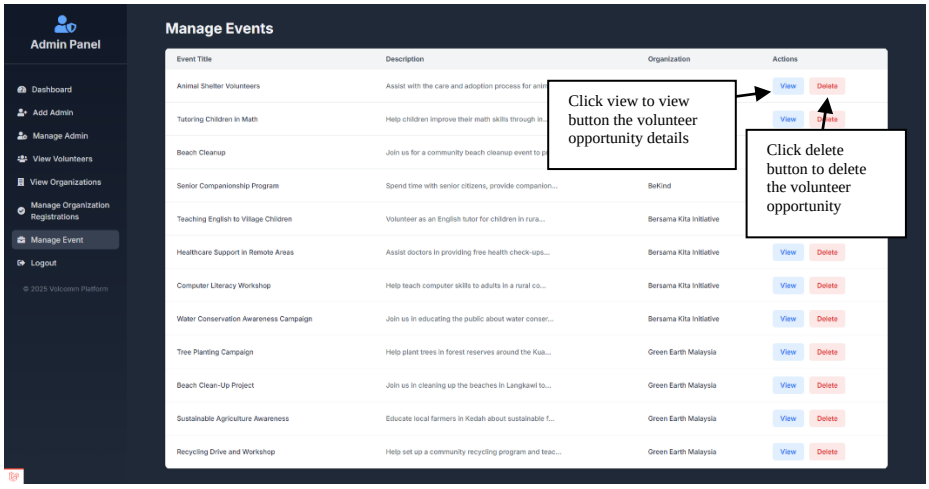


Figure 4.41 Manage volunteer opportunity

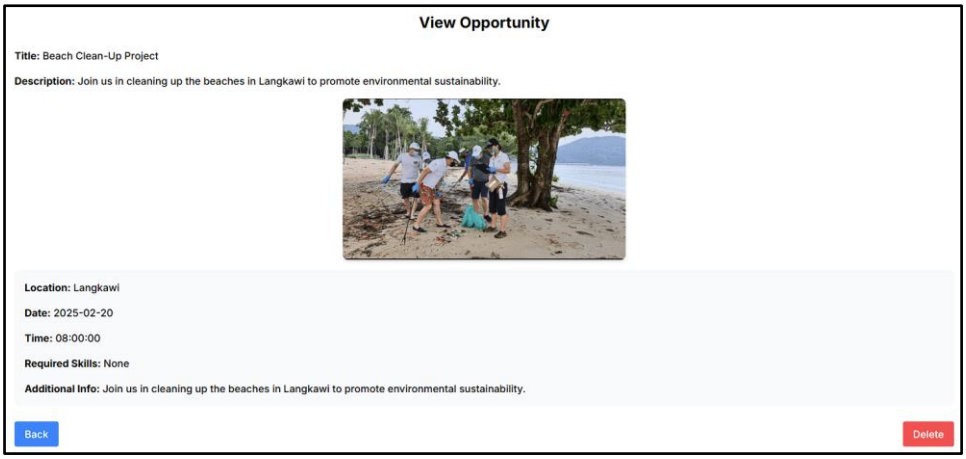


Figure 4.42 Manage volunteer opportunity

4.5 Result Findings

The survey results offer a detailed comparison between the current Volcomm system and the older method of using Facebook groups. The main objective of this evaluation was to assess how effectively the new system enables volunteers to find opportunities that align with their personal interests and preferences. Respondents were asked to evaluate different aspects of both systems, such as ease of use, time efficiency, and overall satisfaction. The results indicate a strong preference for the Volcomm system, as it provides a more organized and streamlined approach for volunteers to search for opportunities. Unlike the Facebook group method, which was often disorganized and difficult to navigate, the new system allows for more personalized and efficient searches. In conclusion, the findings suggest that the VolComm system significantly enhances volunteer management by offering a more user-friendly platform for both volunteers and organizations, improving overall engagement and participation.

4.5.1 Analysis from Volunteer Survey

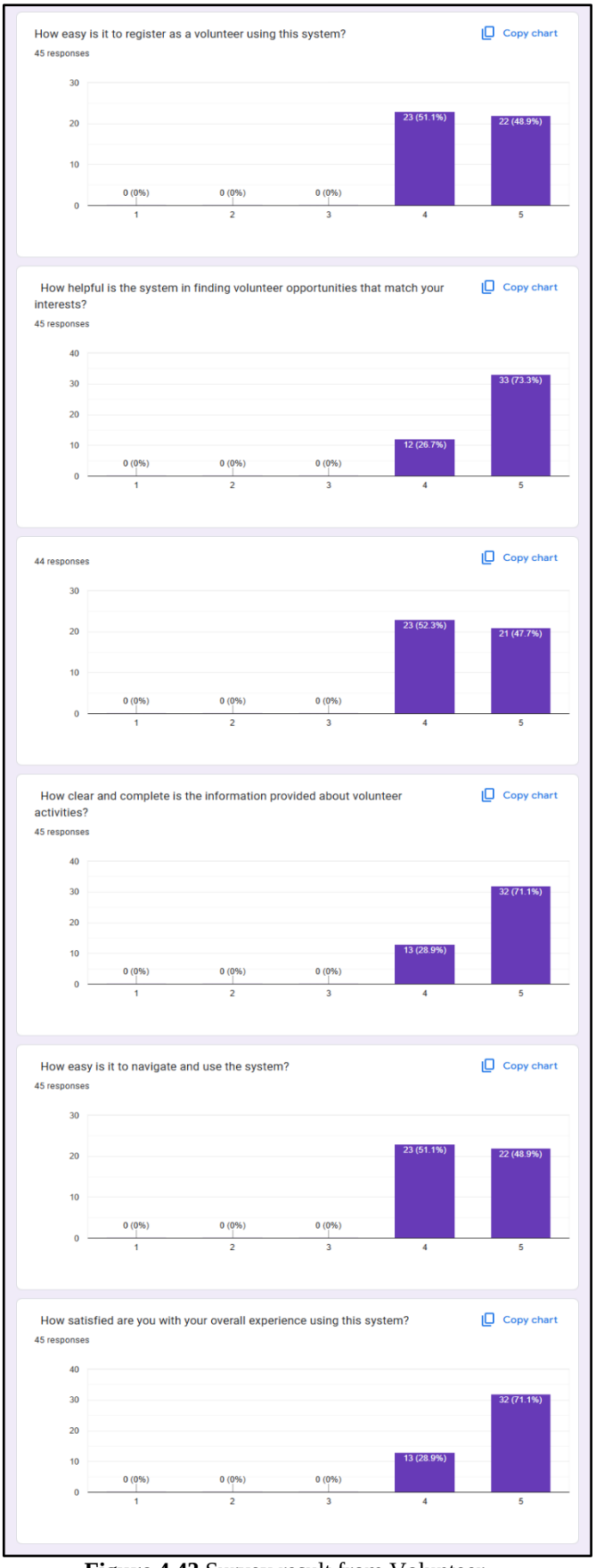


Figure 4.43 Survey result from Volunteer

Figure 4.43 presents the survey results from 45 respondents regarding their experience using VolComm. The first question, “How easy is it to register as a volunteer using VolComm?”, received responses indicating a high level of ease. A total of 23 respondents (51.1%) rated it 4, while 22 respondents (48.9%) rated it 5. This shows that almost all volunteers found the registration process simple and user-friendly, with no respondents selecting lower ratings.

For the second question, “How helpful is VolComm in finding volunteer opportunities that match your interests?”, the feedback was overwhelmingly positive. A total of 31 respondents (73.3%) rated it 5, while 12 respondents (26.7%) rated it 4. This suggests that VolComm effectively connects volunteers to suitable opportunities, making it a valuable tool for those looking to contribute to different causes.

The third question, “How clear and complete is the information provided about volunteer activities?”, also received high satisfaction ratings. The majority, 32 respondents (71.1%), rated it 5, while 13 respondents (28.9%) rated it 4. This indicates that VolComm provides clear and sufficient details about volunteer events, helping users make informed decisions before signing up.

In response to the fourth question, “How easy is it to navigate and use VolComm?”, volunteers expressed high confidence in the platform’s usability. The responses were similar to the previous question, with 23 respondents (51.1%) selecting 4, and 22 respondents (48.9%) selecting 5. This confirms that VolComm is well-structured and easy to navigate, allowing volunteers to access information and register without difficulty.

Lastly, the fifth question, “How satisfied are you with your overall experience using VolComm?”, reflects the general approval of the platform. A total of 32 respondents (71.1%) rated their satisfaction at 5, while 13 respondents (28.9%) rated it 4. With no lower ratings recorded, it is clear that the majority of volunteers had a positive experience using VolComm.

Overall, the results from Figure 4.46 demonstrate that volunteers find VolComm effective, user-friendly, and informative. The consistently high ratings across all five questions indicate that VolComm successfully meets the needs of volunteers by providing an easy registration process, helping them find relevant opportunities, offering clear information, and ensuring smooth navigation.

4.5.2 Analysis from Organizer Survey

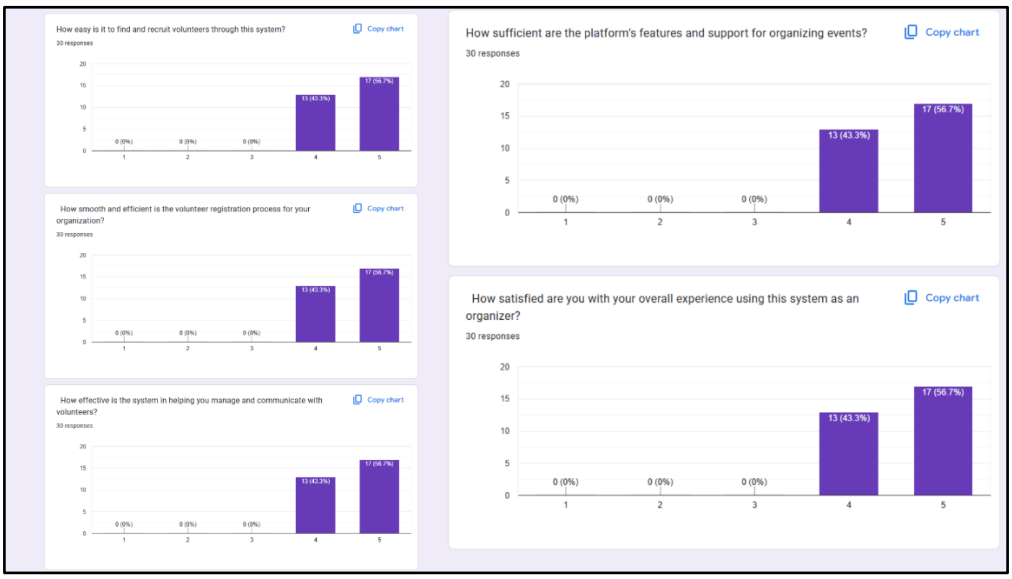


Figure 4.44 Survey results from Organizer

Figure 4.44 presents the survey results from 30 organizers regarding their experience using VolComm. The survey included five questions, and respondents rated their answers on a scale from 1 to 5, where 1 = Not at all and 5 = Very. The following analysis explains the responses for each question.

The first question, “How easy is it to find and recruit volunteers through VolComm?”, received positive responses. A total of 13 organizers (43.3%) rated it 4, while 17 organizers (56.7%) rated it 5. This shows that the majority of organizers find it easy to search for and recruit volunteers using VolComm, with no respondents giving a lower rating.

The second question, “How smooth and efficient is the volunteer registration process for your organization?”, also had high ratings. 13 organizers (43.3%) rated it 4, and 17 organizers (56.7%) rated it 5. This means that the registration process in VolComm is well-structured and does not cause difficulties for organizations managing volunteers.

For the third question, “How effective is VolComm in helping you manage and communicate with volunteers?”, most organizers agreed that VolComm improves communication and volunteer management. 13 organizers (43.3%) rated it 4, while 17 organizers (56.7%) rated it 5. This suggests that VolComm provides useful tools for organizers to interact with their volunteers smoothly.

The fourth question, “How sufficient are VolComm’s features and support for organizing events?”, also received strong approval. 13 organizers (43.3%) gave it a 4, while 17 organizers (56.7%) rated it 5. This means that VolComm includes useful features that help organizers plan and manage events effectively.

The final question, “How satisfied are you with your overall experience using VolComm as an organizer?”, confirms that organizers are happy with the platform. 13 organizers (43.3%) rated their satisfaction as 4, while 17 organizers (56.7%) rated it 5. Since no lower ratings were given, this shows that all organizers had a positive experience using VolComm.

Overall, the responses in Figure 4.47 indicate that VolComm is highly effective for organizers. The platform makes it easy to find volunteers, supports smooth registration, improves communication, and provides useful event management features. With all responses rated 4 or 5, it is clear that VolComm meets the needs of organizers well.

4.6 Summary

This chapter discusses the project's outcomes and conclusions, which were achieved through the earlier phases of requirements gathering, analysis, design, and implementation. It includes screenshots of the user interface, along with explanations of the database design, the implementation of the management system, and how it integrates into the overall system.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

This chapter provides the conclusion of the project, offering an overview of VolComm. It also addresses the project's limitations and provides detailed recommendations for future improvements and enhancements.

5.1 Conclusion

The main aim of this project was to develop the VolComm system, and it has been successfully completed, as all three objectives have been achieved. The objectives of the project were to gather data and analyze the requirements of the VolComm system, design the system based on the analyzed requirements, and develop the system. In conclusion, this study, along with the implementation of email notification and certificate generation features, has successfully met the project's main goals. By gathering detailed information, identifying a comprehensive list of needs, designing the system in detail, and incorporating key features such as email notifications and generate certificate function, the project’s objectives have been fully accomplished.

5.1.1 Objective 1: To gather data and analyze requirements of the VolComm system.

Information gathered from the interviews was carefully analyzed and used to create useful diagrams, such as the Use Case Diagram, Domain Class Diagram, and Activity Diagram. These diagrams were then documented in the Software Requirements Specification (SRS). Additionally, other diagrams like the Multilayer Sequence Diagram (MLSD) and Design Class Diagram (DCD) were created to provide a clearer understanding of how the system would function. The interviews, conducted via WhatsApp, helped gather essential details about the business processes and challenges related to the VolComm system. The information collected was transcribed and

organized to support the development of these diagrams, ensuring a thorough understanding of the system's requirements.

5.1.2 Objective 2: To design the VolComm system based on the analyzed requirements.

The design of the VolComm system was based on the requirements collected and analyzed in the previous phase. The main goal was to ensure a deep understanding of the system's needs by thoroughly reviewing the gathered data. This review shaped the system's design to meet the requirements and address concerns raised by those interviewed. A key step was creating a storyboard using Canva, which illustrated how users would interact with the system and how its features would function. Additionally, Use Case Diagrams (UCD) were created to show user interactions, while the Domain Class Diagram (DCD) outlined the system's structure. Sequence diagrams were developed to model interactions over time, and multi-sequence diagrams depicted more complex processes. Together, these diagrams provided a clear blueprint for the system's design and functionality.

5.1.3 Objective 3: To develop the VolComm system

The goal of this phase was to build the VolComm system based on the completed design. The system was developed with key features, including sending email notifications to keep users informed and generating certificates for successful participants or volunteers. These features were carefully implemented to meet the system's requirements and provide a smooth user experience. The development process ensured that all planned functionalities were included and worked as expected, creating a reliable and user-friendly system.

5.2 Project Strength

The VolComm system offers several strengths that make it an effective tool for managing volunteer activities. It simplifies the process of finding activities by allowing users to search based on their interests, which increases

participation and satisfaction. The system also combines registration and volunteer data management into one platform, eliminating the need for multiple websites and reducing complexity. By centralizing tasks like activity search and registration, the system saves time for both users and administrators, making the process more efficient. Additionally, the automated generation of volunteer certificates ensures accuracy, reduces errors, and provides easy access to certificates.

It offers clear advantages in terms of streamlining volunteer management and enhancing user experience. Its ability to automate procedures and simplify functions results in a system that is more structured and effective. It enables administrators and users to focus on more crucial activities by decreasing manual labor. Higher user happiness, improved participation rates, and more trustworthy record-keeping are all facilitated by the system's strengths. All things considered, the VolComm system provides a workable solution that improves the volunteer experience and facilitates administrator management.

5.3 Limitations

i) Limited Activities and demographic area coverage:

The VolComm system currently offers a limited range of activity categories, such as education, welfare, and the environment. This narrow scope restricts users who may be interested in volunteering for activities in other sectors. Expanding the variety of activity categories could increase the system's usefulness for a broader audience.

ii) No Real-Time Communication Features:

The system does not have an integrated real-time communication feature, meaning users cannot directly contact activity organizers or other participants through the platform. If users have questions or need clarification, they must rely on external messaging apps like WhatsApp. This creates an extra step in communication and can lead to delays.

iii) Did Not Cover Crowdfunding:

The VolComm system currently lacks a crowdfunding feature, which means users cannot raise or contribute funds for specific causes directly through the platform. This limitation restricts the ability to support charitable activities financially within the system.

5.4 Recommendation for Future Enhancement

To enhance the project, the following improvements and additional features are recommended:

i) Expand Activity Categories and Area Coverage:

To address the limited range of activity categories, it is recommended that the VolComm system expands its offerings to include sectors such as sports and arts. This will allow users with different interests and skills to find relevant volunteer opportunities, making the platform more inclusive and useful to a broader audience.

ii) Integrate Real-Time Communication Features:

To improve communication, the system should incorporate real-time messaging features, enabling users to contact activity organizers or other participants directly through the platform. This would eliminate the need for external messaging apps like WhatsApp and make communication faster and more efficient, reducing any delays in receiving responses or clarifications.

iii) Include Crowdfunding Features:

To address the lack of crowdfunding options, it is recommended that the VolComm system integrates a crowdfunding feature with a secure payment gateway. This would allow users to donate or raise funds for various causes directly on the platform, adding a valuable feature that supports charitable activities and enhances the system’s overall functionality.

5.5 Summary

This chapter reviewed how the project achieved its three main objectives. It also discussed the limitations of outsourcing and provided suggestions for improving the project in the future. In the future, the system can be further developed to work more efficiently, using the project's constraints more effectively. The recommendations given in this chapter can serve as helpful guidelines for improving and expanding the system in the future.

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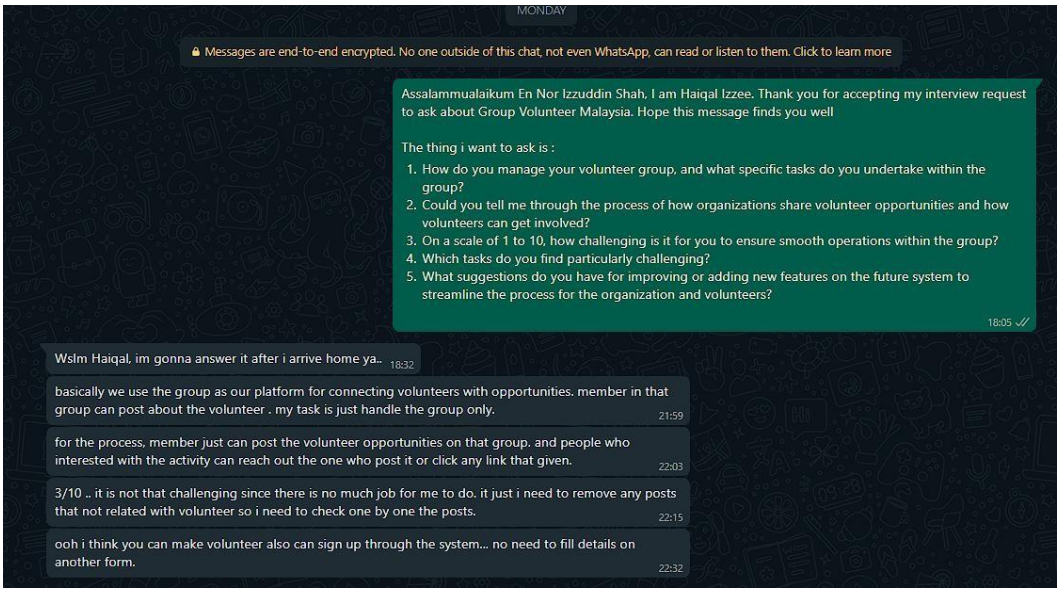
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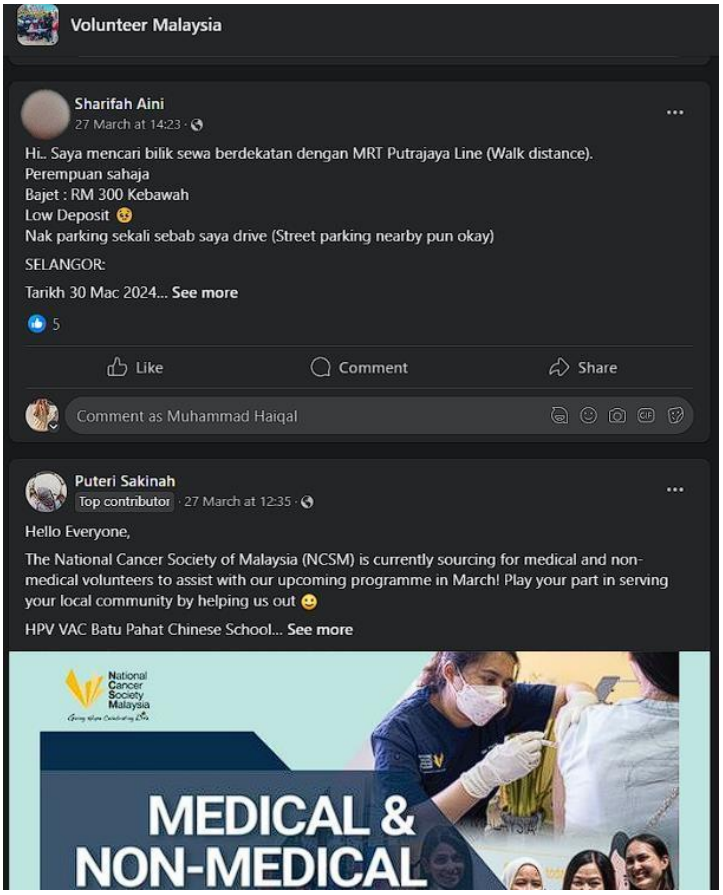
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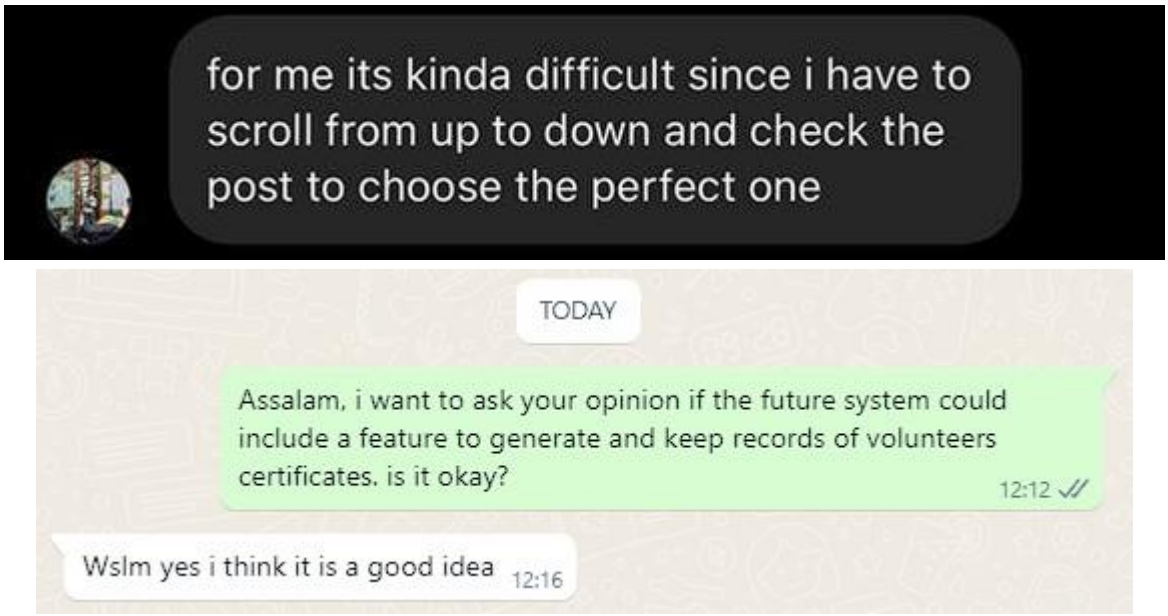
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APPENDIX A


MONDAY
Messages are end-to-end encrypted. No one outside of this chat, not even WhatsApp, can read or listen to them. Click to learn more
Assalammualaikum En Nor Izzuddin Shah, I am Haiqal Izzee. Thank you for accepting my interview request to ask about Group Volunteer Malaysia. Hope this message finds you well
The thing i want to ask is :
1. How do you manage your volunteer group, and what specific tasks do you undertake within the group?
2. Could you tell me through the process of how organizations share volunteer opportunities and how volunteers can get involved?
3. On a scale of 1 to 10, how challenging is it for you to ensure smooth operations within the group?
4. Which tasks do you find particularly challenging?
5. What suggestions do you have for improving or adding new features on the future system to streamline the process for the organization and volunteers?
18:05
Wslm Haiqal, im gonna answer it after i arrive home ya.. 18:32
basically we use the group as our platform for connecting volunteers with opportunities. member in that group can post about the volunteer . my task is just handle the group only. 21:59
for the process, member just can post the volunteer opportunities on that group. and people who interested with the activity can reach out the one who post it or click any link that given. 22:03
3/10 .. it is not that challenging since there is no much job for me to do. it just i need to remove any posts that not related with volunteer so i need to check one by one the posts. 22:15
ooh i think you can make volunteer also can sign up through the system... no need to fill details on another form. 22:32


Volunteer Malaysia
Sharifah Aini
27 March at 14:23 ·
Hi.. Saya mencari bilik sewa berdekatan dengan MRT Putrajaya Line (Walk distance).
Perempuan sahaja
Bajet : RM 300 Kebawah
Low Deposit
Nak parking sekali sebab saya drive (Street parking nearby pun okay)
SELANGOR:
Tarikh 30 Mac 2024... See more
5
Like
Comment
Share
Comment as Muhammad Haiqal
Puteri Sakinah
Top contributor · 27 March at 12:35 ·
Hello Everyone,
The National Cancer Society of Malaysia (NCSM) is currently sourcing for medical and non-medical volunteers to assist with our upcoming programme in March! Play your part in serving your local community by helping us out
HPV VAC Batu Pahat Chinese School... See more
National Cancer Society Malaysia
MEDICAL &
NON-MEDICAL


for me its kinda difficult since i have to scroll from up to down and check the post to choose the perfect one
TODAY
Assalam, i want to ask your opinion if the future system could include a feature to generate and keep records of volunteers certificates. is it okay?
12:12
Wslm yes i think it is a good idea
12:16

APPENDIX B



UNIVERSITI TEKNOLOGI MARA

SOFTWARE REQUIREMENTS SPECIFICATION

Supervisor:
DR.NUR SUHAILAYANI SUHAIMI

Project Title:
VOLCOMM: VOLUNTEER AND COMMUNITY SERVICE HUB FOR NON-GOVERNMENTAL ORGANIZATIONS (NGOS)

Name:
MUHAMMAD HAIQAL BIN MOHD IZZEE (2023115539)

Submit Date:
23 FEBRUARY 2025

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Initial SRS - Version 1.0	Muhammad Haiqal	Added template for SRS	9/8/2024
Added Introduction - Version 1.0	Muhammad Haiqal	Added everything needed and finished the conditions for the introduction of this document.	9/8/2024
Added Overall Description - Version 1.0	Muhammad Haiqal	Added all the overall description.	9/8/2024
Added Use Case Diagram - Version 1.0	Muhammad Haiqal	Added the use case diagram of functionality forthe system.	11/8/2024
Added Use Case Description - Version 1.0	Muhammad Haiqal	Added all of the use case descriptions of theproject.	11/8/2024
Added Other Non-Functional Requirements - Version 1.0	Muhammad Haiqal	Added all other non-functional requirements ofthe project.	12/8/2024
Added Class Diagram - Version 1.0	Muhammad Haiqal	Added the class diagram of the project.	13/8/2024
Added System Sequence Diagram - Version 1.0	Muhammad Haiqal	Added all of the system sequence diagrams ofthe project.	13/8/2024
Added Business Activity Diagram - Version 1.0	Muhammad Haiqal	Added all of the business activity diagrams ofthe project.	13/8/2024
Modified Use Case Descriptions and System Sequence Diagrams - Version 1.0	Muhammad Haiqal	Made changes to the use case descriptions and system sequence diagrams for the project.	29/1/2025

1. Introduction

1.0 Document Purpose

This document describes the software requirements for a specific item, including its version or release number. To identify the product and its version or release number, more information or context is needed. The scope of what the SRS (Software Requirements Specification) covers is not clear without more details. This document explains the purpose and features of the software, its interfaces, what it will do, the limitations it must work under, and how it will respond to external inputs. It is meant for both the users and the developers of the software.

1.1 Product Scope

The scope of this project is to deliver a comprehensive solution that allows volunteers to register, search for, and sign up for volunteer opportunities, while enabling NGOs to post opportunities, manage volunteer registrations, and issue certificates of participation. The platform will feature user-friendly interfaces for both volunteers and NGOs

1.2 Intended Audience and Document Overview

The All stakeholders use the SRS for their specific roles. Developers use the SRS to understand what needs to be built and how it should function. Project managers use it to plan and manage the project, including the schedule and resources. Testers use it to know what needs to be tested and verified. Documentation writers use it to know what needs to be documented and how to do it. This section gives an overview of the software and its functions. It explains what the software must do and how it should work. It also discusses the constraints and requirements the software must meet in terms of performance, security, and other factors. This section also covers the software's appearance, user interface, and how it interacts with other systems. This document is for various roles such as requirements engineers, programmers, developers, UI designers, system analysts, and project managers.

1.3 Definitions, Acronyms and Abbreviations

Table 1.1 Acronyms and Definition

Acronyms	Definition
VolComm	Volunteer And Community Service Hub For Non-Governmental Organizations
SDP	Software Development Plan
SDLC	Software Development Life Cycle
SRS	Software Requirements Specifications
DCD	Design Class Diagram
SDD	Software Design Description
REQ	Requirement
UCD	Use Case Description
SSD	System Sequence Diagram
UC	Use Case

1.4 Document Conventions

Main Header	Font type: Times New Roman Font style: BOLD Size: 14
Sub Header	Font type: Times New Roman Font style: BOLD Size: 12
Contents Header	Font type: Times New Roman Font style: BOLD Size: 12
Contents	Font type: Times New Roman Font style: BOLD Size: 12

1.5 References and Acknowledgments

1. IEEE. (1998). IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830-1998). IEEE.
2. Booch, G., Rumbaugh, J., & Jacobson, I. (2005). The Unified Modeling Language User Guide (2nd ed.). Addison-Wesley.

2. Overall Description

2.0 Product Perspective

VolComm: Volunteer And Community Service Hub For Non-Governmental Organizations is a web-based platform designed to connect volunteers with NGOs looking for support. The system allows volunteers to register, view and sign up for opportunities, and track their participation history. Organizations can post volunteer opportunities, manage registrations, confirm attendance, and generate certificates for volunteers. The platform also includes features for organizations to upload and design certificates, which are automatically generated for volunteers based on their participation. An admin role is included to oversee and approve organizations, ensuring that only verified NGOs can post opportunities. VolComm simplifies the process of volunteer management, making it easier for NGOs to find the help they need and for volunteers

2.1 Product Functionality

The following are the major functions that the VolComm SRS (Software Requirements Specification) document should include:

- User Management: manage user accounts and permissions, including registration and update account.
- Online Access: provide safe and private website access for volunteer, organization, and admin including the ability to register and view volunteering activity information.

2.2 Users and Characteristics

VolComm has three main types of users which are :

- Volunteers: They can register on the platform, browse and sign up for volunteer opportunities, and track their participation history. They can also generate certificates after completing volunteer work.
- Organizations: They can create and post volunteer opportunities, manage volunteer registrations, confirm attendance for volunteers.
- Admins: They oversee the platform, approve or reject organization accounts, and ensure that all posted opportunities and registered organizations meet the required standards.

2.3 Operating Environment

The operating environment refers to the hardware and software settings where the VolComm system functions. VolComm is a web-based application, so it operates primarily within a web server environment. Users access the system via standard web browsers on various devices, including desktops, laptops, tablets, and smartphones. The software requirements are minimal, needing just an online server and Internet connection. This is because the software runs only on an online server.

2.4 Design and Implementation Constraints

The design and implementation of VolComm face several constraints. First, the platform must be easy to use, with a simple and intuitive interface, as users may have varying levels of technical skills. Second, the system needs to handle potentially large volumes of data, especially for volunteer registrations and certificate generation, requiring efficient database management. Third, security is crucial, as personal information and sensitive data, like attendance records and certificates, must be protected. Finally, the platform must be scalable to support future growth, allowing more organizations and volunteers to join without impacting performance.

2.5 User Documentation

i. User Manual
ii. Online Research

2.6 Assumptions and Dependencies

VolComm assumes that users (volunteers, organizations, and admins) have basic internet access and know how to use the platform. Organizations are expected to be legitimate and provide accurate information, while volunteers should provide truthful details when registering.

The system depends on a stable internet connection, working email services for notifications, and external tools for features like file uploads and certificate generation. Regular maintenance and updates are needed to ensure security and smooth operation. These assumptions and dependencies are key for VolComm to work properly.

3.1 Use Case Diagram

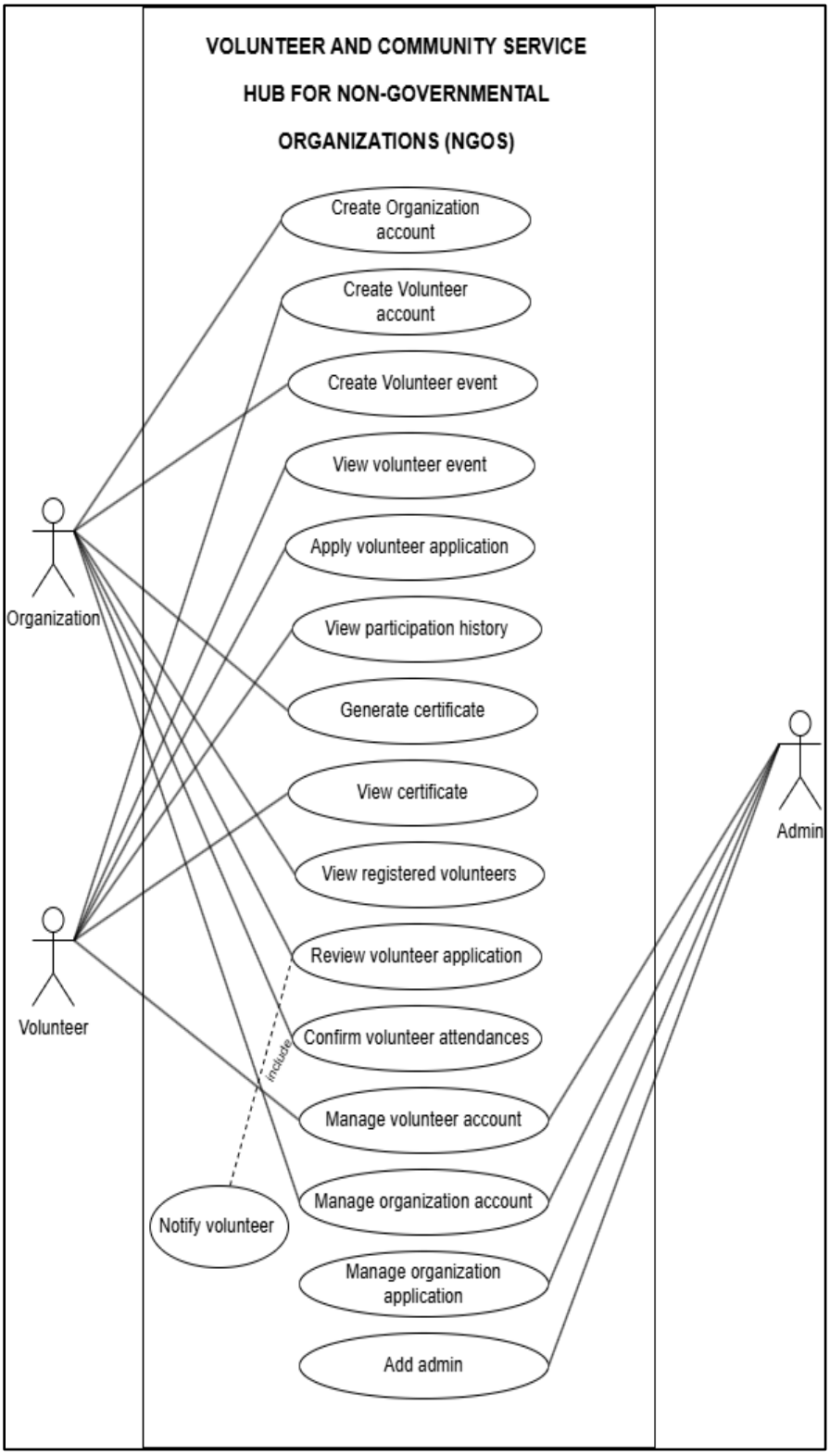


Figure 3.1 VolComm Use Case Diagram

3.2 Use Case Description

3.2.1 UCD001 Create Organization Account

Use Case ID:	UCD001		
Use Case Name:	Create Organization Account		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Organization	
Description:	Register account in the system.	
Preconditions:	User has a valid email address. User has not registered with the system previously.	
Postconditions:	The account is successfully created and saved.	
Priority:	High	
Frequency of Use:	High	
Normal Course of Events:	Actor	System
	1. The user fills in the required details 2. The user submits the form.	1.1 The system validates the input data. 2.1 The system creates a new user account 2.2 The system redirects the user to the login page.
Alternative Courses:	N/A	
Exceptions:	If required fields are left blank, the system will prompt the user to complete them before submitting.	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.2 UCD002 Create Volunteer Account

Use Case ID:	UCD002		
Use Case Name:	Create Volunteer Account		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Volunteer	
Description:	Register account in the system.	
Preconditions:	User has a valid email address. User has not registered with the system previously.	
Postconditions:	The account is successfully created and saved.	
Priority:	High	
Frequency of Use:	High	
Normal Course of Events:	Actor	System
	1. The user fills in the required details 2. The user submits the form.	1.1 The system validates the input data. 2.1 The system creates a new user account 2.2 The system redirects the user to the login page.
Alternative Courses:	N/A	
Exceptions:	If required fields are left blank, the system will prompt the user to complete them before submitting.	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.3 UCD003 Create Volunteer Opportunity

Use Case ID:	UCD003		
Use Case Name:	Create Volunteer Opportunity		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Organization	
Description:	Organization create and post a new volunteer opportunity on the system.	
Preconditions:	The Organization is logged into the system.	
Postconditions:	A new volunteer opportunity is created and visible to volunteers on the platform.	
Priority:	High	
Frequency of Use:	High	
Normal Course of Events:	Actor	System
	1. The Organization selects "Create Volunteer Opportunity" from the sidebar. 2. The Organization fills in the details. 3. The Organization submits the form.	1.1 The system redirect to the opportunity creation form. 2.1 The system validates the input data. 3.1 The system saves the opportunity and makes it visible to volunteers.
Alternative Courses:	N/A	
Exceptions:	If required fields are left blank, the system will prompt the Organization to complete them before submitting.	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.4 UCD004 View Volunteer Opportunity

Use Case ID:	UCD004		
Use Case Name:	View Volunteer Opportunity		
Created By:	Muhammad Haiqal	Last Updated By	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Volunteer	
Description:	Manage account information	
Preconditions:	The user is logged into the system.	
Postconditions:	The system displays the details of the selected volunteering opportunity.	
Priority:	High	
Frequency of Use:	High	
Normal Course of Events:	Actor	System
	1. Volunteer clicks "View Details" for a specific opportunity.	1.1 The system displays the details of the selected opportunity.
Alternative Courses:	N/A	
Exceptions:	If there are no active opportunities, the system displays a message: "No opportunities available at the moment."	
Includes:	N/A.	
Notes and Issues:	N/A	

3.2.5 UCD005 Apply volunteer application

Use Case ID:	UCD005		
Use Case Name:	Register Volunteer Opportunities		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Volunteer	
Description:	Volunteer register for a specific volunteer opportunity.	
Preconditions:	The Volunteer is logged into the system. There are available volunteer opportunities posted by organizations.	
Postconditions:	The Volunteer is successfully registered for the selected volunteer opportunity.	
Priority:	High	
Frequency of Use:	High	
Normal Course of Events:	Actor	System
	1. Volunteer selects a volunteer opportunity from the list. 2. Volunteer clicks "View Details" 3. Volunteer has previously submitted the necessary documents.	1.1 The system displays the details of the selected opportunity. 2.1 The system checks if the opportunity requires specific expertise or qualifications. 2.2 If no expertise is required, the system proceeds to 3.2. 3.1 The system verifies the documents and proceeds with registration. 3.2 The system registers the Volunteer and updates the list of participants. 3.3 The system sends a confirmation email or notification to the Volunteer.
Alternative Courses:	N/A	
Exceptions:	If the opportunity has reached its maximum number of participants, the system will display a message indicating that registration is closed.	
Includes:	N/A	

Notes and Issues:	

3.2.6 UCD006 View Participation History

Use Case ID:	UCD006		
Use Case Name:	View Participation History		
Created By:	Muhammad Haiqal	Last Updated By	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Volunteer	
Description:	Volunteer view their participation history in various volunteer opportunities.	
Preconditions:	The Volunteer has participated in one or more volunteer opportunities.	
Postconditions:	The Volunteer's participation history is displayed on the screen.	
Priority:	High	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. Volunteer selects "View Participation History" from the dashboard. 2. Volunteer views the list of past volunteer opportunities.	1.1 The system retrieves the Volunteer's participation history from the database. 2.1 The system displays the list, including details such as event name, date, and participation status.
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A.	
Notes and Issues:	N/A	

3.2.7 UCD007 Generate Certificates

Use Case ID:	UCD007		
Use Case Name:	Generate Certificates		
Created By:	Muhammad Haiqal	Last Updated By	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Organization	
Description:	Volunteer generate certificates they have earned from participating in volunteer opportunities	
Preconditions:	The volunteers must have a status of "attended" for the selected opportunity.	
Postconditions:	Certificates are generated for the selected volunteers and saved in the system.	
Priority:	High	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. Organization selects a specific opportunity from the dashboard. 2. Organization selects volunteers by clicking "Select All" or choosing specific volunteers. 3. Organization uploads a logo, inputs signer name, and adds a digital signature and click “Generate certificate”	1.1 The system displays the list of volunteer who attend the volunteering 2.1 The system confirms the selected volunteers and displays certificate customization options. 3.1 The system generates certificates for the selected volunteers.
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A.	
Notes and Issues:	N/A	

3.2.8 UCD008 View Certificate

Use Case ID:	UCD008		
Use Case Name:	View Certificate		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Volunteer	
Description:	Allows the volunteer to view certificates they have earned for specific volunteer opportunities	
Preconditions:	The volunteer must have completed the opportunity and have a generated certificate associated with it.	
Postconditions:	The system displays the certificate for the selected opportunity.	
Priority:	High	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. Volunteer clicks "View Certificate" from the sidebar. 2. Volunteer clicks "View Certificate" on selected opportunity	1.1 The system displays a list of opportunities with generated certificates. 2.1 The system retrieves and displays the certificate for the selected opportunity.
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.9 UCD009 View Registered volunteers

Use Case ID:	UCD009		
Use Case Name:	Confirm Volunteer Attendance		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Organization	
Description:	Allows the organization to view a list of volunteers who have registered for a specific volunteer opportunity.	
Preconditions:	The selected opportunity must exist and have volunteers registered for it.	
Postconditions:	The system displays the list of volunteers registered for the selected opportunity.	
Priority:	High	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. Organization selects an opportunity and clicks “More Details”. 2. Organization scrolls down to the "Registered Volunteers" section.	1.1 The system retrieves and displays the details of the selected opportunity. 2.1 The system displays the list of volunteers who have registered for the opportunity.
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.10 UCD010 Review volunteer application

Use Case ID:	UCD010		
Use Case Name:	Review volunteer application		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Organization	
Description:	Allows the organization to review and accept/reject volunteer applications for opportunities that require qualifications..	
Preconditions:	1. The system updates the status of the application to either "approved" or "rejected." 2. The volunteer is notified of the application status.	
Postconditions:	The system displays the list of volunteers registered for the selected opportunity.	
Priority:	High	
Frequency of Use:	High	
Normal Course of Events:	Actor	System
	1. Organization clicks "Validate Registration" from the sidebar.	1.1 The system retrieves and displays a list of pending applications for opportunities requiring qualifications.
	2. Organization reviews the list of applications.	2.1 The system displays volunteer details and their submitted qualifications.
	3. Organization clicks "Approve" for an application.	3.1 The system updates the application status to "approved."
	4. Organization clicks "Reject" for an application.	4. Organization clicks "Reject" for an application.
	5. System notifies the volunteer of their application status.	5. The system sends an email to the volunteer about their updated application status.
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.11 UCD011 Confirm Volunteer Attendance

Use Case ID:	UCD011		
Use Case Name:	Confirm Volunteer Attendance		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Organization	
Description:	Allows the organization to view a list of volunteers who have registered for a specific volunteer opportunity.	
Preconditions:	The selected opportunity must exist and have volunteers registered for it.	
Postconditions:	The system displays the list of volunteers registered for the selected opportunity.	
Priority:	High	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. Organization clicks "Confirm Attendance" from the sidebar. 2. Organization selects an opportunity and clicks "View Volunteer List." 3. Organization clicks "Select All" or selects specific volunteers. 4. Organization clicks the "Confirm Attendance" button.	1.1 The system displays a list of all opportunities with registered volunteers. 2.1 The system displays a list of all volunteers registered for the selected opportunity. 3.1 The system confirms the selected volunteers for attendance. 4.1 The system updates the attendance status of the selected volunteers to "attended."
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.12 UCD012 Manage Volunteer Account

Use Case ID:	UCD012		
Use Case Name:	Manage Volunteer Account		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Volunteer, Admin	
Description:	Allows volunteers to manage their account details, including editing their profile, adding or removing qualifications, and changing their interests. Admin can view volunteer profiles and terminate accounts.	
Preconditions:	Volunteer accounts must exist in the system.	
Postconditions:	For volunteers: The account details are updated in the system. For admins: The volunteer account is terminated	
Priority:	High	
Frequency of Use:	High	
Normal Course of Events:	Actor	System
	1. Volunteer selects "Edit Profile" from the dashboard. 2. Volunteer can use the toggle to edits profile details, adds or removes qualifications, and updates interests. 3. Volunteer submits the changes.	1.1 The system displays the volunteer's current account details. 2.1 The system displays the list of volunteers who have registered for the opportunity. 3.1 The system updates the volunteer's account details and displays a success message.
	1. Admin selects "Manage Volunteers" from the admin dashboard. 2. Admin selects a volunteer profile to view. 3. Admin clicks "Terminate Account" for a specific volunteer.	1.1 The system displays a list of all volunteers. 2.1 The system displays the volunteer's account details, including their profile, qualifications, and interests. 3.1 The system confirms the action and terminates the volunteer's account.

Alternative Courses:	N/A
Exceptions:	N/A
Includes:	N/A
Notes and Issues:	N/A

3.2.13 UCD013 Manage Organization Account

Use Case ID:	UCD013		
Use Case Name:	Manage Organization Account		
Created By:	Muhammad Haiqal	Last Updated By:	Muhammad Haiqal
Date Created:	11/8/2024	Date Last Updated:	29/1/2025

Actor:	Organization, Admin	
Description:	Allows organizations to manage their account details, including changing their logo, and updating their password. Admin also can view and terminate organization accounts	
Preconditions:	The organization account must exist in the system.	
Postconditions:	For organizations: The updated profile details, logo, and password are saved in the system. For admins: The selected organization account is terminated.	
Priority:	High	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. Organization clicks "Edit Profile" from the sidebar. 2. Organization updates profile details. 3. Organization clicks "Update Profile."	1.1 he system displays the organization's current profile details and logo. 2.1 The system validates the updated details 3.1 The system saves the updated profile details and displays a success message

	1. Admin selects "Manage Organizations" from the admin dashboard. 2. Admin selects an organization account to view. 3. Admin clicks "Terminate Account" for a specific organization.	1.1 The system displays a list of all organization accounts. 2.1 The system displays the organization's details. 3.1 The system confirms the action and terminates the organization's account.
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.14 UCD014 Manage Organization Registration

Use Case ID:	UCD014		
Use Case Name:	Manage Organization Registration		
Created By:	Muhammad Haiqal	Last Updated By:	
Date Created:	11/8/2024	Date Last Updated:	

Actor:	Organization	
Description:	This use case allows the admin to approve or reject organizations that have registered but require approval.	
Preconditions:	The system displays list of organizations with status pending approval.	
Postconditions:	If approved they can access the system and if rejected,they cannot access the system.	
Priority:	High	
Frequency of Use:	Medium	
	Actor	System

Normal Course of Events:	1. Admin clicks "Manage Organization Registration" from the sidebar. 2. Admin reviews the list of pending organizations. 3. Admin clicks "Accept" to approve a registration. 4. Admin clicks "Reject" to reject a registration.	1.1 The system displays a list of organizations pending approval. 2.1 The system displays organization details. 3.1 The organization status changes to approved. 4.1 The organization status changes to rejected.
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A	
Notes and Issues:	N/A	

3.2.15 UCD015 Add Admin

Use Case ID:	UCD015		
Use Case Name:	Add Admin		
Created By:	Muhammad Haiqal	Last Updated By:	
Date Created:	11/8/2024	Date Last Updated:	

Actor:	Admin	
Description:	This use case allows an admin to add another admin	
Preconditions:	The new admin's email must be unique in the system.	
Postconditions:	A new admin account is created and stored in the system.	
Priority:	High	
Frequency of Use:	Low	
	Actor	System

Normal Course of Events:	1. Admin clicks "Add Admin" from the sidebar. 2. Admin fills out the form with the new admin's credentials 3. Admin clicks the "Add" button to create the new admin account	1.1 The system displays a form for adding a new admin. 2.1 The system validates the entered data. 3.1 The system creates the new admin account in the database.
Alternative Courses:	N/A	
Exceptions:	N/A	
Includes:	N/A	
Notes and Issues:	N/A	

4 Other Non-functional Requirements

4.0 Performance Requirements

VolComm should provide quick response times and handle a growing number of users and data efficiently. It must be designed to maintain high performance even as the number of simultaneous users increases, ensuring that all operations are executed smoothly and without significant delays.

4.1 Safety and Security Requirements

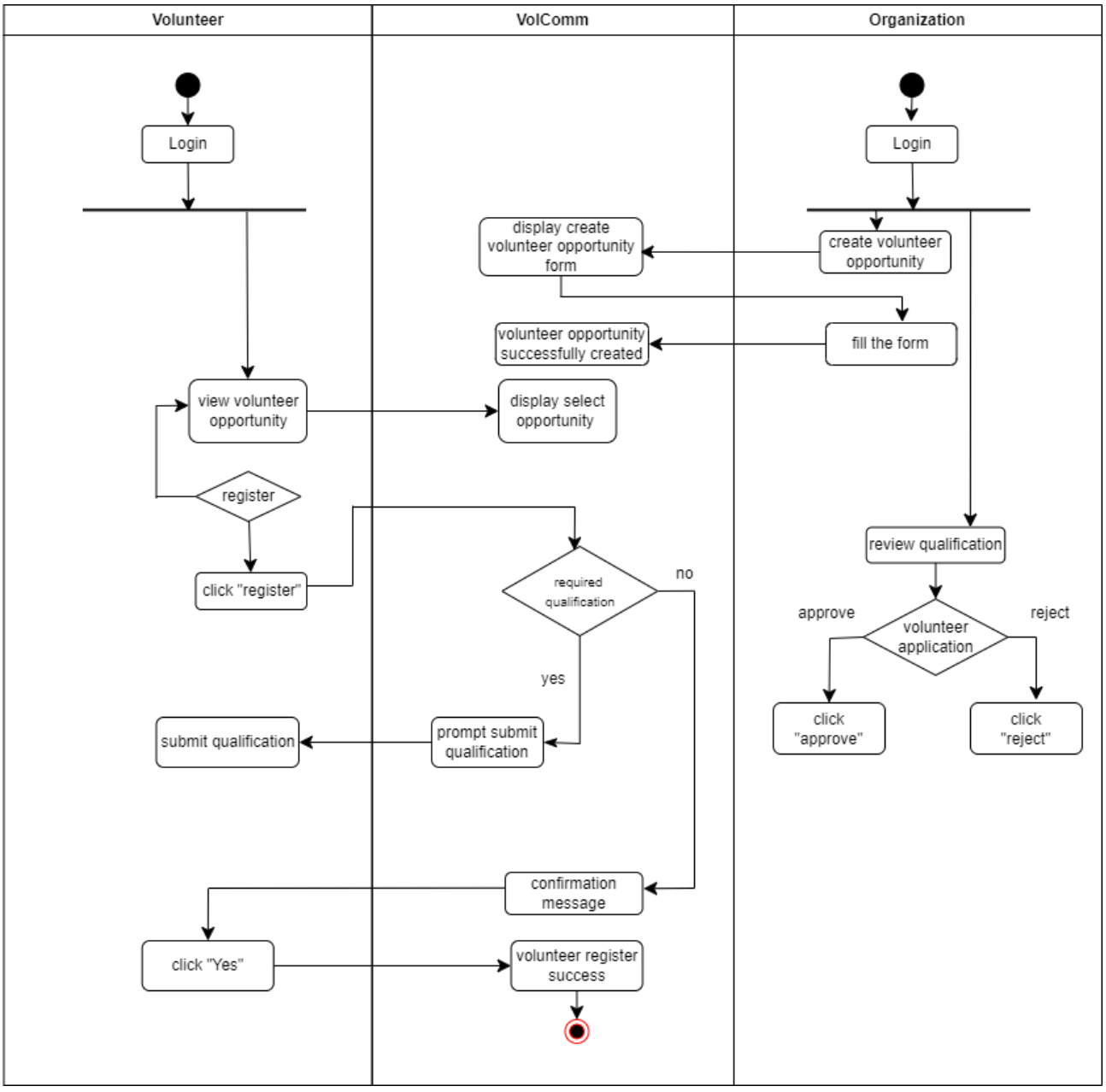
VolComm must protect sensitive data through encryption, both during transmission and while stored. Strong user authentication must be enforced, with options for multi-factor authentication to enhance security. Access to different parts of the system should be controlled based on user roles, and regular backups should be securely stored to prevent data loss. Additionally, the system should keep detailed logs of user actions and changes for security audits.

4.2 Software Quality Attributes

The system should be user-friendly with an intuitive interface, making it easy for users to navigate and perform tasks. It must be reliable, functioning consistently under expected conditions with minimal downtime. Maintainability is crucial, with well-documented code and a modular design to facilitate updates and fixes. The system should be scalable to handle increased load, interoperable with other software used by the organization, portable across different platforms and devices, and efficient in its use of resources to ensure smooth operation.

Appendix A – REQUIREMENT MODELS

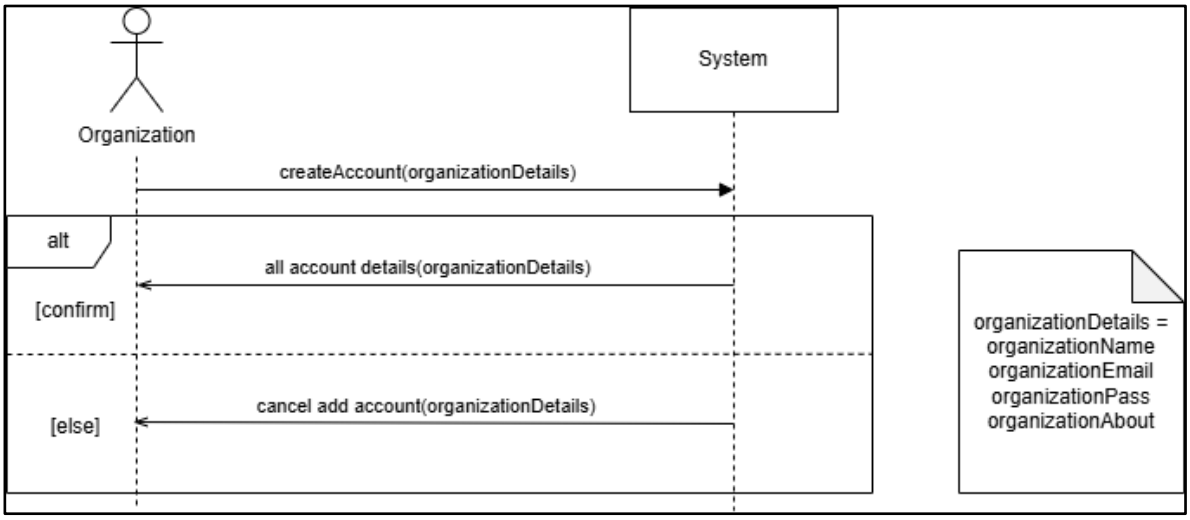
1. Business Activity Diagram



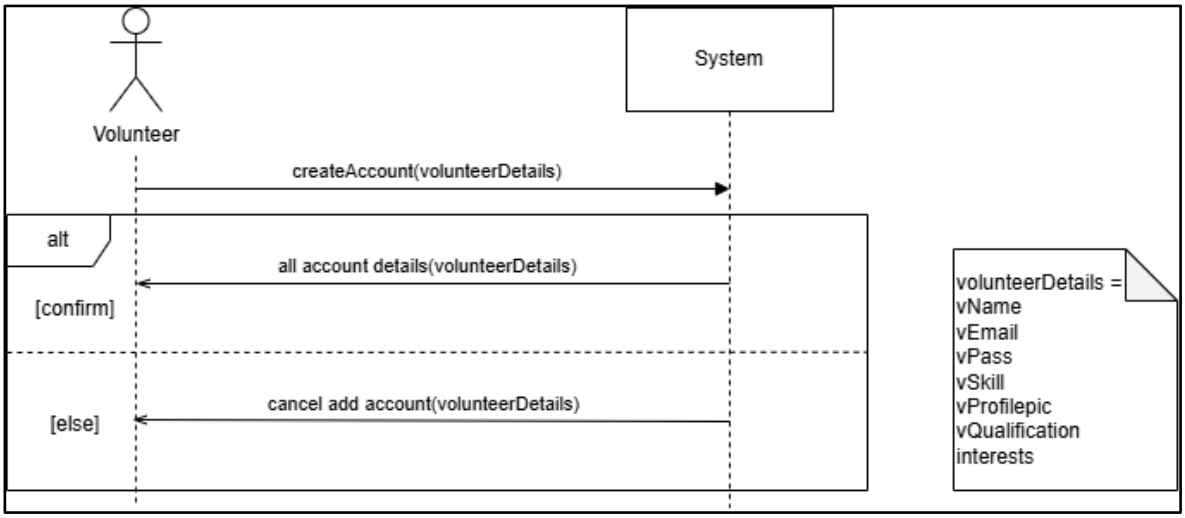
2. Class Diagram

3. System Sequence Diagram

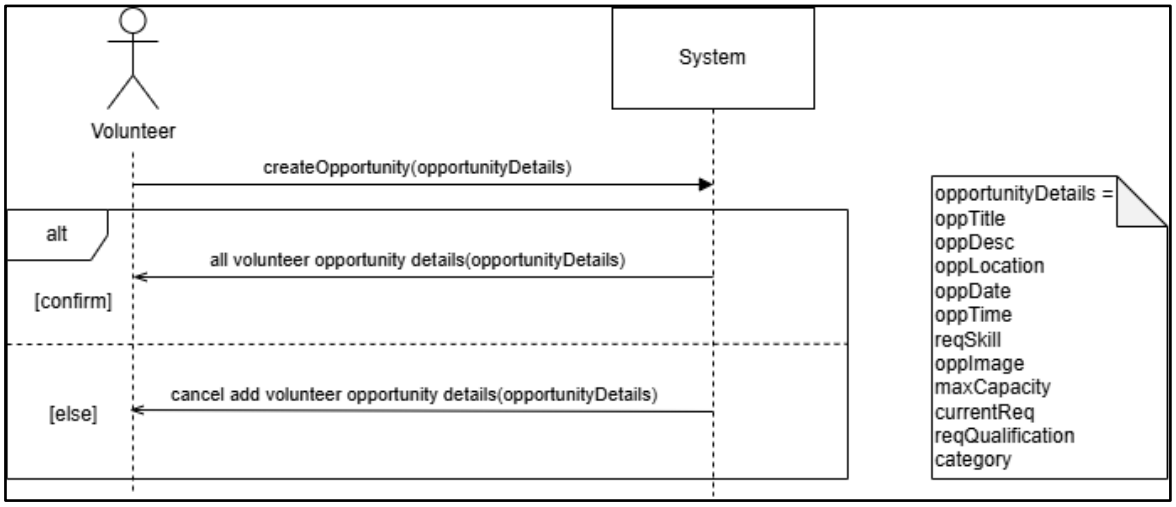
Create Organization Account



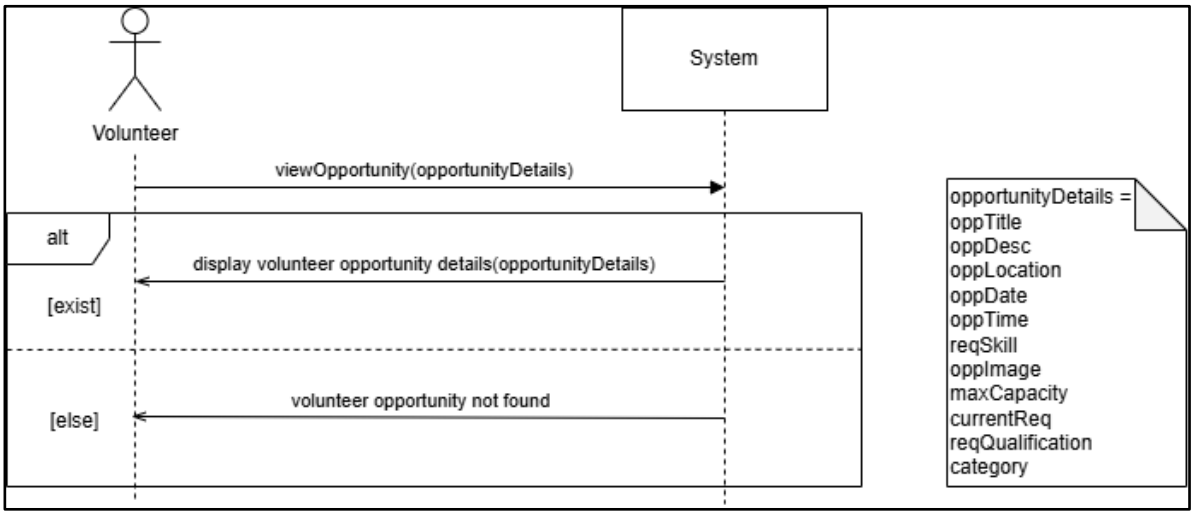
Create Volunteer Account



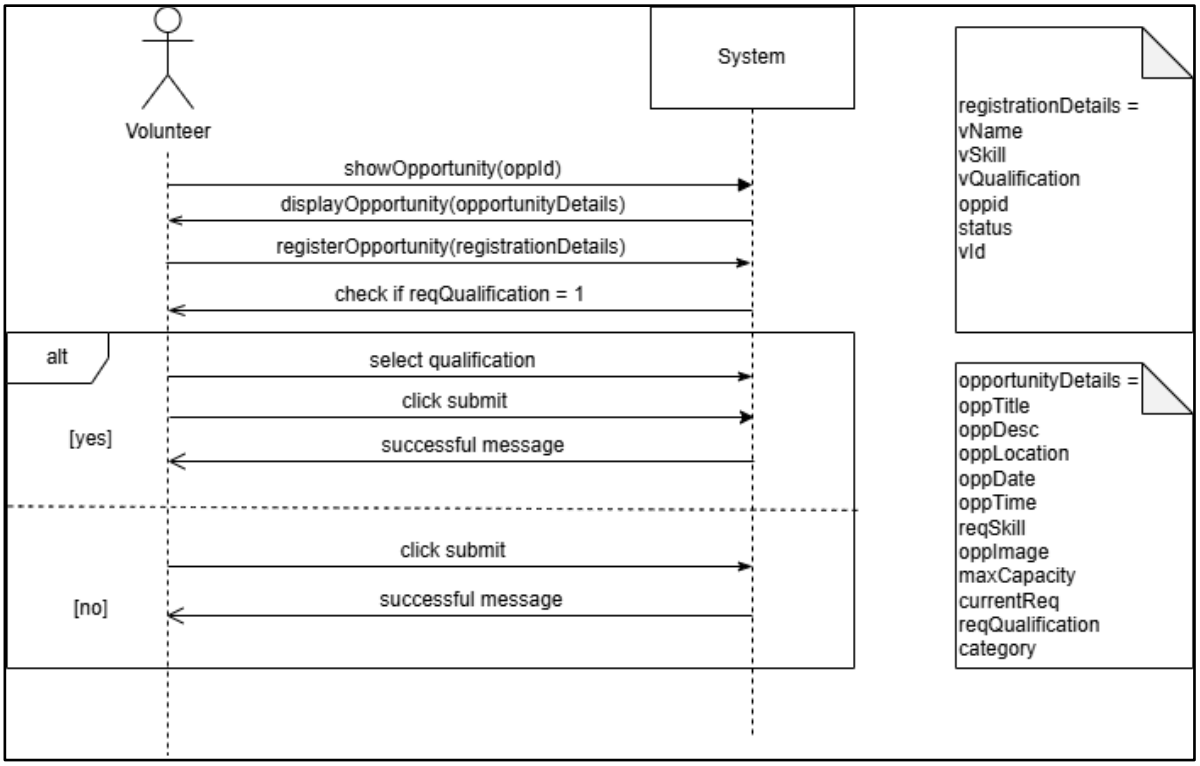
Create Volunteer opportunity



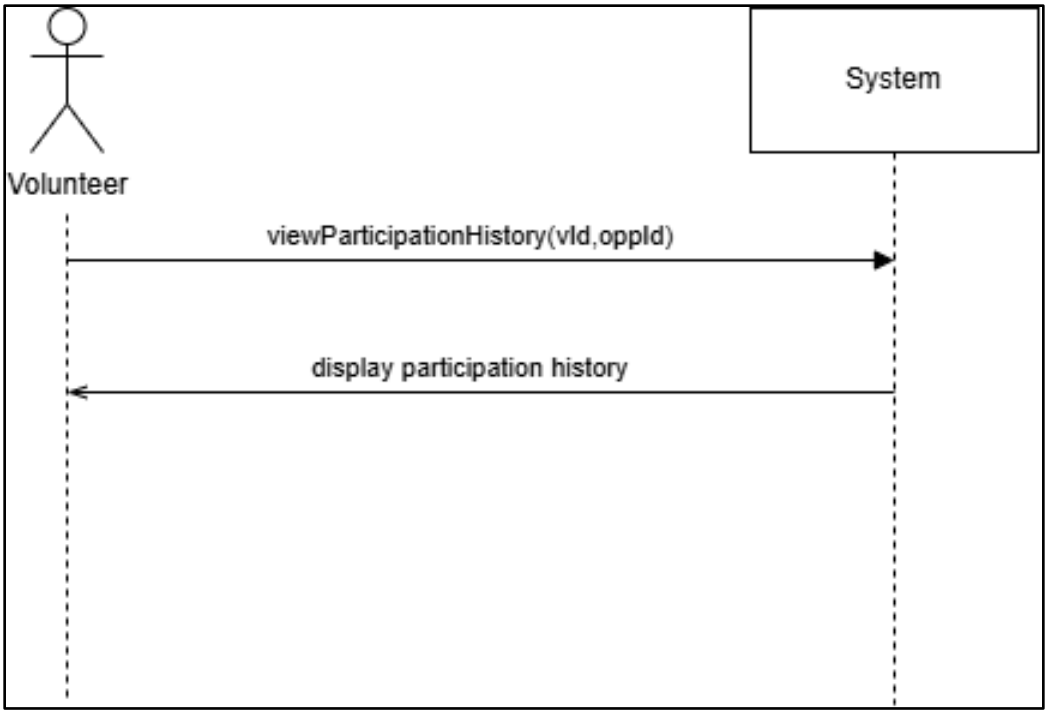
View volunteer opportunity



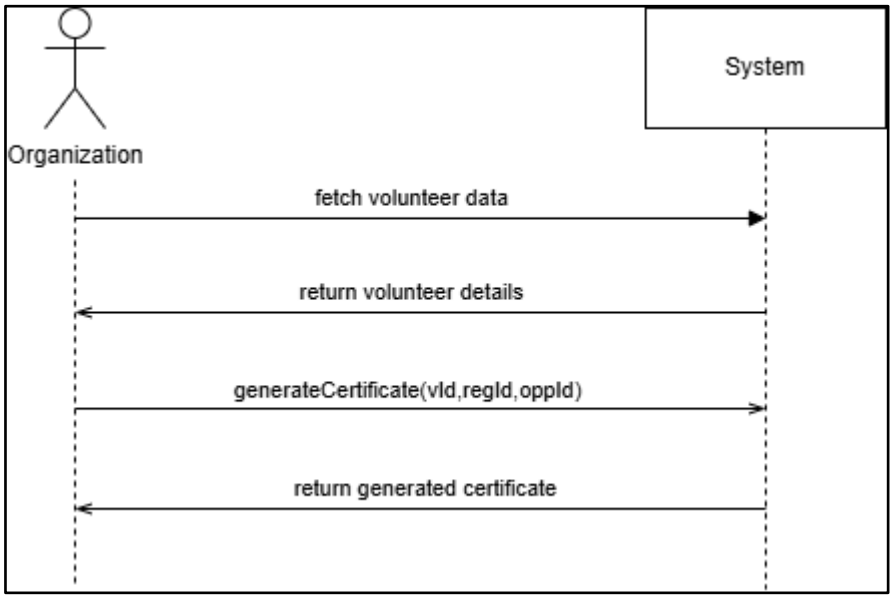
Apply volunteer application



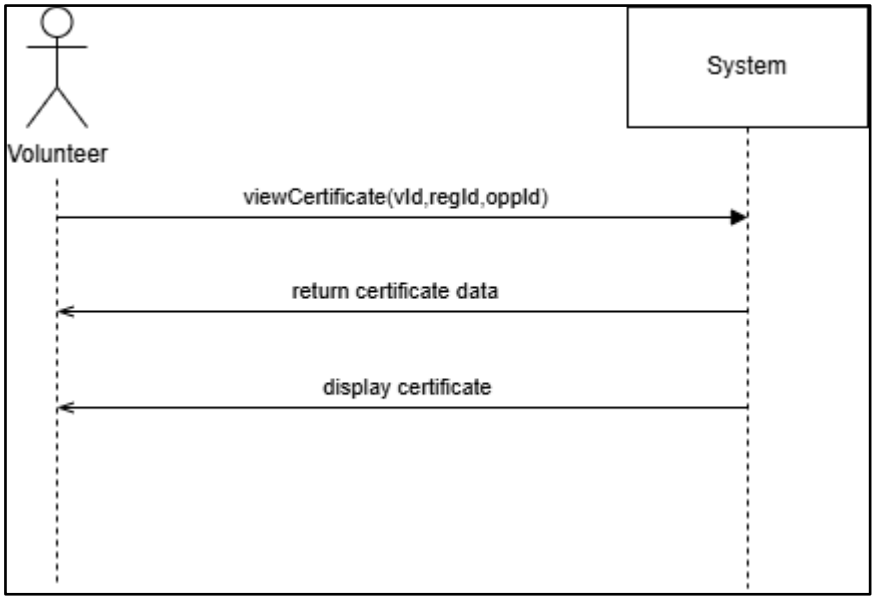
View participation history



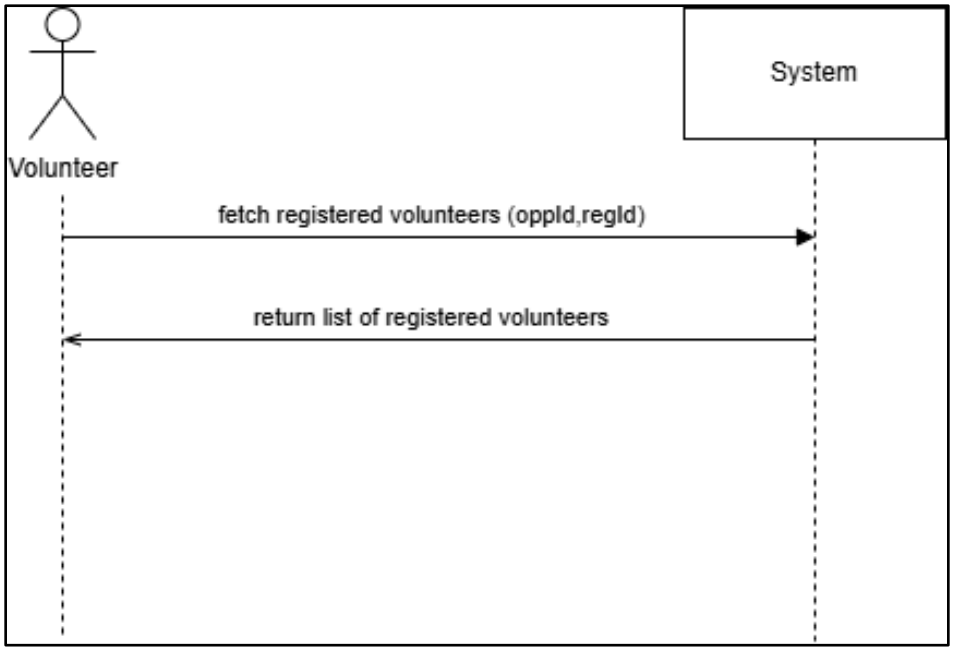
Generate certificate



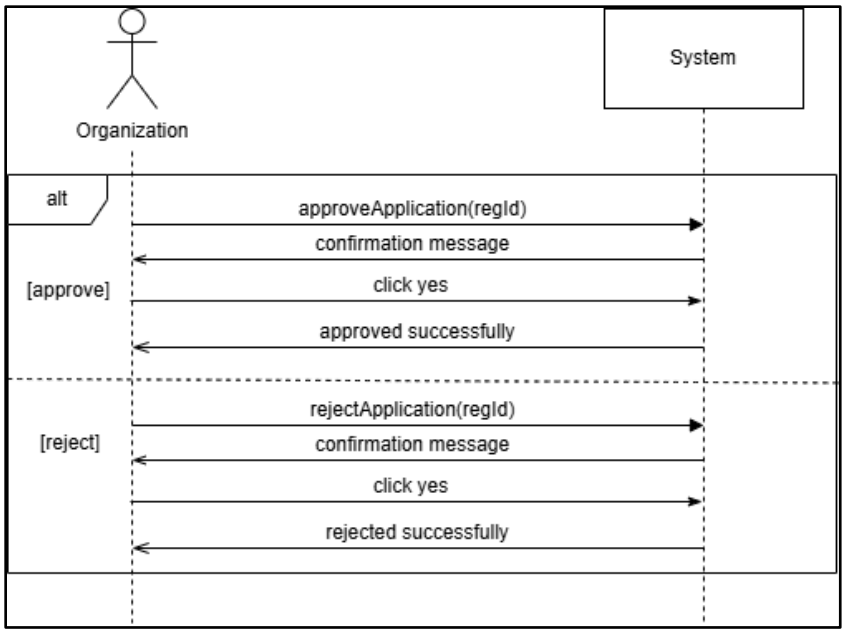
View certificate



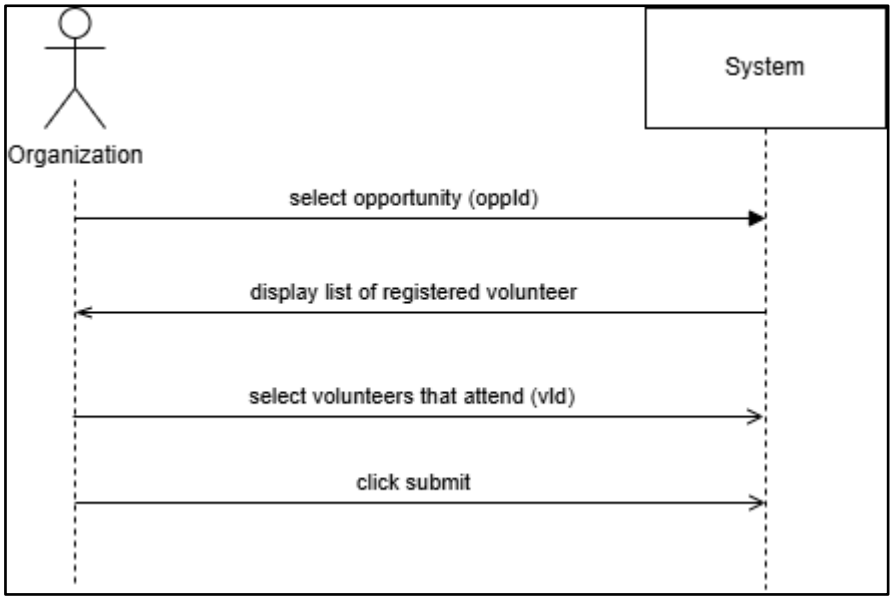
View registered volunteers



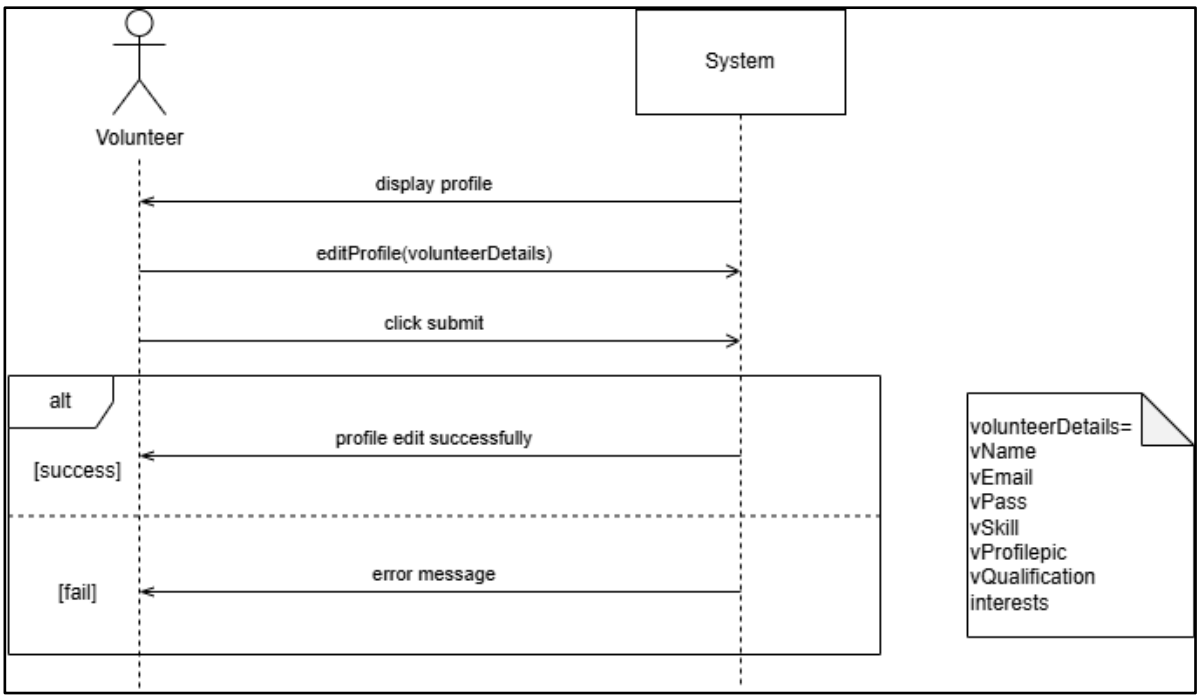
Review volunteer application



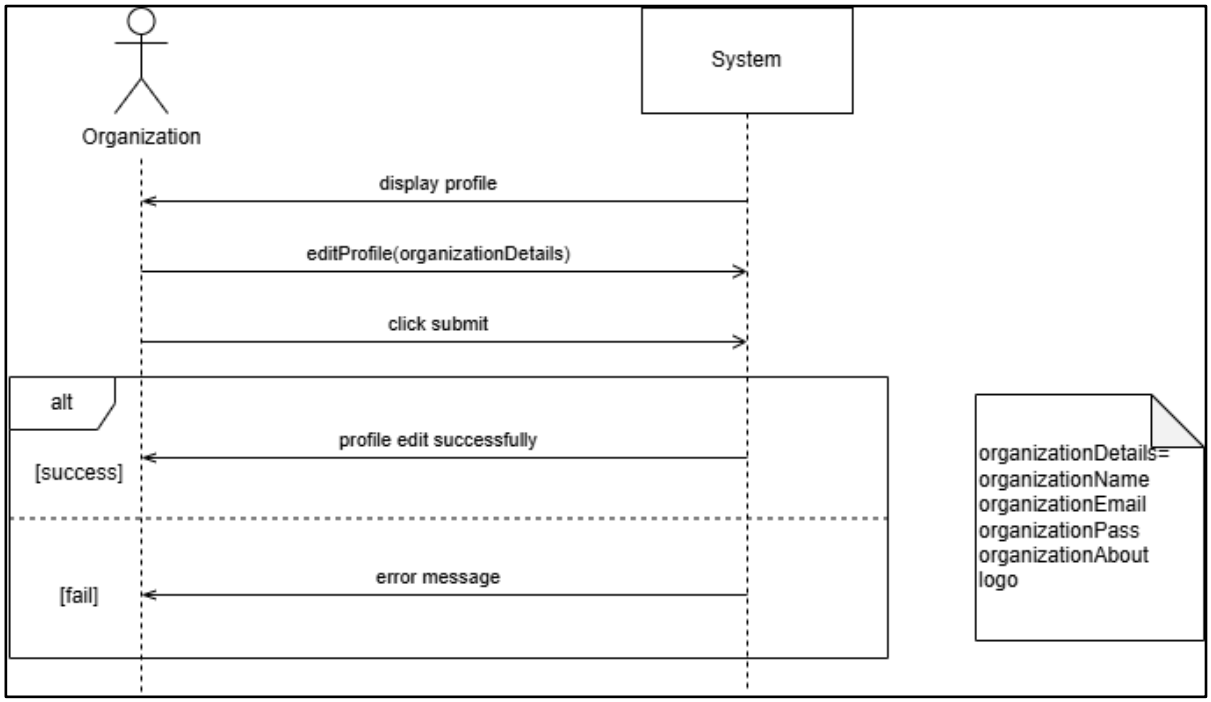
Confirm volunteer attendances



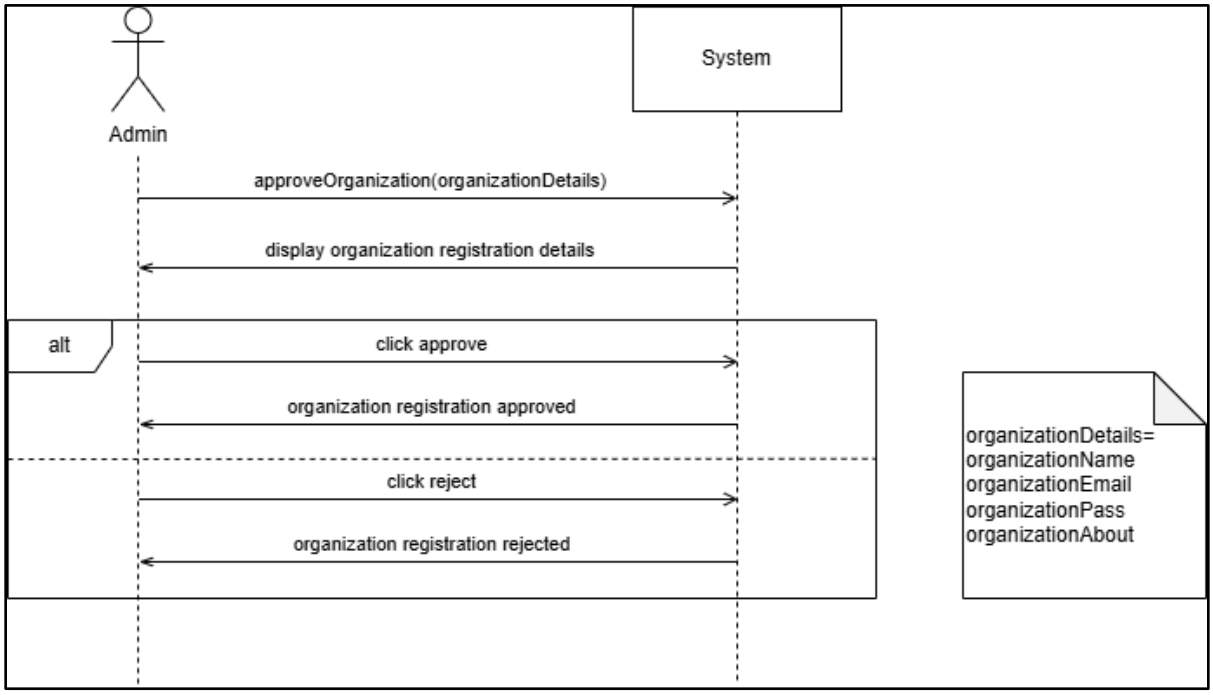
Manage volunteer account



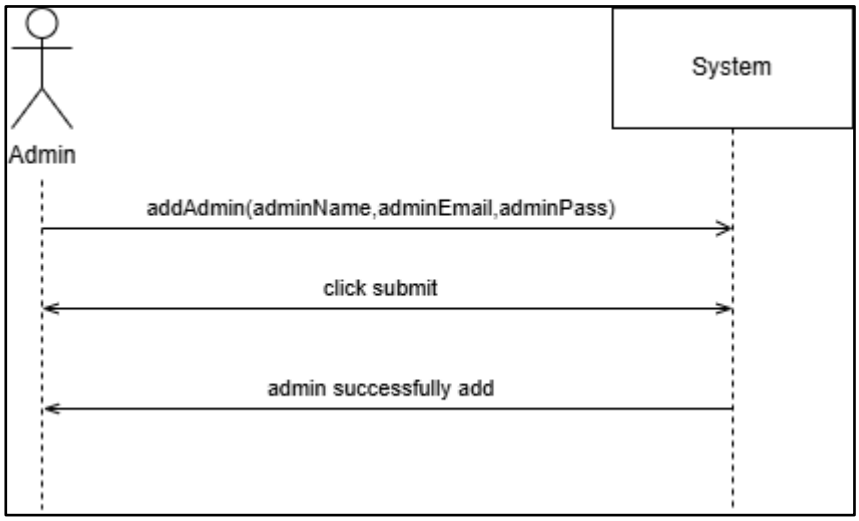
Manage organization account



Approve organization



Add admin


System
Admin
addAdmin(adminName,adminEmail,adminPass)
click submit
admin successfully add

APPENDIX C



SOFTWARE DESIGN DOCUMENT

Supervisor:
DR.NUR SUHAILAYANI SUHAIMI

Project Title:
VOLCOMM: VOLUNTEER AND COMMUNITY SERVICE HUB FOR NON-GOVERNMENTAL ORGANIZATIONS (NGOS)

Name:
MUHAMMAD HAIQAL BIN MOHD IZZEE (2023115539)

Submit Date:
23 FEBRUARY 2025

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1. Introduction

1.1 Purpose

The purpose of this Software Design Document (SDD) is to outline the detailed architecture and design for the VOLCOMM system, a volunteer management platform. This document describes how different components of the system, such as volunteers, organizations, opportunities, and admin roles, interact with one another. It also includes use cases, class diagrams, and the logical flow of various processes like volunteer registration, opportunity management, and certificate generation. By defining these structures, the SDD ensures that the Volcomm system is developed efficiently and meets the functional requirements for managing volunteers, creating opportunities, and handling registrations. This document provides developers with a clear roadmap to implement and maintain the system effectively.

1.2 Scope

The scope of this project includes several key functionalities. First, the system enables volunteers to register, log in, update their profiles, view available opportunities, and track their volunteer certificates. Second, organizations can create, update, view, and delete volunteering opportunities, as well as manage volunteer registrations for each opportunity. Additionally, organizations can register and manage their profiles, ensuring smooth interactions with volunteers. A key feature of the system is its automated email notifications, which inform volunteers and organizations of important updates such as successful registrations. Lastly, after completing an opportunity, the system automatically generates certificates for volunteers, which they can view and download.

1.3 References.

- Arsan, T., Başkan, E. B., Ar, E., & Bozkuş, Z. (2012). A Software Architecture for Inventory Management System. In Springer eBooks (pp. 15–27), Reasearchgate.net. Retrieved May 22, 2023, from https://www.researchgate.net/publication/291696392_A_Software_Architecture_for_Inventory_Management_System

- Fonyuy Taryuni Isidore. "Design of Inventory Management System." *IDEmpiere*, 20 Aug. 2014.Retrieved May 22, 2023, from https://wiki.idempiere.org/en/Design_of_Inventory_Management_System

- Ramadeen,Deivendran,Yousuf & Sridhar. (2001) "Proposed Design of an Inventory Database System at Process Research ORTECH." toronto.edu. Retrieved May 22, 2023, from https://www.cs.toronto.edu/~sme/CSC340F/2002/samples/Asst3-5.pdf

- "Inventory Management Software - a Complete Guide!!" *Inventory Management Software - a Complete Guide!!*, Retrieved May 22, 2023, from https://www.binaryfolks.com/blog/inventory-management-software-a-complete-guide

- Wisdom, Afrika. "Design and Implementation of Inventory Management System." (DOC) Design and Implementation of Inventory Management System | Afrika Wisdom - Academia.edu. Retrieved May 22, 2023, from www.academia.edu/12898167/Design_and_implementation_of_inventory_management_system.

1.4 Document Structure

The Software Design Document (SDD) for the Volcomm system is organized into sections that explain the overall architecture and design of the system. It starts with an introduction that outlines the purpose and scope of the project, providing a clear understanding of what the system will achieve. Following this, the document includes use cases, which describe the different user interactions. It is also shows class diagrams that show the relationships between different components like volunteers, organizations, and opportunities. The document also includes sequence diagrams to explain the flow of processes and how different parts of the system communicate with each other. Finally, the SDD details the technical specifications and system architecture, guiding developers on how to implement and maintain the system.

1.5 Definitions, Acronyms and Abbreviations

- VOLCOMM : VOLUNTEER AND COMMUNITY SERVICE HUB FOR NON-GOVERNMENTAL ORGANIZATIONS (NGOS)
- UI : User Interface
- DB : Database
- CRUD : Create, Read, Update, Delete
- SDD : Software Design Description

1.6 System Overview

VOLCOMM is a system or hub that brings together volunteers and non-governmental organizations (NGOs) to support community service initiatives effectively and efficiently. It offers volunteers the chance to sign up easily and explore volunteer or community service opportunities to get involved in meaningful projects while also providing certificates upon completion of tasks. For organizations looking for help or support from volunteers for their causes or projects can register on VOLCOMM platform to create and manage their volunteer opportunities seamlessly along with tracking applications from potential volunteers. With its user interface and streamlined processes in place to connect motivated individuals with impactful causes supported by organizations in need of assistance. VOLCOMM also aims to make a positive difference, in society through community service collaborations. The platform includes dashboards, for volunteers and organizations to facilitate communication and coordination of community service initiatives.

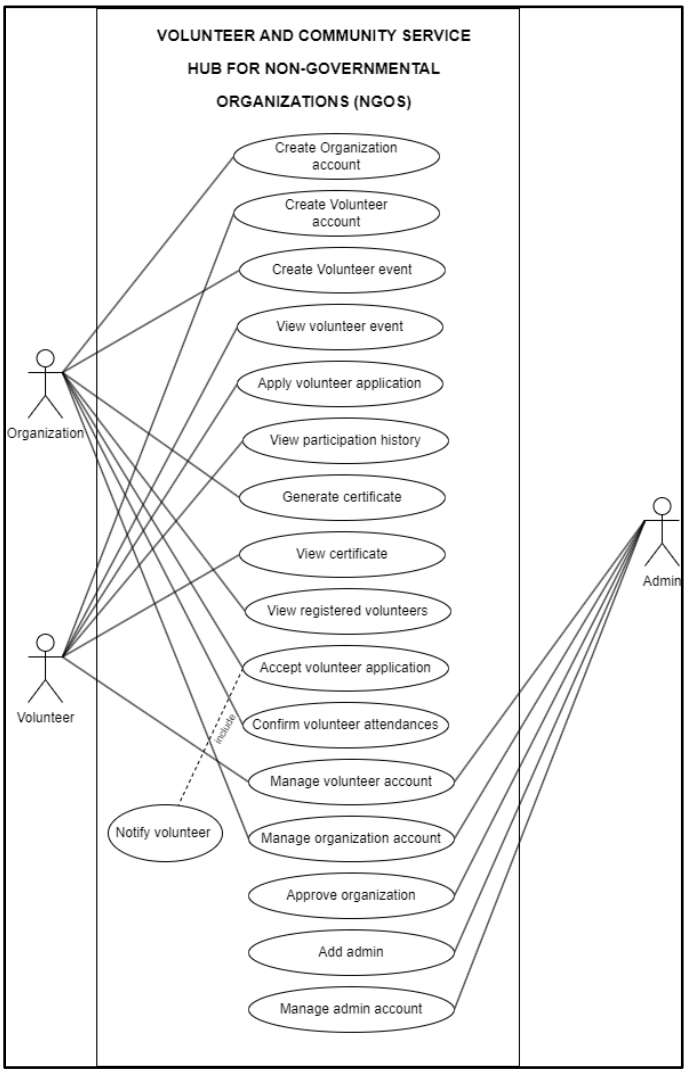
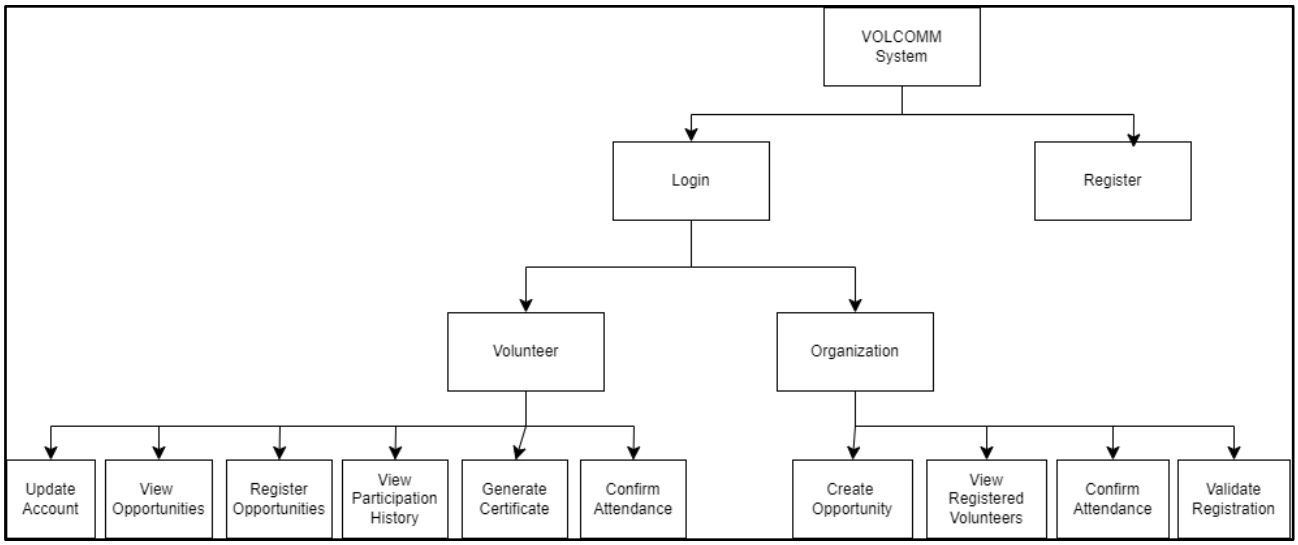


Figure 1.6.1 shows Use Case Diagram

2. System Architecture

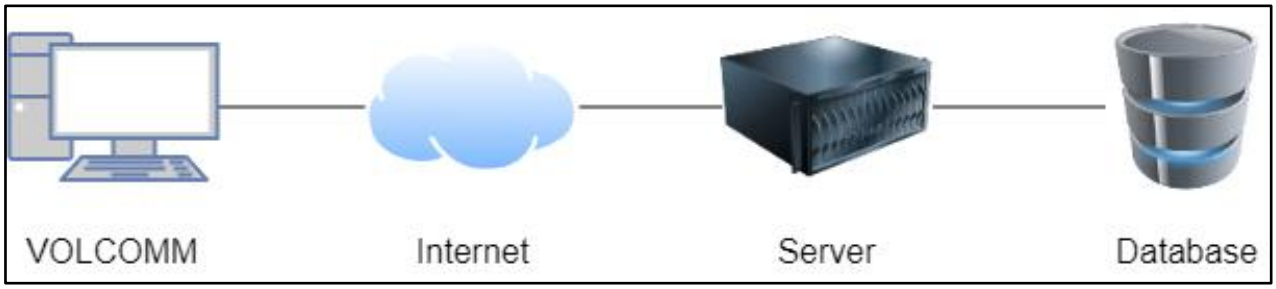
2.1 Architectural Description



2.1.1 shows Block Diagram of the major subsystem

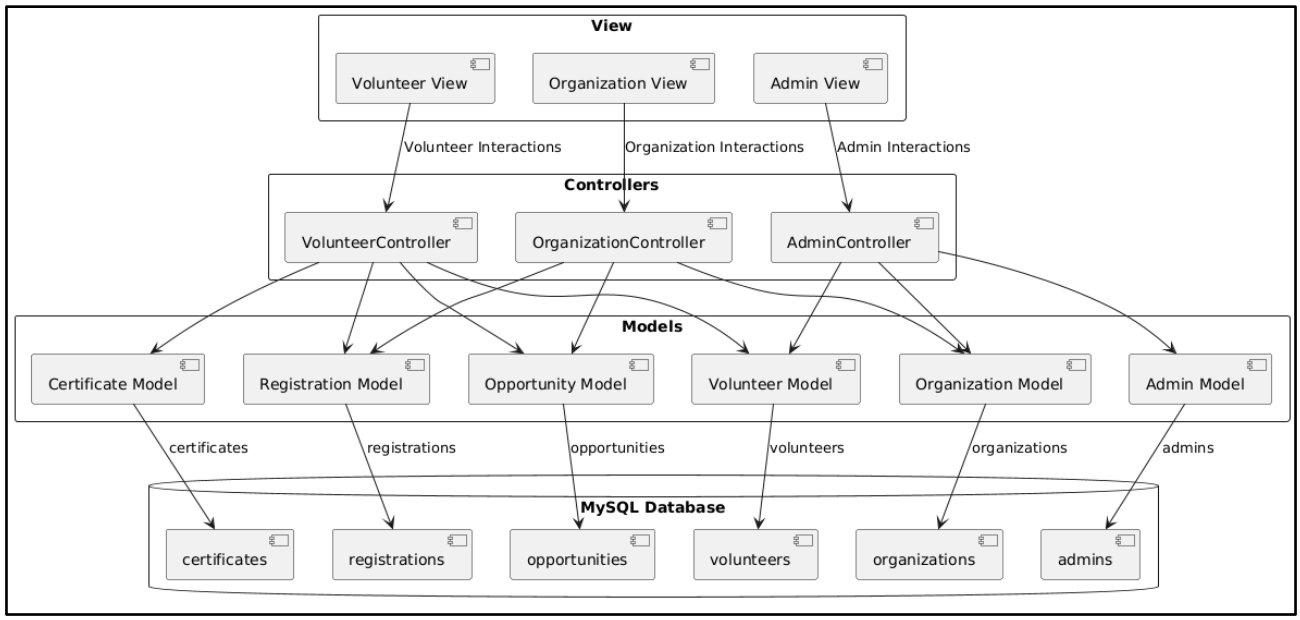
This is a high-level overview of how responsibilities of the systems were partitioned and then assigned to subsystems.

The system uses MySQL as its database to store and manage data, such as volunteer profiles, opportunities, and registrations. It runs on a server, which handles requests from users, processes data, and delivers results. This setup ensures smooth data management and efficient access for volunteers and organizations.



2.1.2 shows Network Diagram

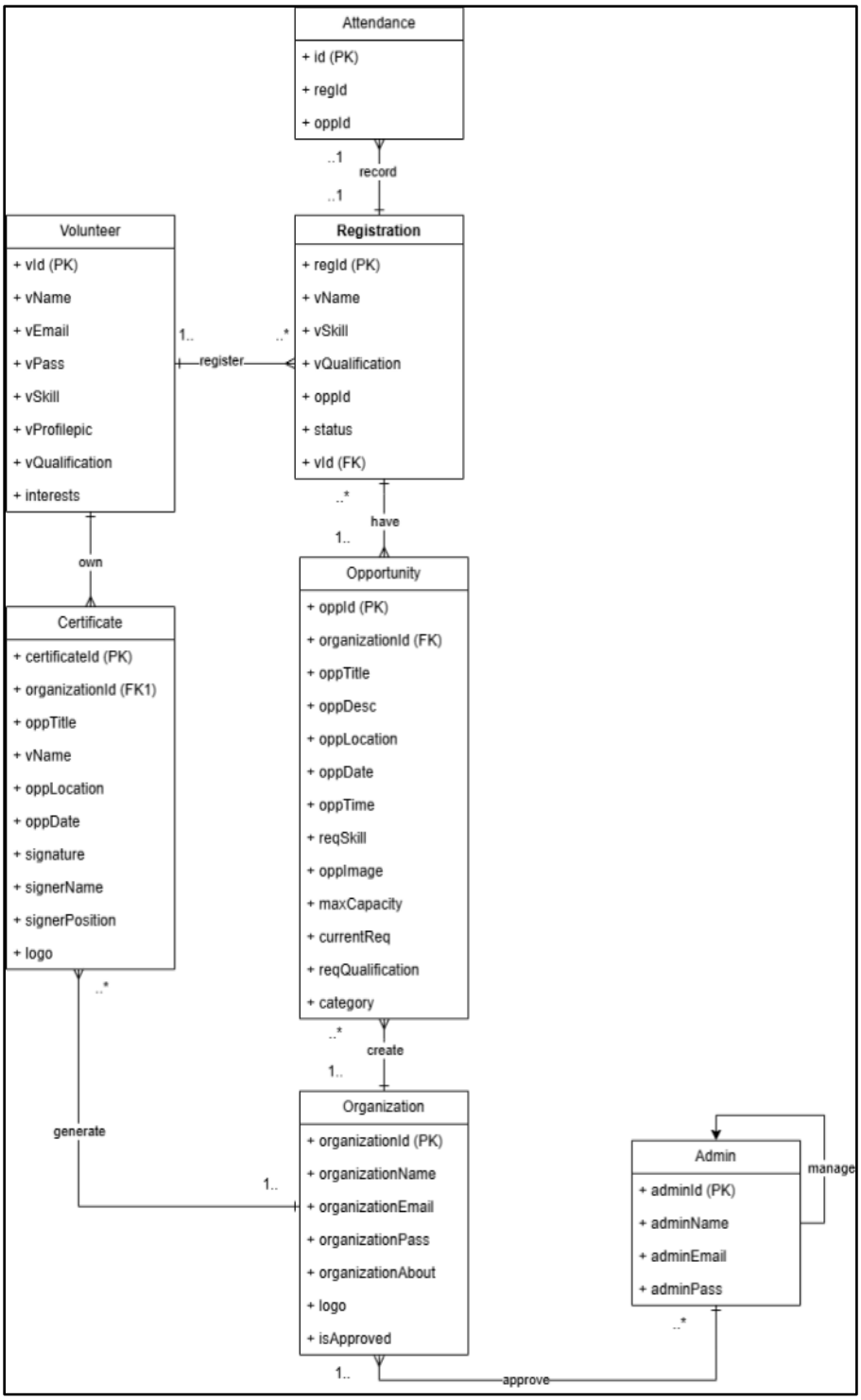
2.2 Design Rationale



2.2.1 shows Design Rationale Diagram.

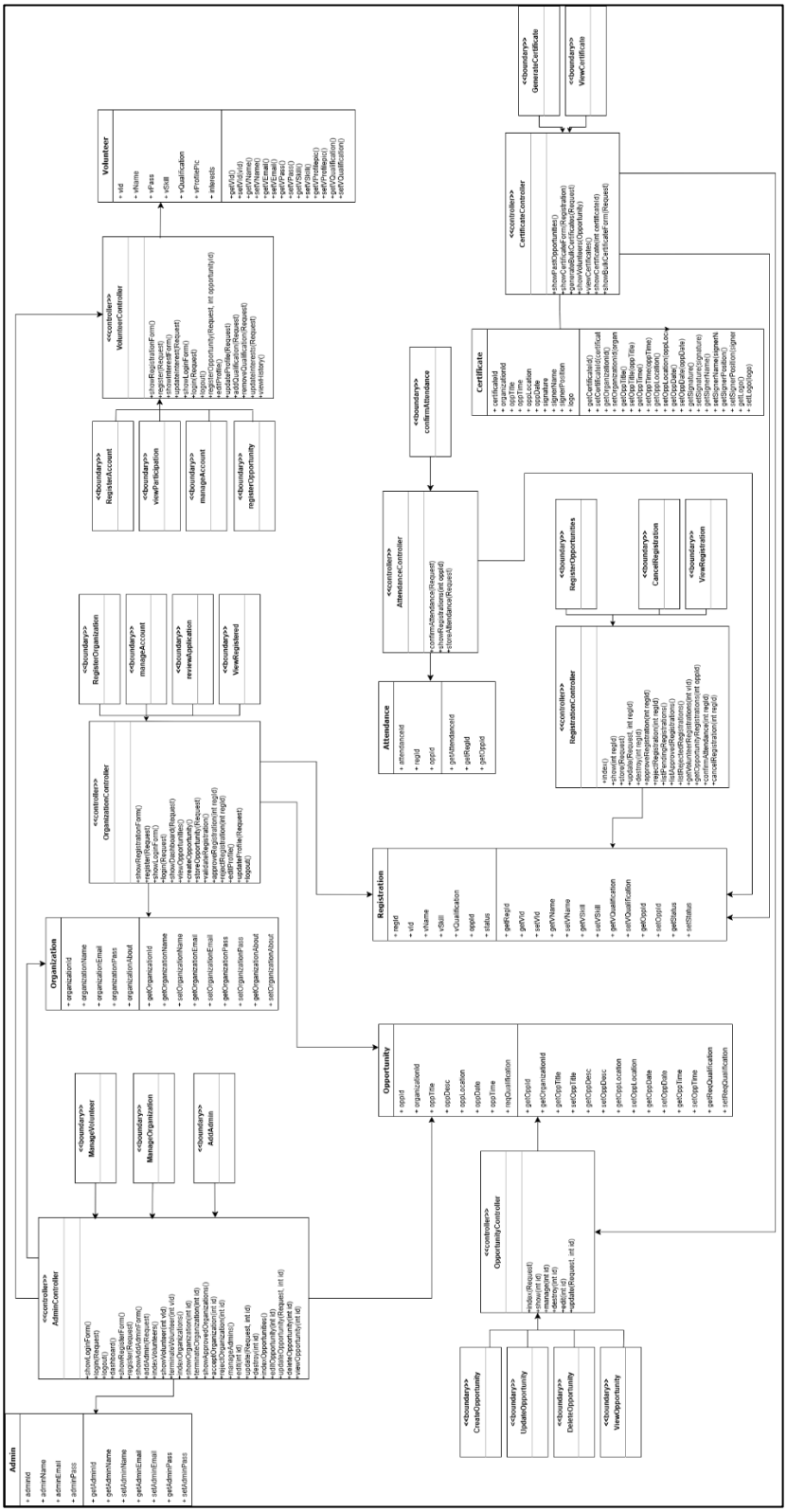
The Design Rationale Diagram illustrates the reasons behind the choices made in a system's design. It explains why certain decisions were made about how data is organized and managed. This diagram helps developers understand the logic and rationale behind the system's structure, making it easier to see how different components fit together and function effectively.

2.3 Domain Class Diagram



2.3.1 shows Domain Class Diagram.

2.4 Design Class Diagram



2.4.1 shows a Design class diagram.

3. Data Design

3.1 Database Description

The database that this system used to store the data is relational database. The database is located on cloud and is secured using a combination of physical and logical security measures. Moreover, the size of the database is a medium-sized database with 10,000 rows of data. Hence, the database is able to handle the expected load and is affordable. (M=Mandatory, U=Unique)

3.2 Data Dictionary

Volunteer						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
vId	Volunteer ID	Integer	Primary Key (PK)		Yes	Yes
vName	Volunteer Name	String			Yes	No
vEmail	Volunteer Email	String			Yes	Yes
vPass	Volunteer Password	String			Yes	No
vSkill	Volunteer Skills	String			No	No
vProfilepic	Volunteer Profile Picture	String	URL or path		No	No
vQualification	Volunteer Qualification	String			No	No
interests	Volunteer Interests	String			No	No

Registration						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
regId	Registration ID	Integer	Primary Key (PK)		Yes	Yes
vName	Volunteer Name	String			Yes	No
vSkill	Volunteer Skill	String			No	No
vQualification	Volunteer Qualification	String			No	No
oppId	Opportunity ID	Integer	Foreign Key (FK)		Yes	No
status	Registration Status	String			No	No

vId	Volunteer ID	Integer	Foreign Key (FK)		Yes	No

Opportunity						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
oppId	Opportunity ID	Integer	Primary Key (PK)		Yes	Yes
organizationId	Organization ID	Integer	Foreign Key (FK)		Yes	No
oppTitle	Opportunity Title	String			Yes	No
oppDesc	Opportunity Description	String			No	No
oppLocation	Opportunity Location	String			No	No
oppDate	Opportunity Date	Date			No	No
oppTime	Opportunity Time	Time			No	No
reqSkill	Required Skill	String			No	No
oppImage	Opportunity Image	String	URL or path		No	No
maxCapacity	Maximum Capacity	Integer			No	No
currentReq	Current Requirement	Integer			No	No
reqQualification	Required Qualification	String			No	No
category	Opportunity Category	String			No	No

Certificate						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
certificateId	Certificate ID	Integer	Primary Key (PK)		Yes	Yes
organizationId	Organization ID	Integer	Foreign Key (FK)		Yes	No
oppTitle	Opportunity Title	String			Yes	No
vName	Volunteer Name	String			Yes	No
oppLocation	Opportunity Location	String			Yes	No

oppDate	Opportunity Date	Date			Yes	No
signature	Signature	String			Yes	No
signerName	Signer's Name	String			Yes	No
signerPosition	Signer's Position	String			Yes	No

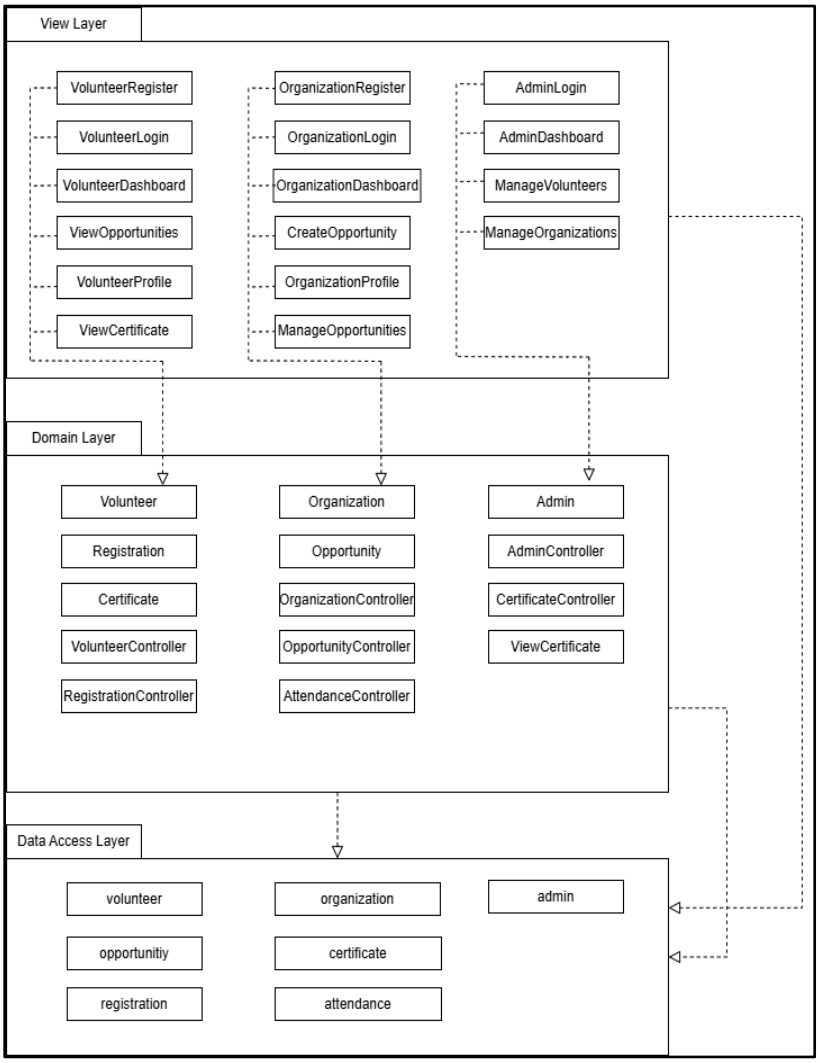
Organization						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
organizationId	Organization ID	Integer	Primary Key (PK)		Yes	Yes
organizationName	Organization Name	String			Yes	No
organizationEmail	Organization Email	String			Yes	Yes
organizationPass	Organization Password	String			Yes	No
organizationAbout	About Organization	String			No	No
logo	Logo	String	URL or path		No	No
isApproved	Is Approved	Boolean			Yes	No
organizationId	Organization ID	Integer	Primary Key (PK)		Yes	Yes

Admin						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
adminId	Admin ID	Integer	Primary Key (PK)		Yes	Yes
adminName	Admin Name	String			Yes	No
adminEmail	Admin Email	String			Yes	Yes
adminPass	Admin Password	String			Yes	No

Attendance						
Attribute Name	**Description**	**Type**	**Additional Type Information**	**Default Value**	**M**	**U**
id	Attendance ID	Integer	Primary Key (PK)		Yes	Yes
regId	Registration ID	Integer	Foreign Key (FK)		Yes	No
oppId	Opportunity ID	Integer	Foreign Key (FK)		Yes	No

4. Package Diagram

4.1 Package Diagram



4.1.1 shows Package diagram.

4.2 Package Functions

Table 4.2.1 Package functions

Package	Package Functions
View Layer (SDD_PKG_100)	The view layer is where users engage with the interface by seeing data and interacting through actions, like filling out forms or clicking buttons.
Domain Layer (SDD_PKG_200)	The domain layer houses the business logic of the application handling data and executing the rules and functions that are central, to the softwares main objectives.
Data Access Layer (SDD_PKG_300)	The data access layer is responsible for handling data storage and retrieval of records from the database. It retrieves, saves, and updates data so that the application can easily read and write information.

4.3 Package Content

Table 4.3.1 Package Content

Package	Class ID	Class Name
SDD_PKG_100	SDD_PKG_101	VolunteerRegister
	SDD_PKG_102	VolunteerLogin
	SDD_PKG_103	VolunteerDashboard
	SDD_PKG_104	ViewOpportunities
	SDD_PKG_105	VolunteerProfile
	SDD_PKG_106	ViewCertificate
	SDD_PKG_107	OrganizationRegister
	SDD_PKG_108	OrganizationLogin
	SDD_PKG_109	OrganizationDashboard
	SDD_PKG_110	CreateOpportunity
	SDD_PKG_111	OrganizationProfile
	SDD_PKG_112	ManageOpportunities
	SDD_PKG_113	AdminLogin
	SDD_PKG_114	AdminDashboard
	SDD_PKG_115	ManageVolunteers
	SDD_PKG_116	ManageOrganizations
SDD_PKG_200	SDD_PKG_201	VolunteerController
	SDD_PKG_202	OrganizationController
	SDD_PKG_203	OpportunityController
	SDD_PKG_204	RegistrationController
	SDD_PKG_205	CertificateController
	SDD_PKG_206	AdminController

	SDD_PKG_207	AttendanceController
SDD_PKG_300	SDD_PKG_301	volunteers
	SDD_PKG_302	organizations
	SDD_PKG_303	opportunities
	SDD_PKG_304	registrations
	SDD_PKG_305	certificates
	SDD_PKG_306	admins
	SDD_PKG_307	attendance

5. Class Description

5.1 Class ID & NAME

(SDD_PKG_201 - VolunteerController)

The purpose of the VolunteerController is to handle volunteer registration, login, and profile management, ensuring users can create accounts, access their profiles, and engage with the platform effectively.

(SDD_PKG_202 - OrganizationController)

The purpose of the OrganizationController is to manage NGO accounts by overseeing registration, profile updates, and the management of opportunities, facilitating smooth operations and visibility for organizations.

(SDD_PKG_203 - OpportunityController)

The purpose of the OpportunityController is to handle the creation, updating, deletion, and viewing of volunteer opportunities, ensuring that current and relevant opportunities are available to volunteers.

(SDD_PKG_204 - RegistrationController)

The purpose of the RegistrationController is to manage the process of volunteers signing up for specific opportunities, connecting them with projects and ensuring their participation is recorded.

(SDD_PKG_205 - CertificateController)

The purpose of the CertificateController is to generate and manage certificates for completed volunteer work, providing recognition and documentation of volunteers' contributions.

(SDD_PKG_206 - AdminController)

The purpose of the AdminController is to oversee administrative tasks, such as approving opportunities and managing user accounts, ensuring the platform runs smoothly and efficiently.

(SDD_PKG_207 - AttendanceController)

The purpose of the AttendanceController is to generate and manage certificates for volunteers who have successfully attended and completed volunteer opportunities.

5.1.2 Class Attributes & Methods

(SDD_PKG_207 - VolunteerController)

- register() – Allows a new volunteer to create an account.
- editProfile() – Enables a volunteer to update their profile details (name, skills, qualifications, and interests).
- viewParticipationHistory() – Retrieves and displays the volunteer's past activities.
- applyForOpportunity($opportunityId) – Handles volunteer applications for a specific opportunity.
- viewCertificates() – Lists the certificates earned by the volunteer.

(SDD_PKG_208 - OrganizationController)

- editProfile() – Allows an organization to update its details, including name, logo, email, and password.
- confirmAttendance($opportunityId, $volunteerIds) – Confirms the attendance of volunteers who participated in an event.
- generateCertificate($volunteerIds, $opportunityId, $certificateDetails) – Creates certificates for volunteers who successfully attended an event.

(SDD_PKG_209 - OpportunityController)

- createOpportunity($request) – Enables organizations to post a new volunteer opportunity.
- updateOpportunity($opportunityId, $request) – Allows organizations to modify an existing opportunity.
- deleteOpportunity($opportunityId) – Removes an opportunity from the system.

(SDD_PKG_210 - RegistrationController)

- registerForOpportunity($volunteerId, $opportunityId) – Submits a volunteer’s application for an opportunity.
- approveApplication($applicationId) – Changes the status of a pending application to "approved."
- rejectApplication($applicationId) – Marks an application as "rejected" if the volunteer does not meet the qualifications.

- viewApplicationsForOpportunity($opportunityId) – Displays all volunteer applications for a specific opportunity.
- getAllOpportunities() – Retrieves all active volunteer opportunities for display.
- getOpportunityDetails($opportunityId) – Fetches and displays details of a specific opportunity.

(SDD_PKG_211 - CertificateController)

- generateCertificate($volunteerId, $opportunityId, $details) – Creates a certificate for a volunteer after successful event participation.
- viewCertificates($volunteerId) – Retrieves a list of certificates awarded to a volunteer.
- downloadCertificate($certificateId) – Allows volunteers to download their certificates as a PDF.

(SDD_PKG_212 - AdminController)

- approveOrganization($organizationId) – Changes the organization's status to "approved," granting them access to the system.
- rejectOrganization($organizationId) – Rejects an organization's registration request.
- terminateVolunteerAccount($volunteerId) – Deletes a volunteer's account permanently.
- terminateOrganizationAccount($organizationId) – Removes an organization from the system.
- addAdmin($request) – Allows an existing admin to create a new admin account.
- viewAllOrganizations() – Lists all registered organizations and their status.

6. Human Interface Design (Screens)

6.1 Overview of the User Interface

The system's screens are designed based on the VOLCOMM use case diagram, making it easy for users to interact with the system. The user interface acts like a language that helps users understand and use the system.

6.1.2 Screen Images

UCD001 - Create Organization account

Figure 6.1 shows organization register page where new organizations can create an account on the platform.

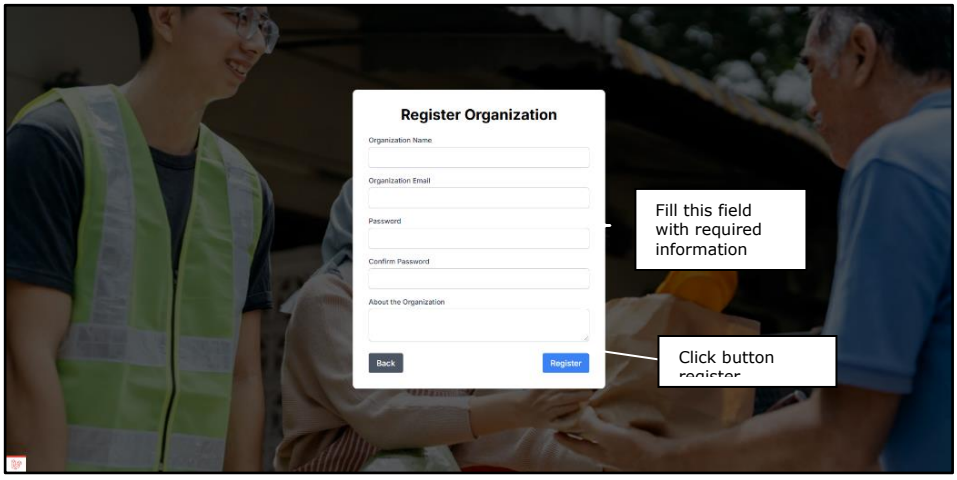


Figure 6.1 Organization register page

UCD002 - Create Volunteer account

Figure 6.2 illustrates a registration page designed for new volunteers to sign up on the platform. Users are required to enter their personal details, including their name, email, and password, to complete the registration process.

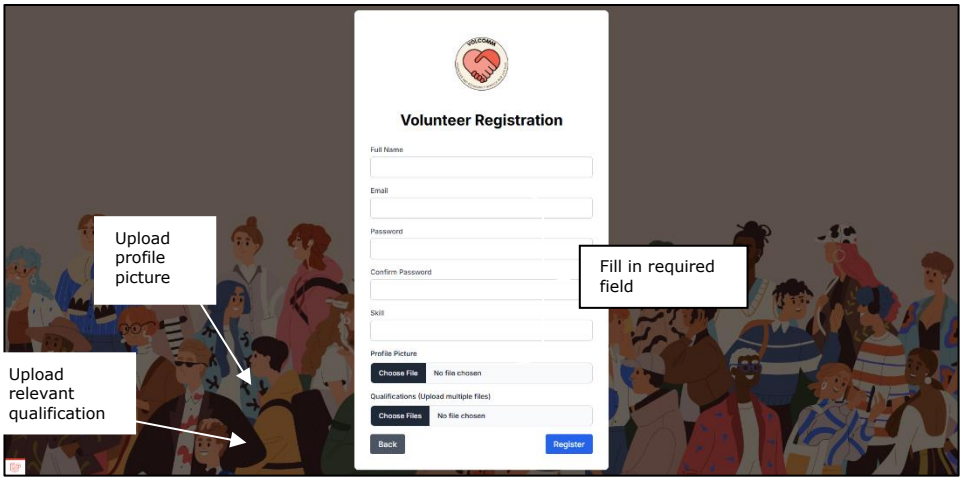


Figure 6.2 Volunteer create account page.

UCD003 - Create Volunteer opportunity

Figure 6.3 shows page that allows organizations to create and post volunteer opportunities. They can provide details such as role descriptions, required skills, and the number of volunteers needed.

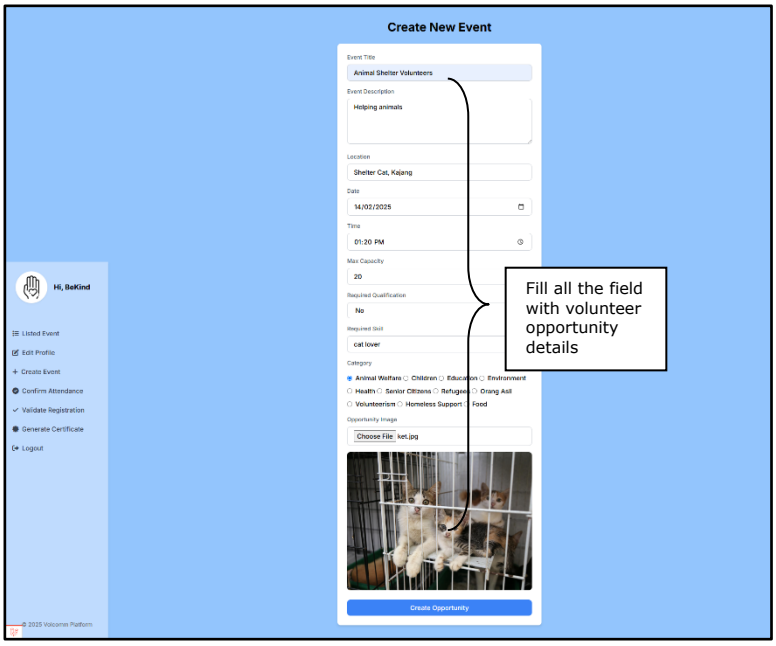


Figure 6.3 Create volunteer opportunity

UCD004 - View volunteer event

Figure 6.4 shows the opportunity page displays available volunteer opportunities. Volunteers can browse through various opportunities, view details such as location, date, and interest, and apply for positions that match their interests.

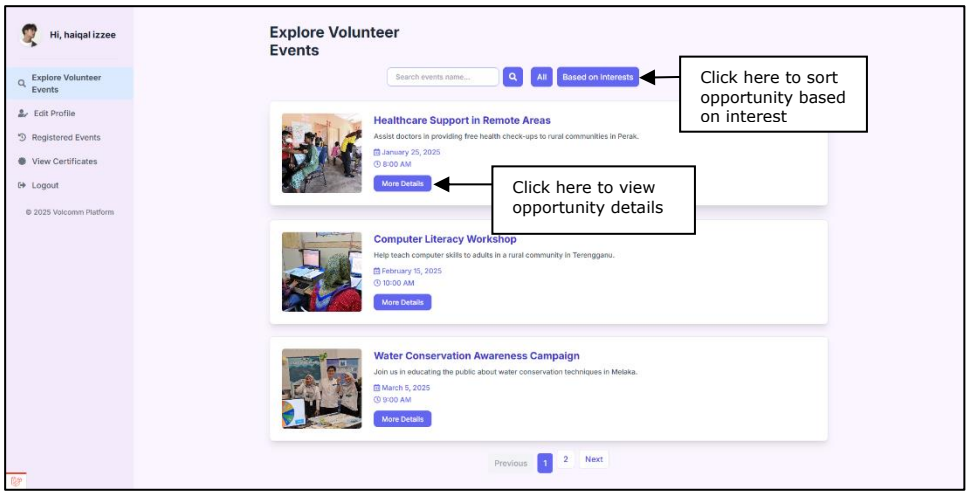


Figure 6.4 View opportunity page

UCD005 - Apply volunteer application

Figure 6.5 shows the selected opportunity details after the user clicks the "More Details" button. It displays information such as the current number of registered volunteers, required skills, and other relevant details.

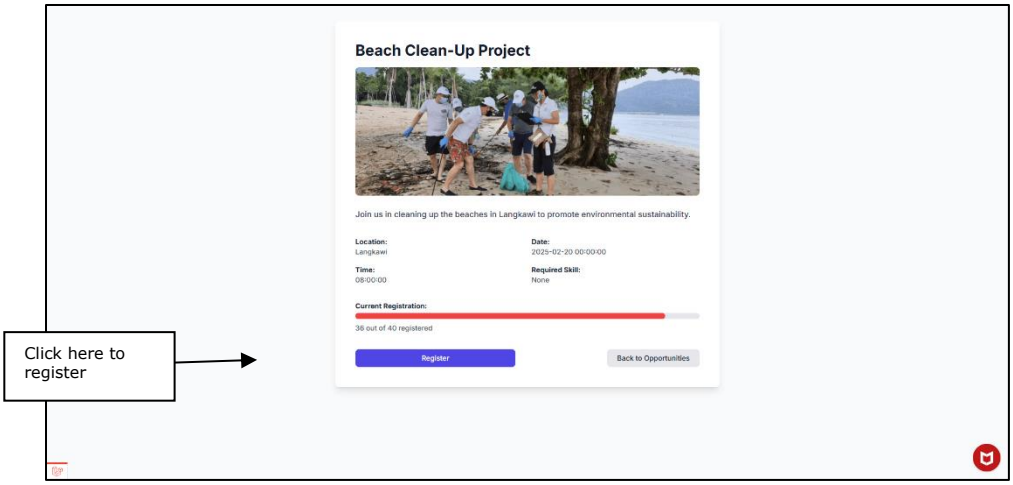


Figure 6.5 Display selected opportunity page

UCD006 - View participation history

Figure 6.6 shows list of volunteer that are already registered under an opportunity that organization gave access to.

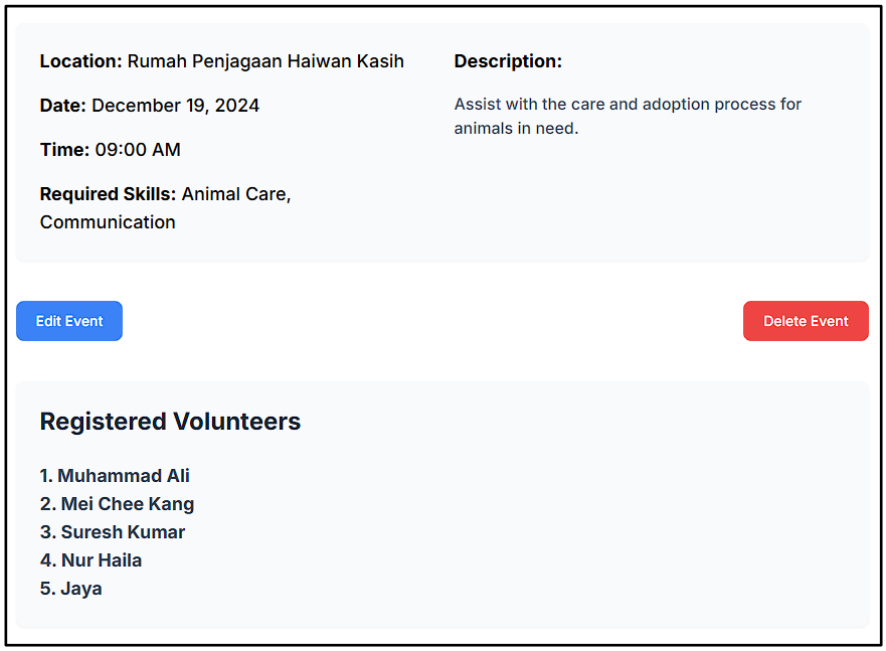


Figure 6.6 Display selected opportunity page

UCD007 - Generate certificate

Figure 6.7 shows the storyboard on how organization generate certificates for volunteers who have completed an opportunity. First,

select the completed opportunity, click "Select All" to tick all volunteers, upload the logo and signature, and enter the signer's name and position.

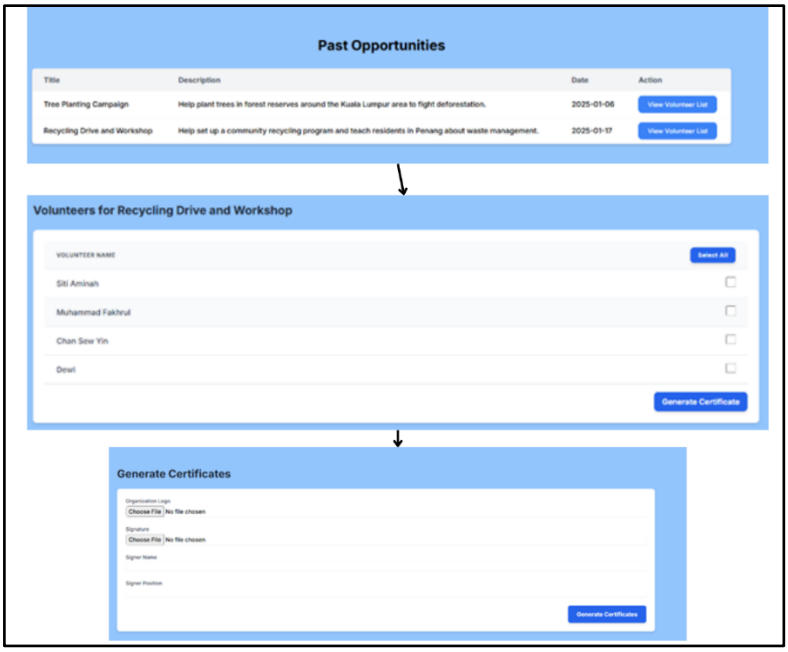


Figure 6.7 Generate certificate storyboard

UCD008 - View certificate

Figure 6.8 shows the "View Certificate" page, displaying certificates earned by volunteers after completing their service. They can click "View Certificate" to access it.



Figure 6.8 View certificate page

UCD009 - View registered volunteers

Figure 6.9 shows after click view details on selected opportunity, organization can see list of volunteers that already registered.

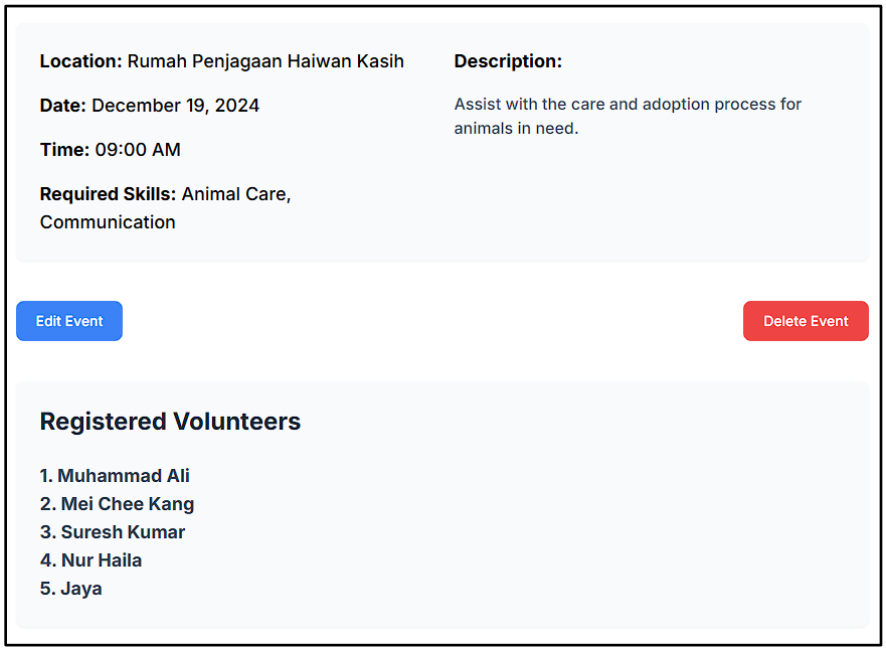


Figure 6.9 View registered volunteers

UCD010 - Review volunteer registration

Figure 6.10 shows the page used by organizations to validate

volunteer registrations for specific opportunities. Before approving or rejecting, organizations can click the qualification file to view the submitted qualifications. Volunteers can then be approved or rejected based on the organization's requirements.



Figure 6.10 Review volunteer registration

UCD011 - Confirm volunteer attendances

Figure 6.11 shows the process of confirming volunteer attendance. Organization can only confirm their attendance on the volunteering day.



Figure 6.11 Storyboard for Confirm attendance

UCD012 - Manage volunteer account

Figure 6.12 displays a page that allows administrators to manage volunteer accounts, including the option to delete an account. Figure 6.13 shows the profile page that appears when the user clicks the "View" button.

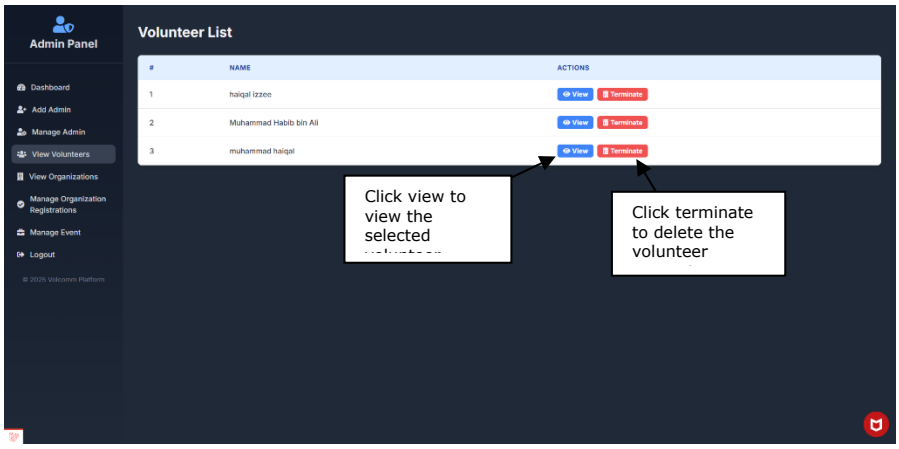


Figure 6.12 Manage volunteer

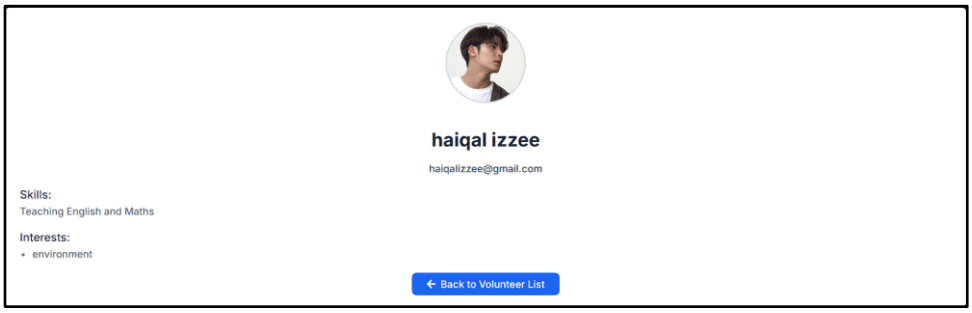


Figure 6.13 Volunteer Profile

UCD013 - Manage organization account

Figure 6.14 displays a page where admin can manage organization profile. It allows them to remove profiles. Figure 6.15 shows the details of a selected organization when the user clicks the "View" button.

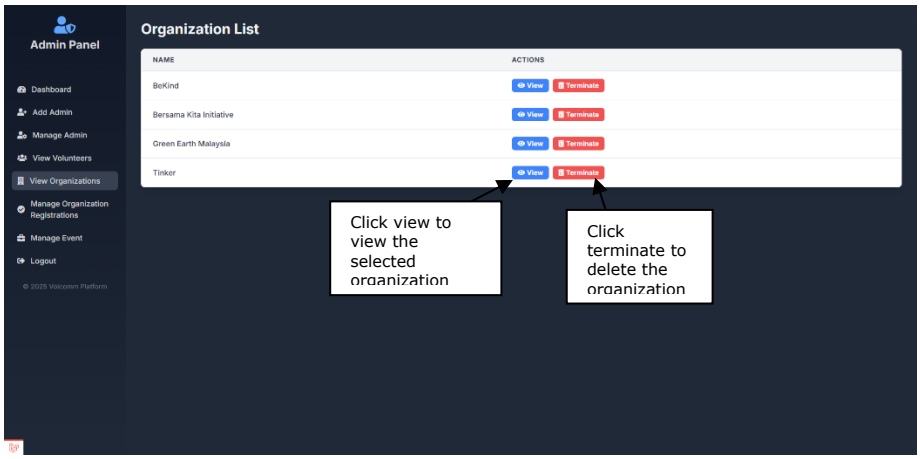


Figure 6.14 Manage organization

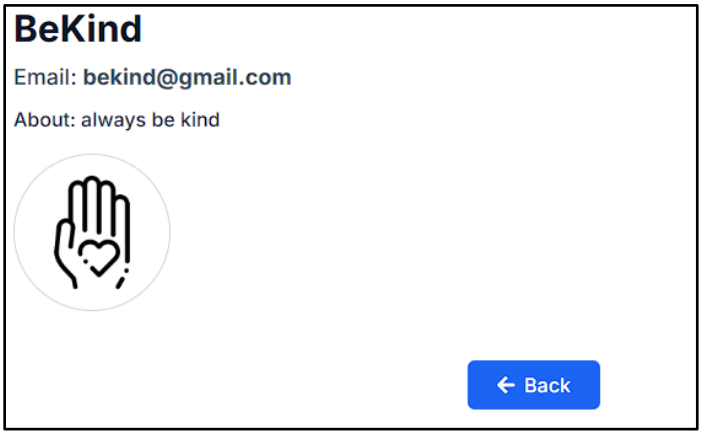


Figure 6.15 View organization page

UCD014 - Manage organization registration

Figure 6.16 displays a page where administrators can approve or reject registration requests submitted by new organizations. They can make a decision to accept or reject the request using the available buttons.

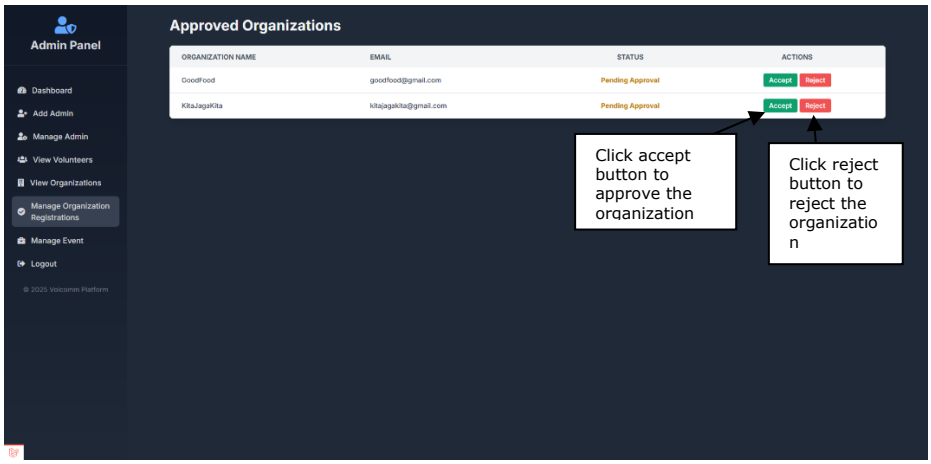


Figure 6.16 Manage organization registration

UCD015 - Add admin

Figure 4.37 shows page that allows an existing admin to add new administrators to the system.

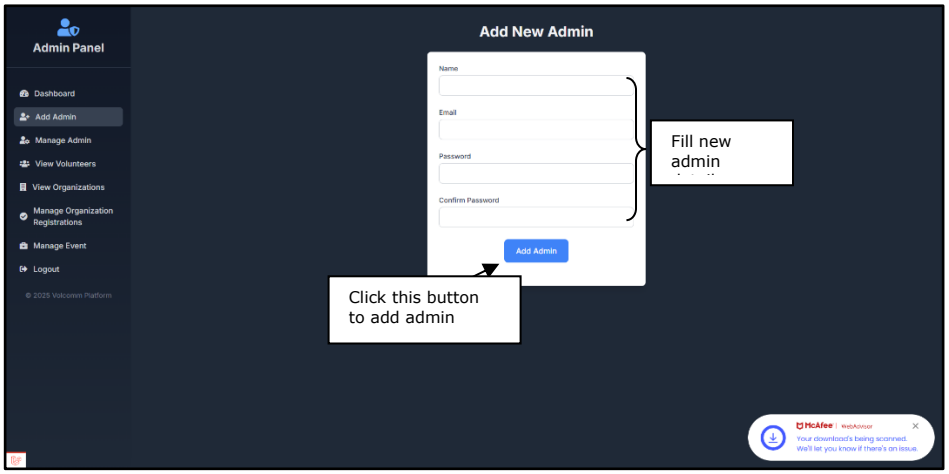


Figure 6.17 Add admin

7. Traceability Requirements Matrix

Table 7.1 Package Content

Package	Class ID	Class Name
SDD_PKG_100	SDD_PKG_101	VolunteerRegister
	SDD_PKG_102	VolunteerLogin
	SDD_PKG_103	VolunteerDashboard
	SDD_PKG_104	ViewOpportunities
	SDD_PKG_105	VolunteerProfile
	SDD_PKG_106	ViewCertificate
	SDD_PKG_107	OrganizationRegister
	SDD_PKG_108	OrganizationLogin
	SDD_PKG_109	OrganizationDashboard
	SDD_PKG_110	CreateOpportunity
	SDD_PKG_111	OrganizationProfile
	SDD_PKG_112	ManageOpportunities
	SDD_PKG_113	AdminLogin
	SDD_PKG_114	AdminDashboard
	SDD_PKG_115	ManageVolunteers
	SDD_PKG_116	ManageOrganizations
SDD_PKG_200	SDD_PKG_201	VolunteerController
	SDD_PKG_202	OrganizationController
	SDD_PKG_203	OpportunityController
	SDD_PKG_204	RegistrationController
	SDD_PKG_205	CertificateController
	SDD_PKG_206	AdminController
	SDD_PKG_207	AttendanceController
SDD_PKG_300	SDD_PKG_301	volunteers
	SDD_PKG_302	organizations
	SDD_PKG_303	opportunities
	SDD_PKG_304	registrations
	SDD_PKG_305	certificates
	SDD_PKG_306	admins
	SDD_PKG_307	attendance

Table 7.2 Use Case Content

Package	Class ID	Class Name
UCD	UCD001	Create Organization account
	UCD002	Create Volunteer account
	UCD003	Create Volunteer event
	UCD004	View volunteer event
	UCD005	Apply volunteer application
	UCD006	View participation history
	UCD007	Generate certificate
	UCD008	View certificate
	UCD009	View registered volunteers
	UCD010	Review volunteer application
	UCD011	Confirm volunteer attendances
	UCD012	Manage volunteer account
	UCD013	Manage organization account
	UCD014	Manage organization application
	UCD015	Add admin

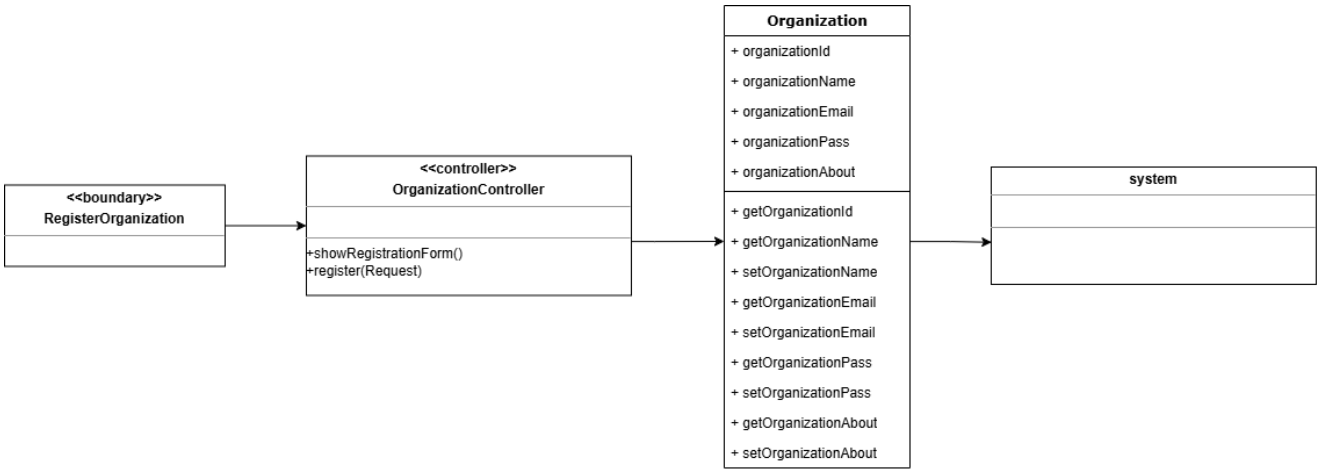
	SDD_PKG_101	SDD_PKG_102	SDD_PKG_103	SDD_PKG_104	SDD_PKG_105	SDD_PKG_106	SDD_PKG_107	SDD_PKG_108	SDD_PKG_109	SDD_PKG_110	SDD_PKG_111	SDD_PKG_112	SDD_PKG_113	SDD_PKG_114	SDD_PKG_115	SDD_PKG_116	SDD_PKG_201	SDD_PKG_202	SDD_PKG_203	SDD_PKG_204	SDD_PKG_205	SDD_PKG_206	SDD_PKG_207	SDD_PKG_301	SDD_PKG_302	SDD_PKG_303	SDD_PKG_304	SDD_PKG_305	SDD_PKG_306	SDD_PKG_307
UCD001							✓											✓							✓						
UCD002	✓																✓							✓						
UCD003										✓									✓							✓					
UCD004				✓															✓							✓					
UCD005																				✓							✓				
UCD006			✓														✓										✓				
UCD007																					✓							✓			
UCD008						✓															✓							✓			
UCD009												✓						✓										✓			
UCD010																		✓		✓								✓			
UCD011																							✓							✓	
UCD012					✓												✓							✓							
UCD013											✓							✓							✓						
UCD014																✓						✓			✓						
UCD015																						✓							✓		

8. Appendices

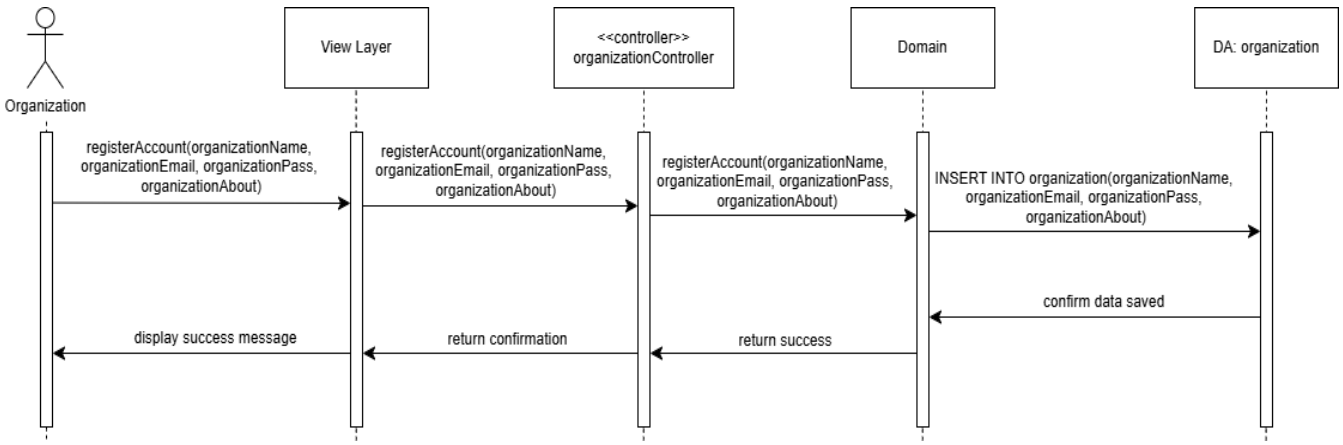
8.1 Create Organization Account



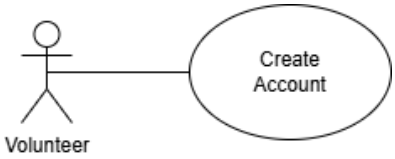
8.1.1 Detail Class Diagram



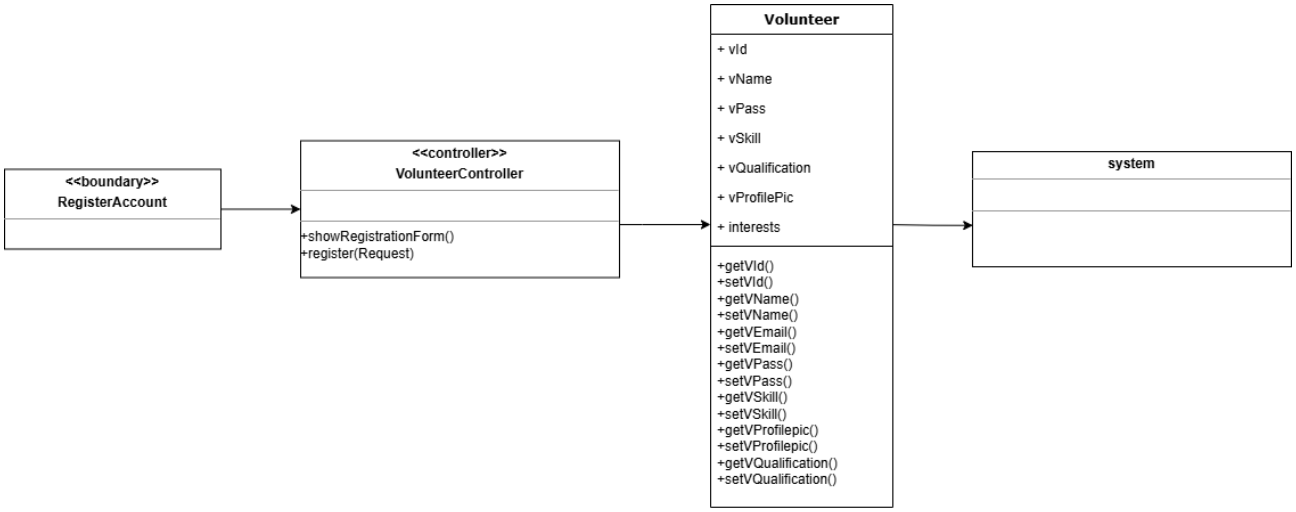
8.1.2 Multilayer Sequence Diagram



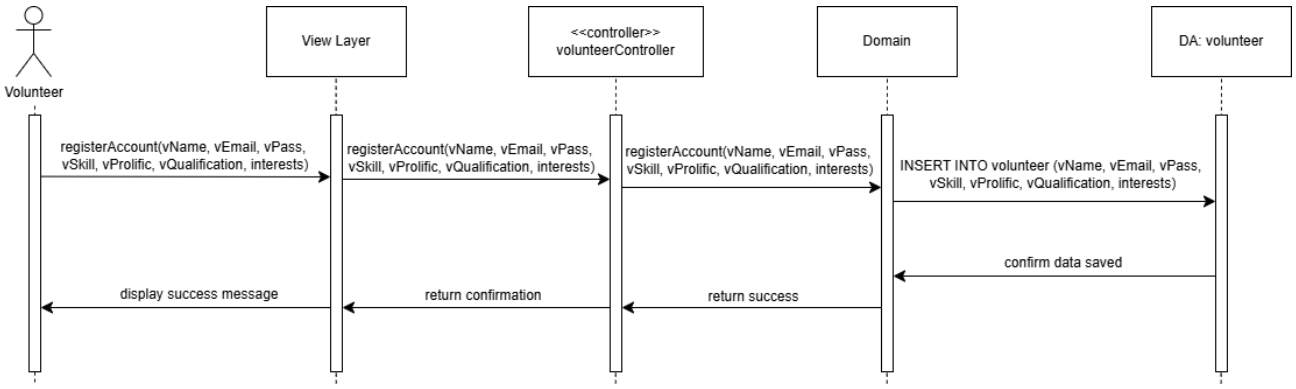
8.2 Create Volunteer Account



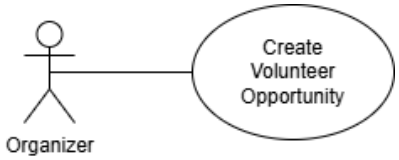
8.2.1 Detail Class Diagram



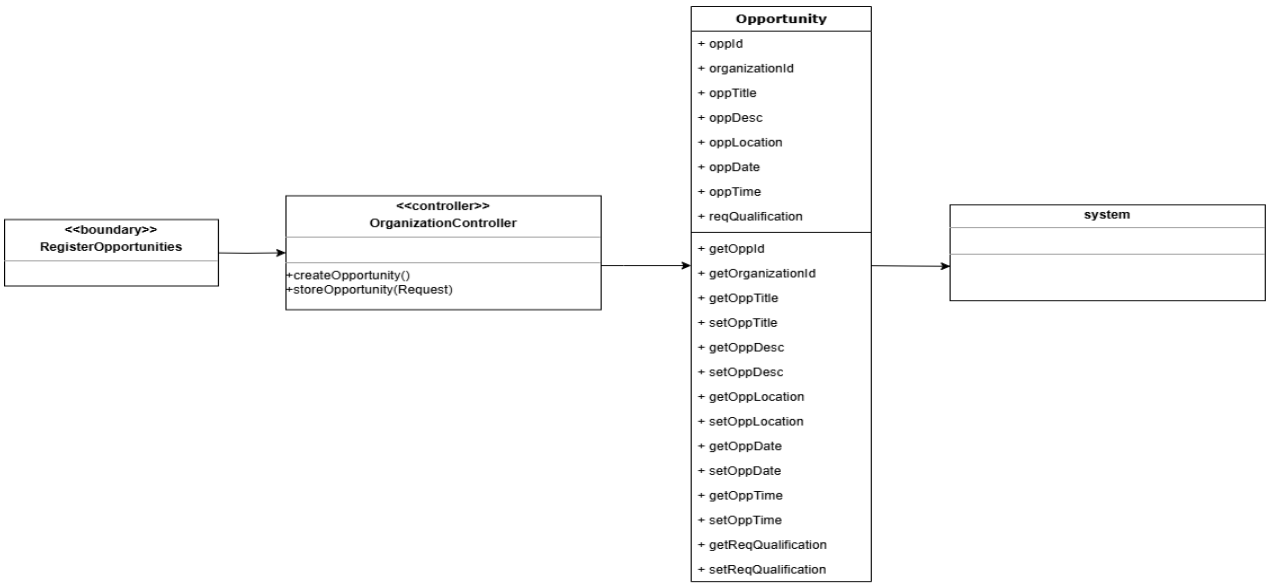
8.2.2 Multilayer Sequence Diagram



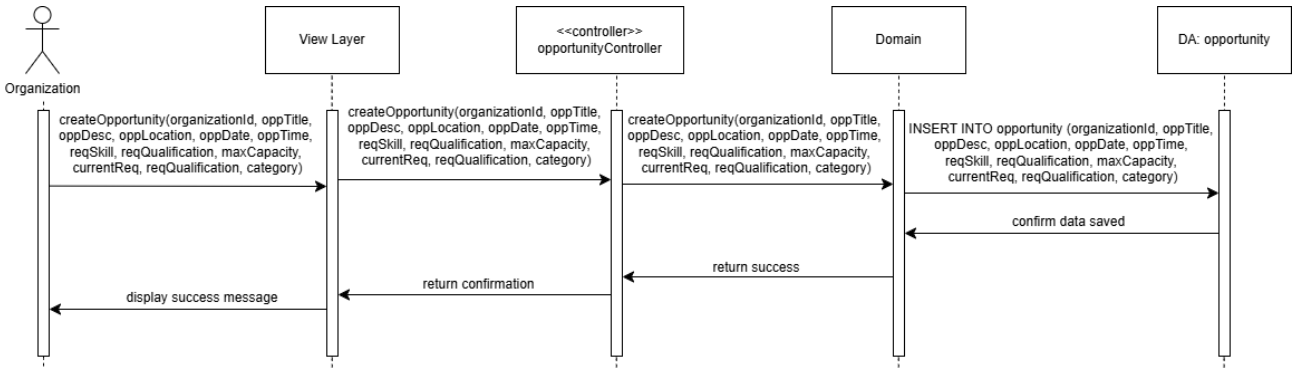
8.3 Create volunteer opportunity



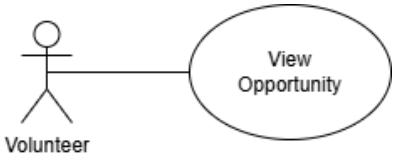
8.3.1 Detail Class Diagram



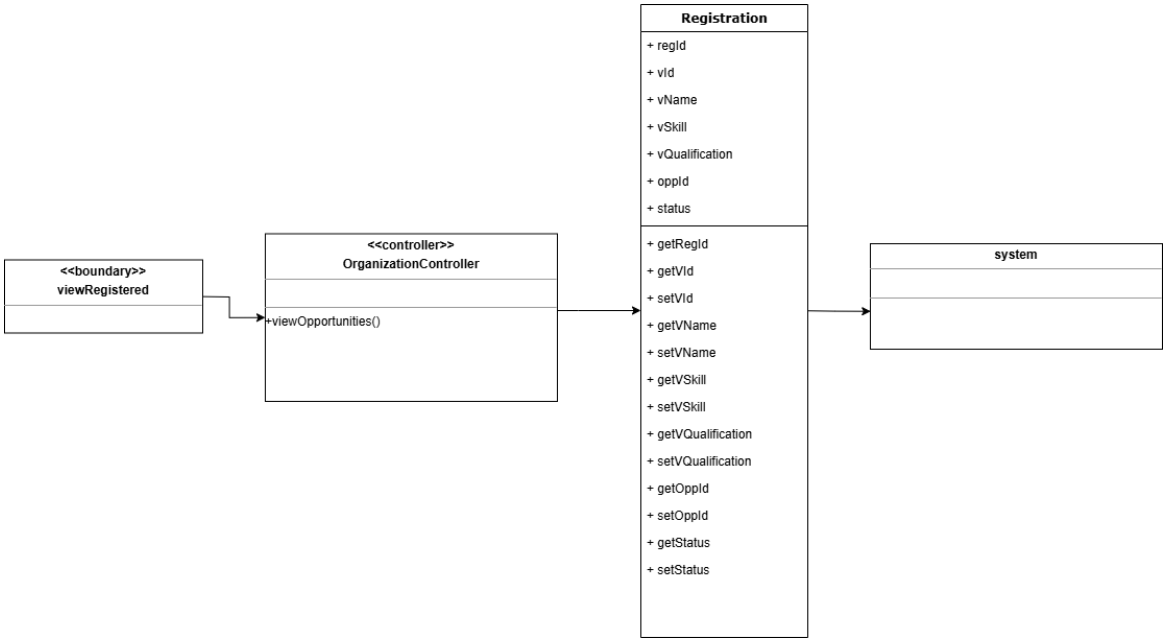
8.3.2 Multilayer Sequence Diagram



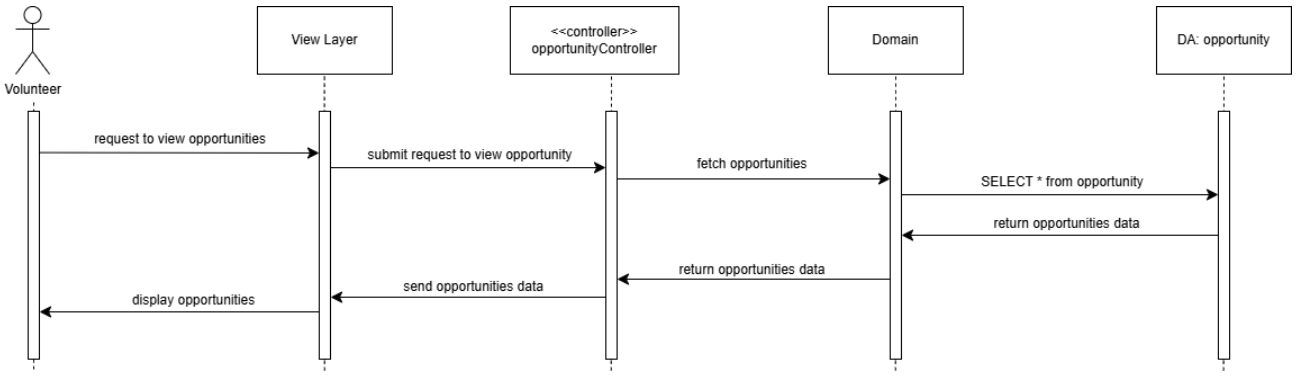
8.4 View volunteer opportunity



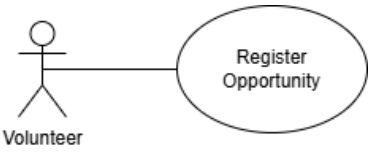
8.4.1 Detail Class Diagram



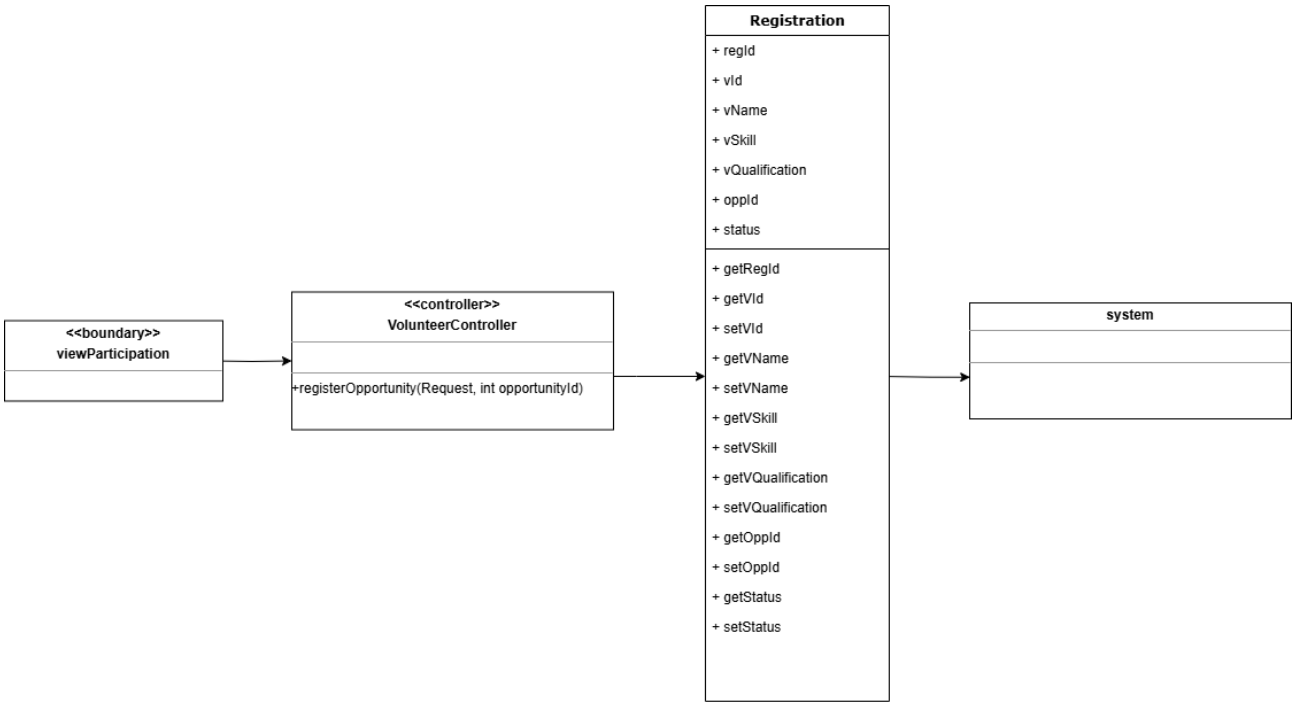
8.4.2 Multilayer Sequence Diagram



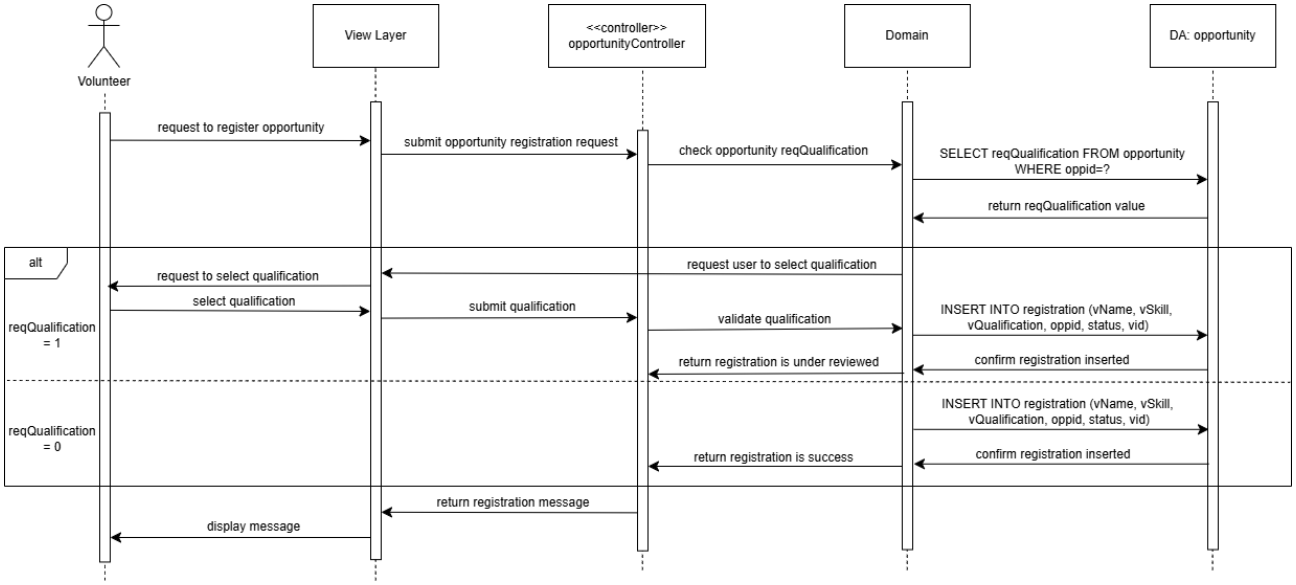
8.5 Apply volunteer application



8.5.1 Detail Class Diagram



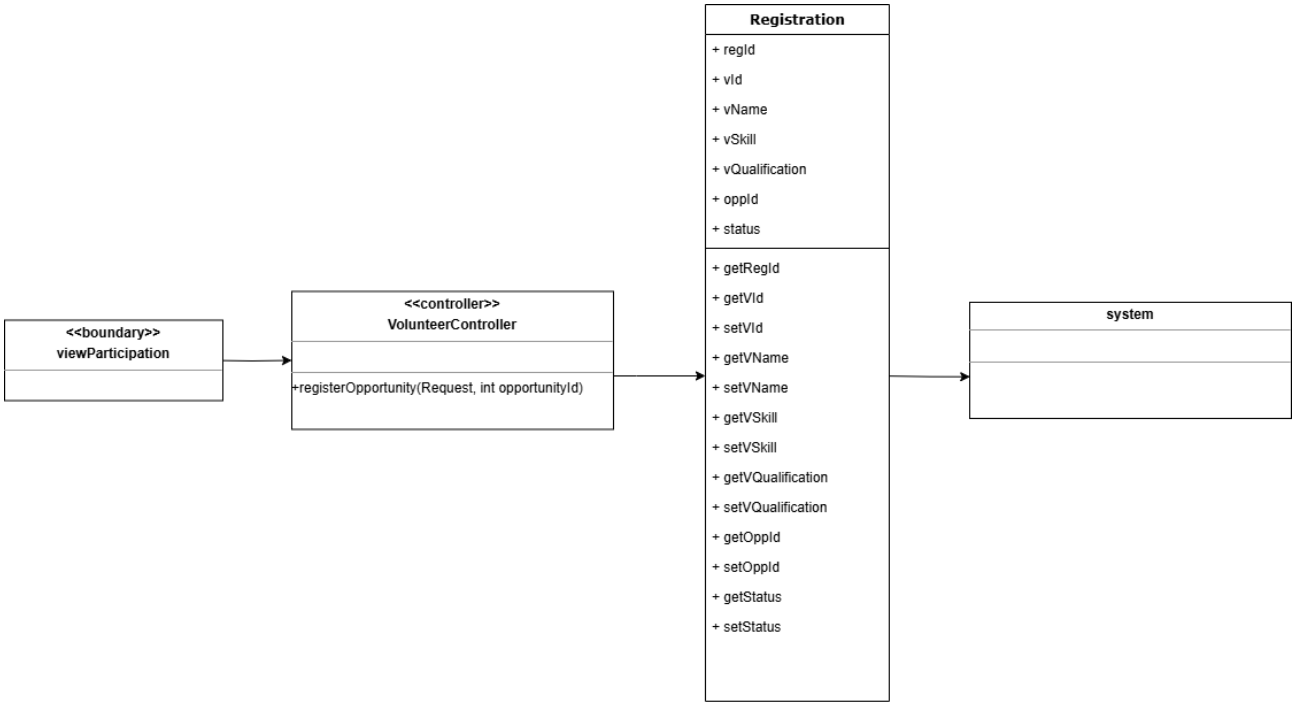
8.5.2 Multilayer Sequence Diagram



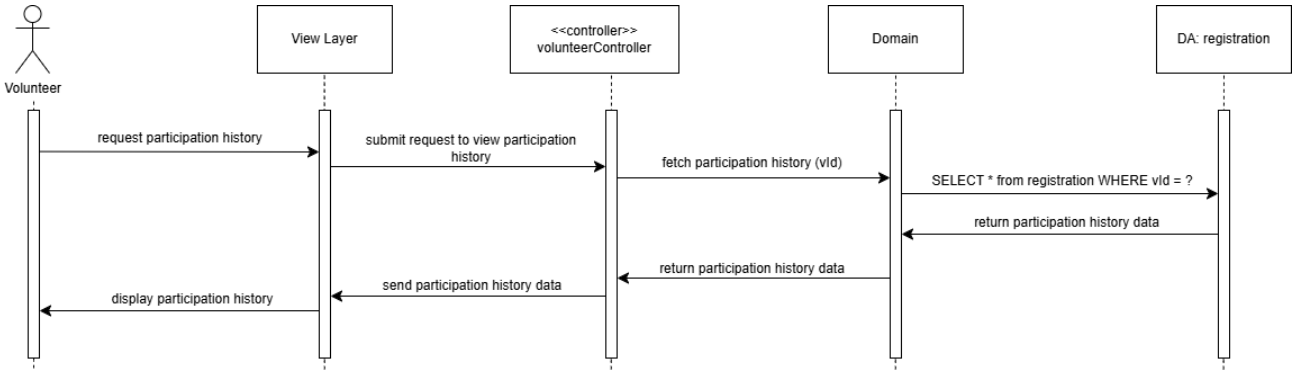
8.6 View Participation history



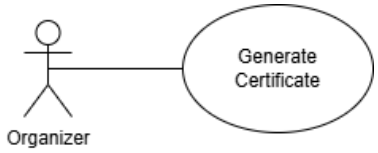
8.6.1 Detail Class Diagram



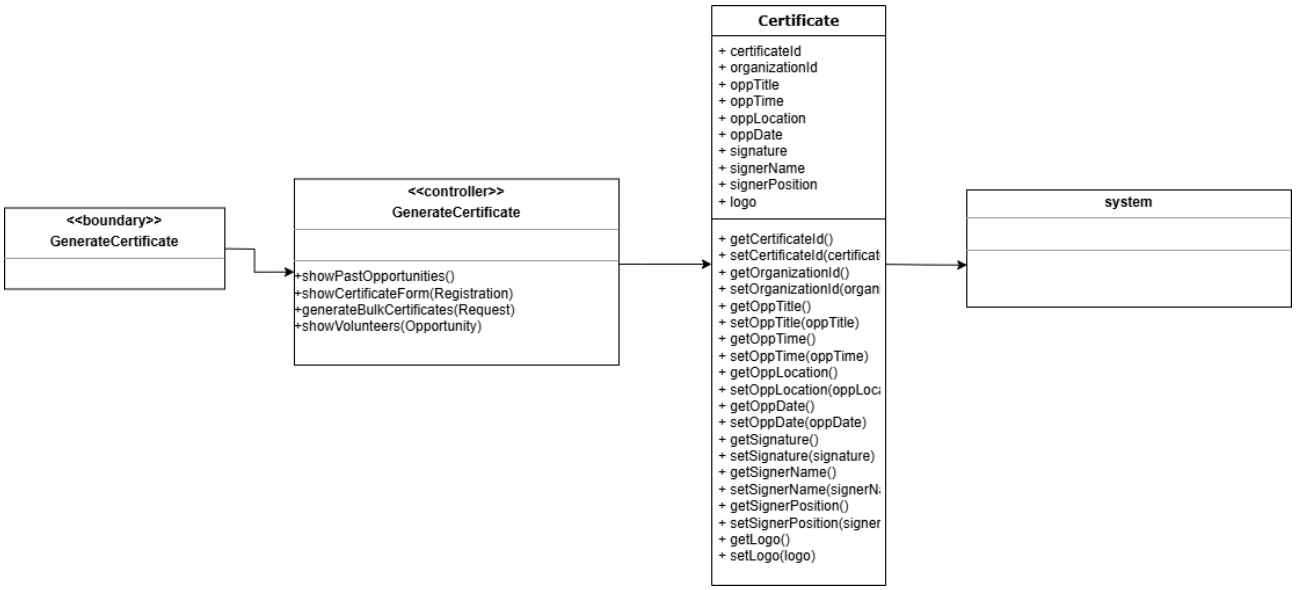
8.6.2 Multilayer Sequence Diagram



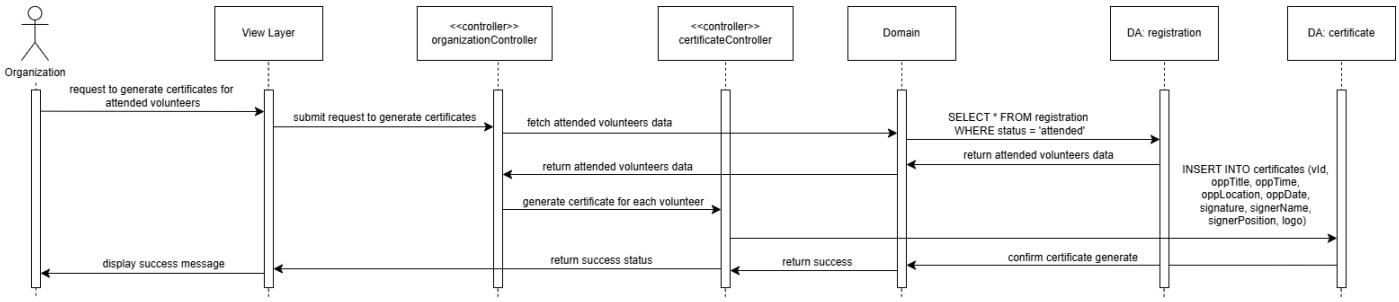
8.7 Generate certificate



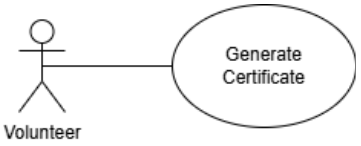
8.7.1 Detail Class Diagram



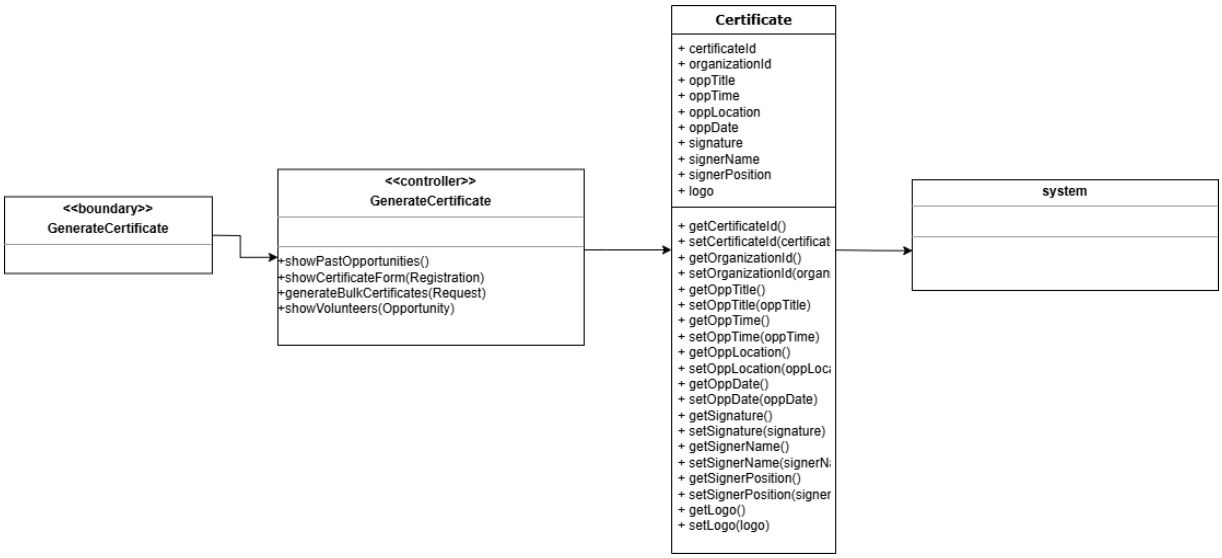
8.7.2 Multilayer Sequence Diagram



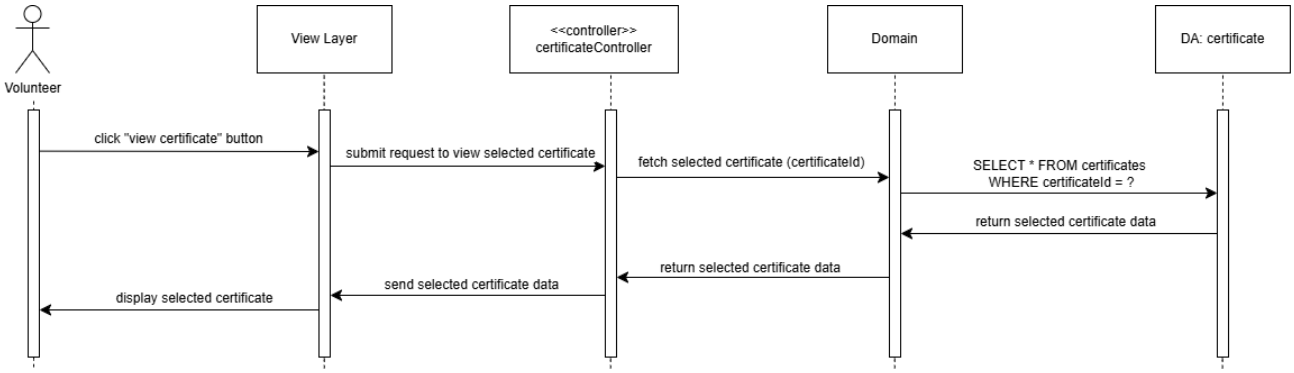
8.8 View Certificate



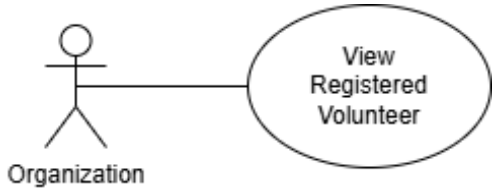
8.8.1 Detail Class Diagram



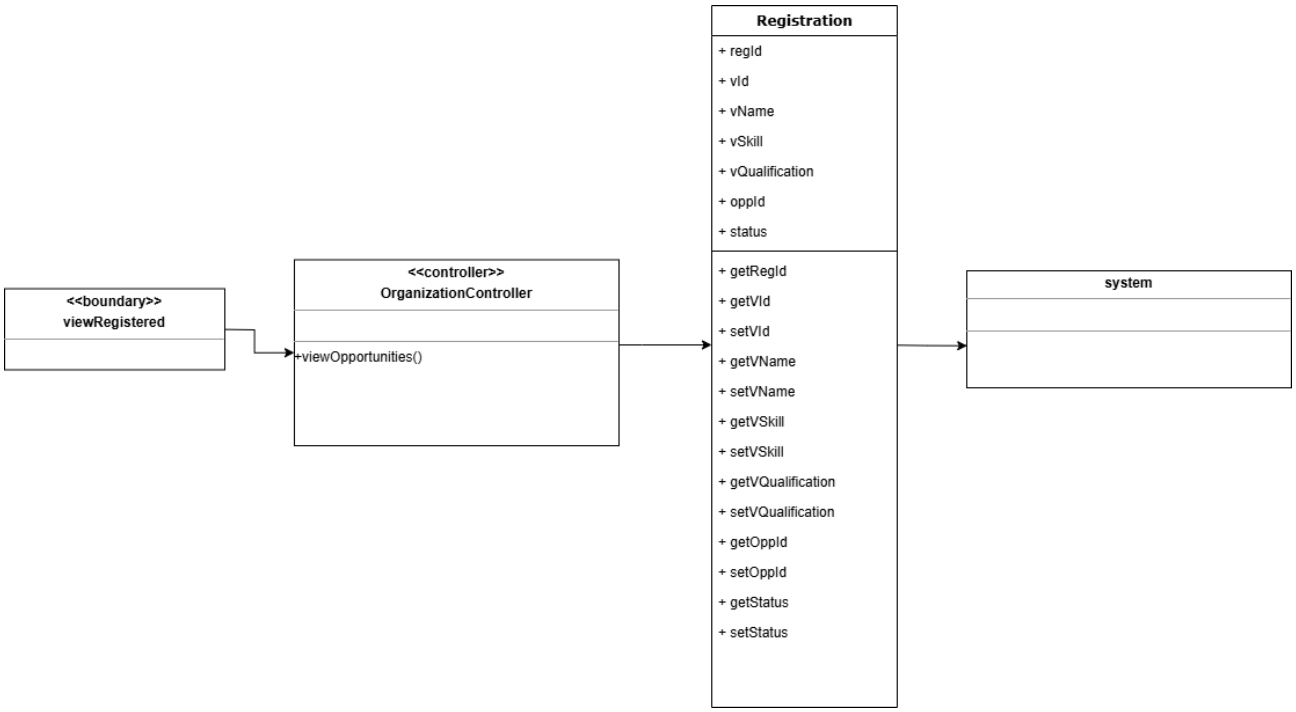
8.8.2 Multilayer Sequence Diagram



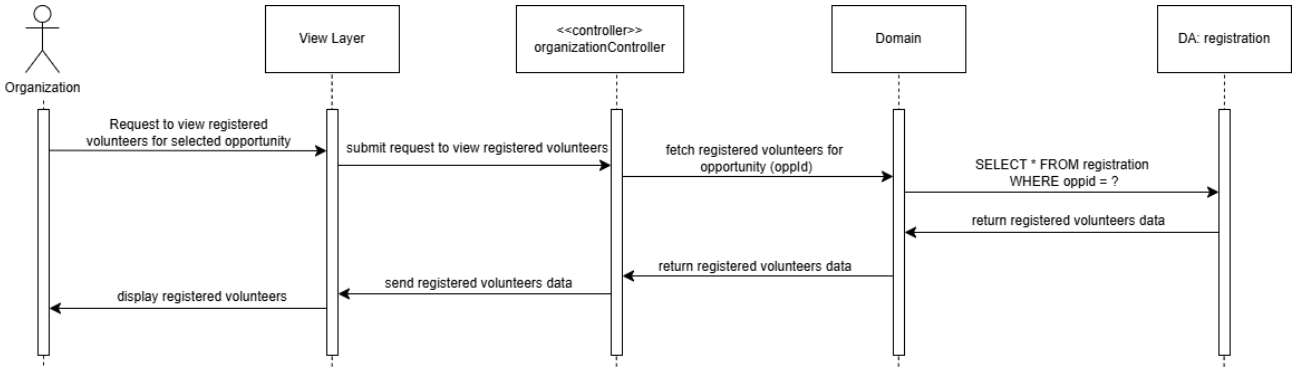
8.9 View registered volunteer



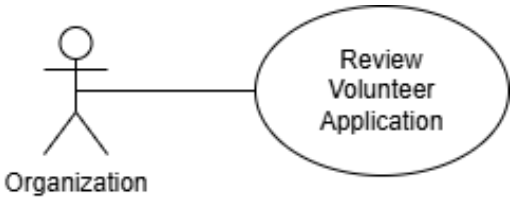
8.9.1 Detail Class Diagram



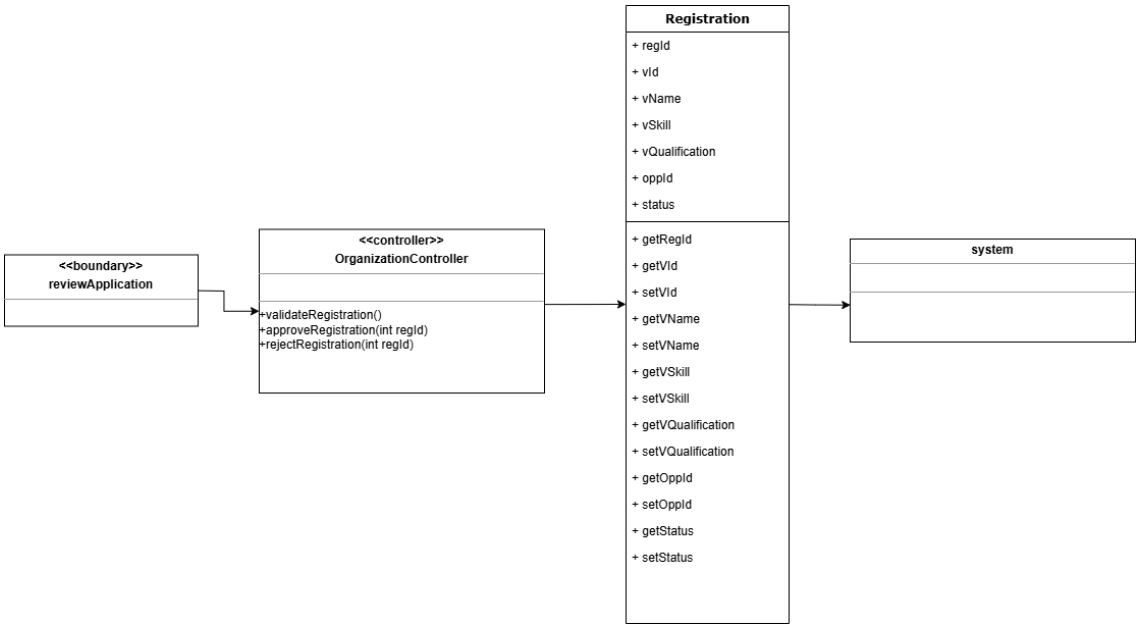
8.9.2 Multilayer Sequence Diagram



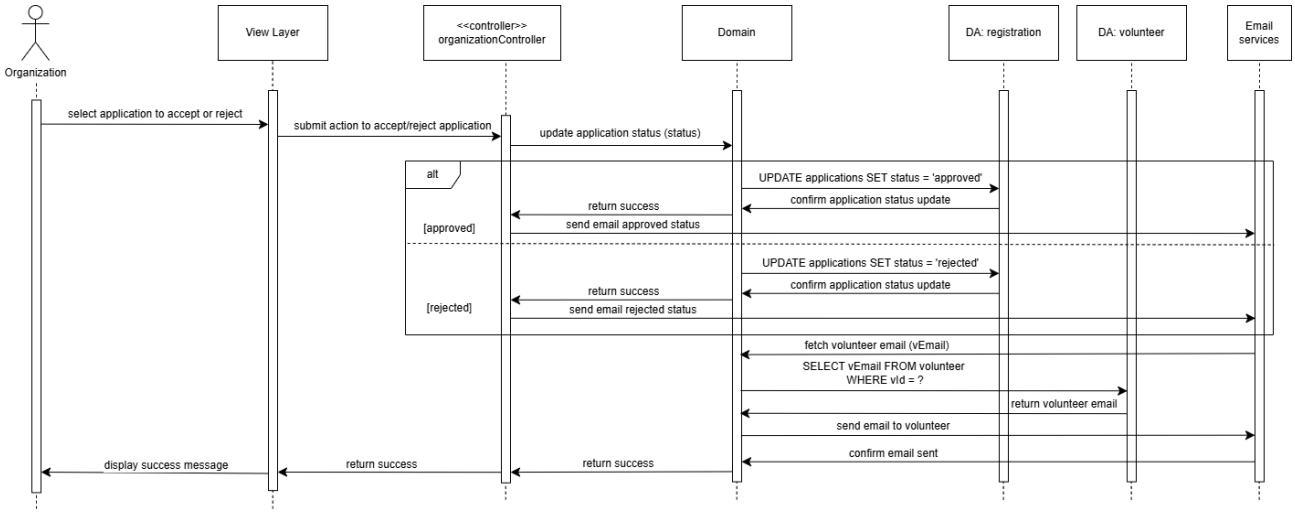
8.10 Review volunteer application



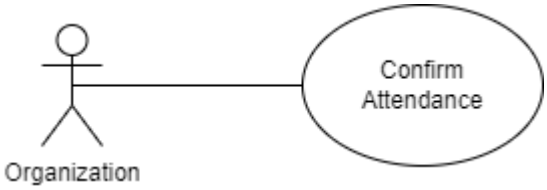
8.10.1 Detail Class Diagram



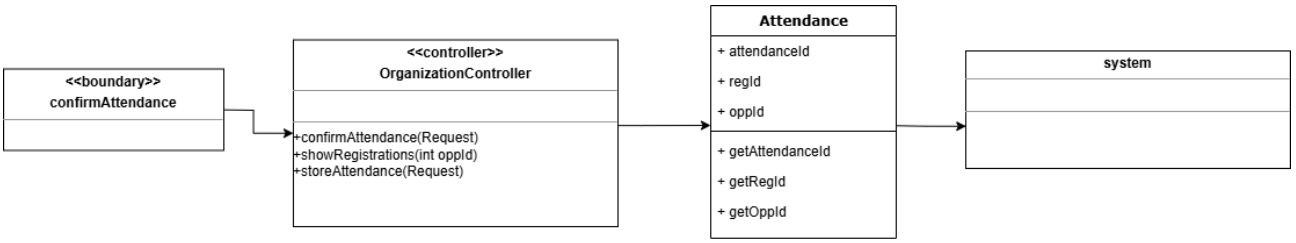
8.10.2 Multilayer Sequence Diagram



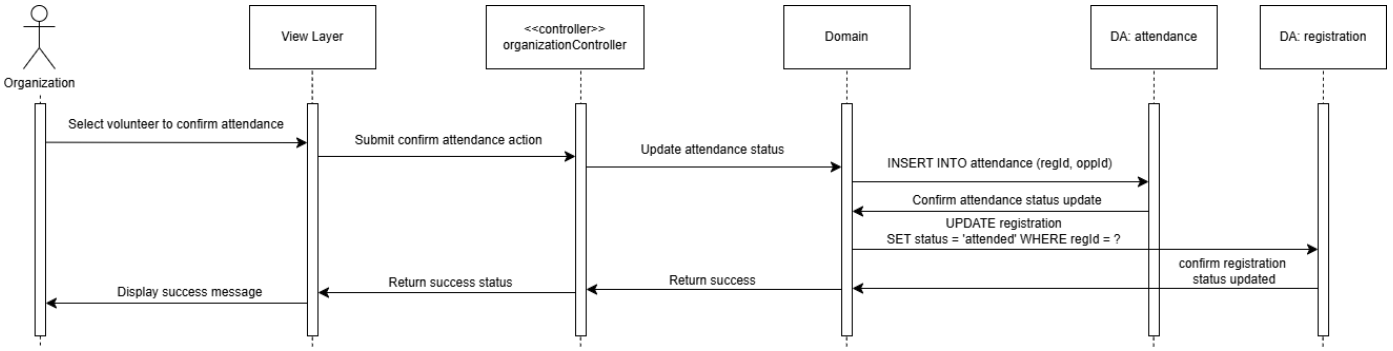
8.11 Confirm Volunteer Attendance



8.11.1Detail Class Diagram



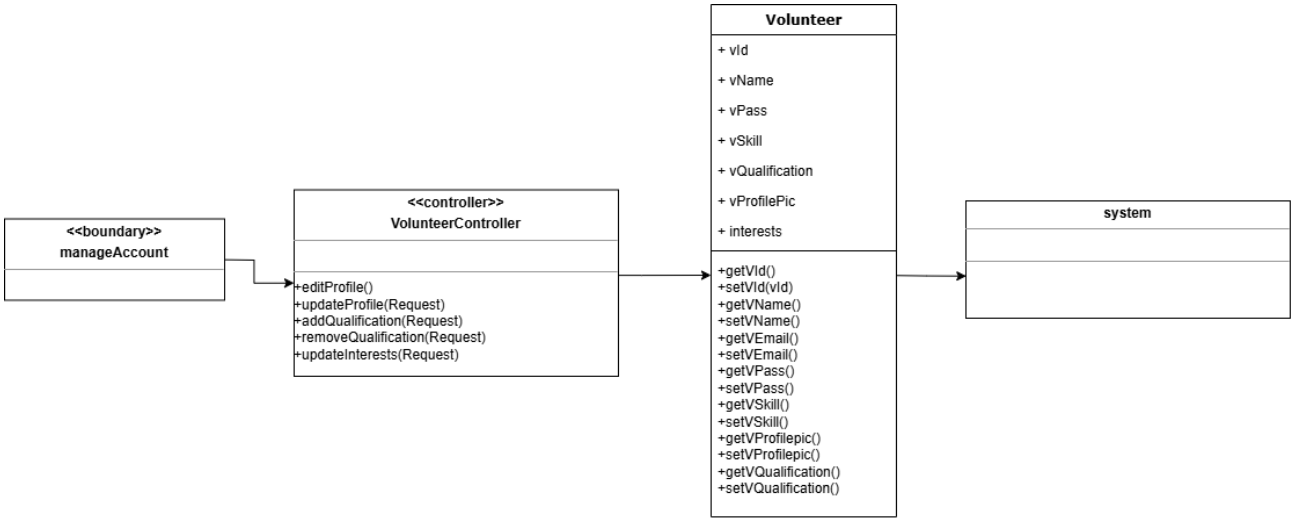
8.11.2 Multilayer Sequence Diagram



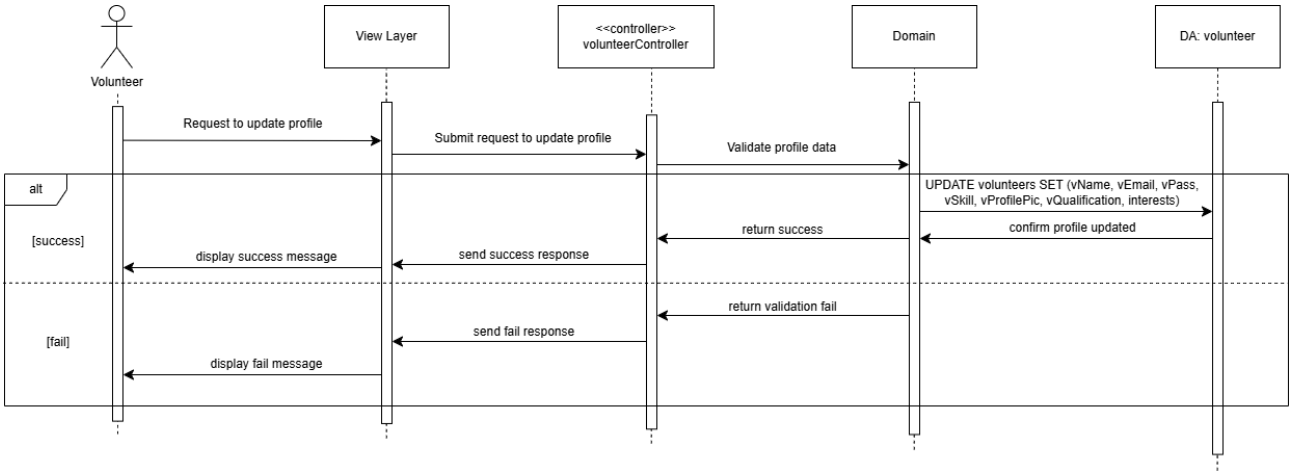
8.12 Manage Volunteer Account



8.12.1 Detail Class Diagram



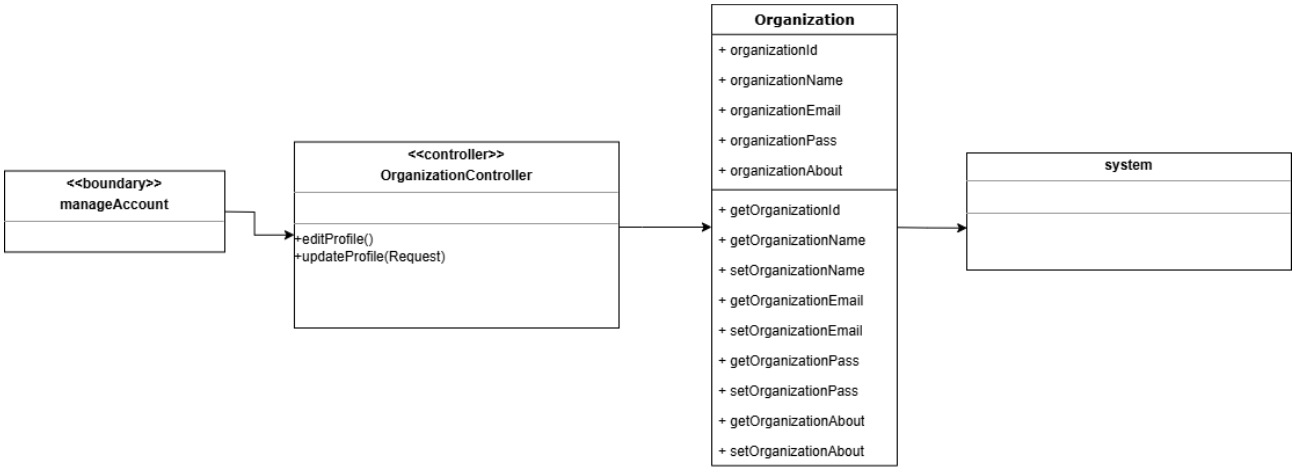
8.12.2 Multilayer Sequence Diagram



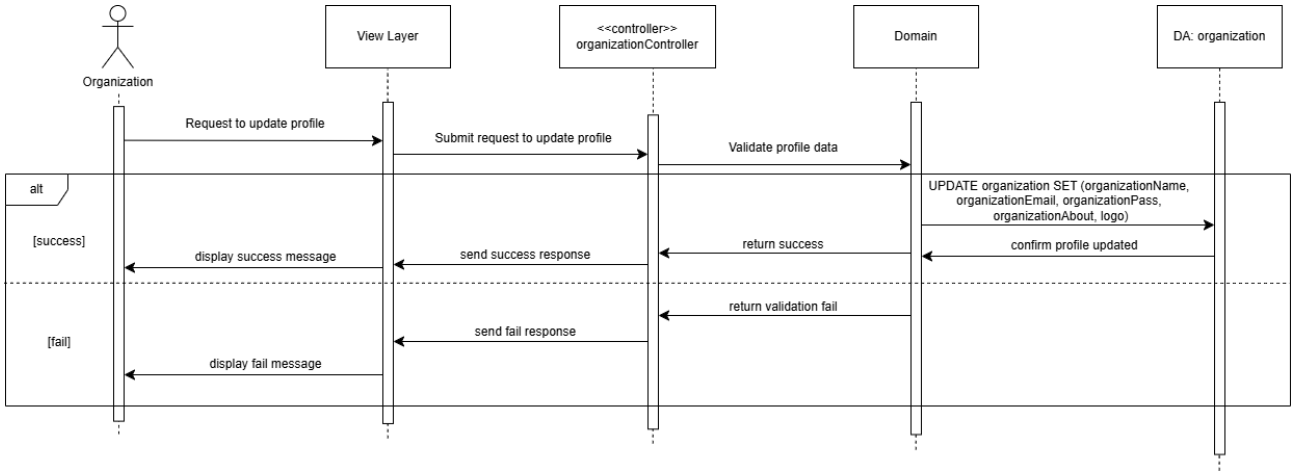
8.13 Manage organization account



8.13.1 Detail Class Diagram



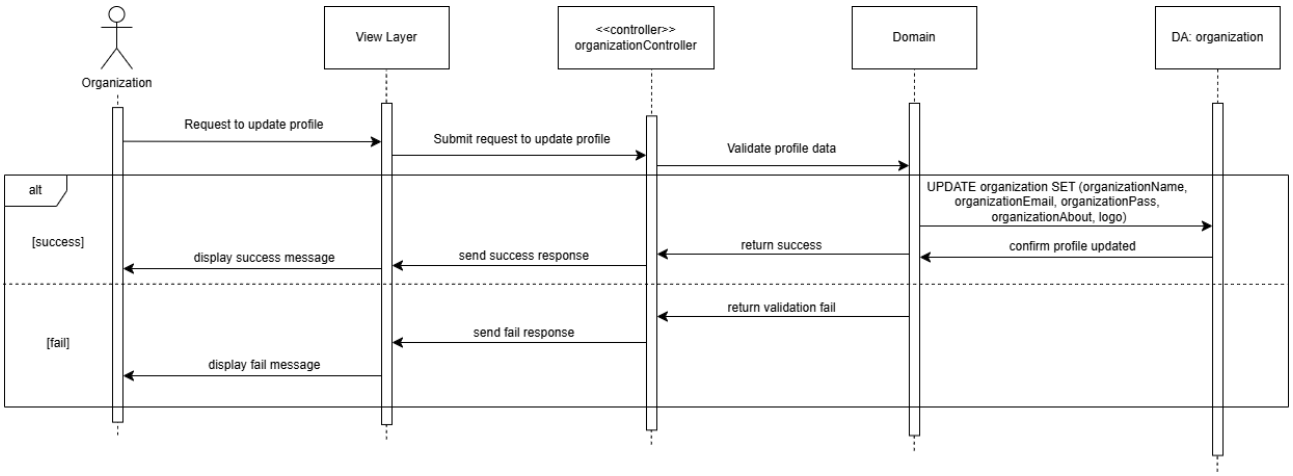
8.13.2 Multilayer Sequence diagram



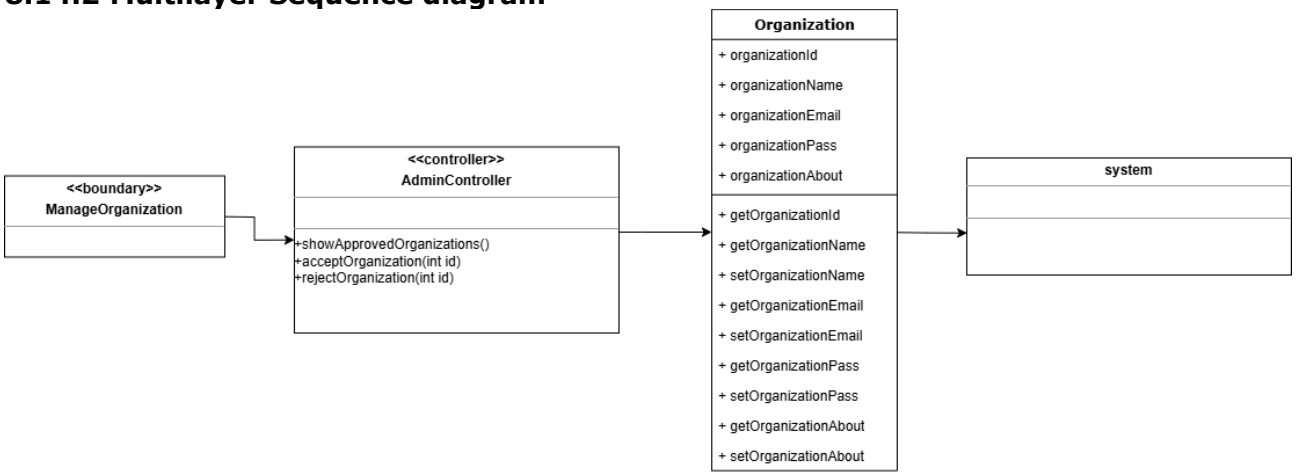
8.14 Manage organization registration



8.14.1Detail Class Diagram



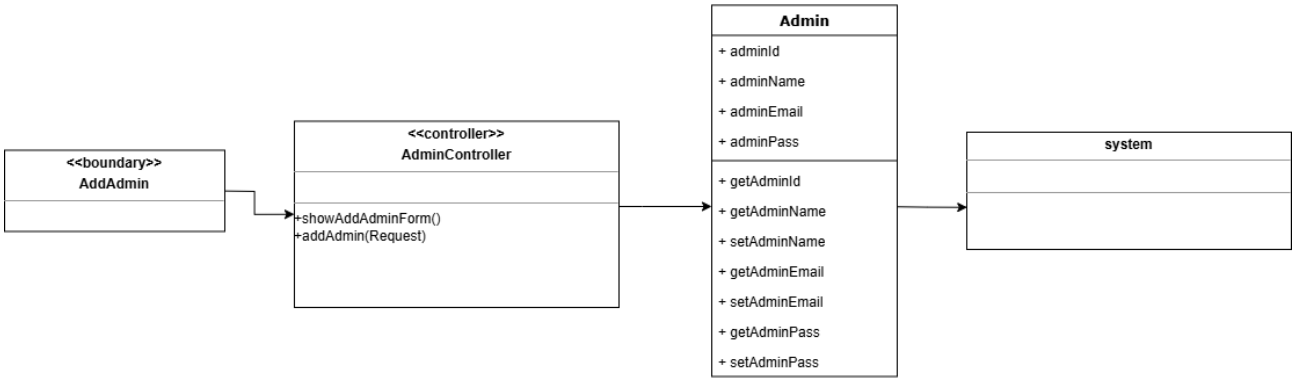
8.14.2 Multilayer Sequence diagram



8.15 Add admin



8.15.1Detail Class Diagram



8.15.2 Multilayer Sequence diagram

